Section 1: Groups, intuitively

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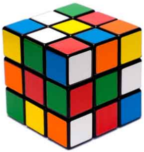
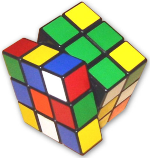
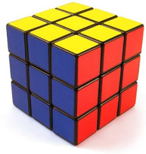
Math 4120, Modern Algebra

A famous toy

Our introduction to group theory will begin by discussing the famous Rubik's Cube.

It was invented in 1974 by Ernő Rubik of Budapest, Hungary.

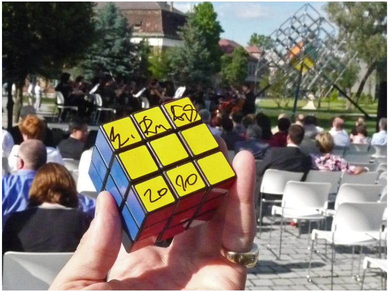
Ernő Rubik is a Hungarian inventor, sculptor and professor of architecture.

According to his Wikipedia entry:

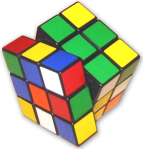
> *He is known to be a very introverted and hardly accessible person, almost impossible to contact or get for autographs.*

A famous toy

- The cube comes out of the box in the solved position:

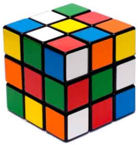


- But then we can scramble it up by consecutively rotating one of its 6 faces:



A famous toy

- The result might look something like this:



- The goal is to return the cube to its original solved position, again by consecutively rotating one of the 6 faces.

Since Rubik's Cube does not seem to require any skill with numbers to solve it, you may be inclined to think that this puzzle is not mathematical.

Big idea

Group theory is not primarily about numbers, but rather about patterns and symmetry; something the Rubik's Cube possesses in abundance.

A famous toy

Let's explore the Rubik's Cube in more detail.

In particular, let's identify some key features that will be recurring themes in our study of patterns and symmetry.

First, some questions to ponder:

- How did we scramble up the cube in the first place? How do we go about unscrambling the cube?
- In particular, what actions, or moves, do we *need* in order to scramble and unscramble the cube? (There are many correct answers.)
- How is Rubik's Cube different from checkers?
- How is Rubik's Cube different from poker?

Four key observations

Observation 1

There is a predefined list of moves that never changes.

Observation 2

Every move is reversible.

Observation 3

Every move is deterministic.

Observation 4

Moves can be combined in any sequence.

In this setting, a *move* is a twist of one of the six faces, by 0°, 90°, 180°, or 270°.

We could add more to our list, but as we shall see, these 4 observations are sufficient to describe the aspects of the mathematical objects that we wish to study.

What does group theory have to do with this?

Group theory studies the mathematical consequences of these 4 observations, which in turn will help us answer interesting questions about symmetrical objects.

Group theory arises everywhere! In puzzles, visual arts, music, nature, the physical and life sciences, computer science, cryptography, and of course, all throughout mathematics.

Group theory is one of the most beautiful subjects in all of mathematics!

Instead of considering our 4 observations as descriptions of Rubik's Cube, let's rephrase them as rules (axioms) that will define the boundaries of our objects of study.

Advantages of our endeavor:

1. We make it clear what it is we want to explore.
2. It helps us speak the same language, so that we know we are discussing the same ideas and common themes, though they may appear in vastly different settings.
3. The rules provide the groundwork for making logical deductions, so that we can discover new facts (many of which are surprising!).

Our informal definition of a group

Rule 1

There is a predefined list of actions that never changes.

Rule 2

Every action is reversible.

Rule 3

Every action is deterministic.

Rule 4

Any sequence of consecutive actions is also an action.

Definition (informal)

A group is a set of actions satisfying Rules 1–4.

Observations about the "Rubik's Cube group"

Frequently, two sequences of moves will be "indistinguishable." We will say that two such moves are *the same*. For example, rotating a face (by 90°) once has the same effect as rotating it five times.

Fact

There are 43,252,003,274,489,856,000 distinct configurations of the Rubik's cube.

While there are infinitely many possible sequences of moves, starting from the solved position, there are 43,252,003,274,489,856,000 "truly distinct" moves.

All 4.3×10^{19} moves are generated by just 6 moves: a 90° clockwise twist of one of the 6 faces.

Let's call these generators a, b, c, d, e, and f. Every word over the alphabet $\{a, b, c, d, e, f\}$ describes a unique configuration of the cube (starting from the solved position).

Summary of the big ideas

Loosing speaking a group is a **set of actions** satisfying some mild properties: deterministic, reversibility, and closure.

A generating set for a group is a **subcollection of actions** that together can produce all actions in the group – like a spanning set in a vector space.

Usually, a generating set is *much smaller* than the whole group.

Given a generating set, the individual actions are called generators.

The set of all possible ways to scramble a Rubik's cube is an example of a group. Two actions are the same if they have the *same "net effect"*, e.g., twisting a face 1 time vs. twisting a face 5 times.

Note that the group is the set of actions one can perform, *not* the set of configurations of the cube. However, there is a bijection between these two sets.

The Rubik's cube group has 4.3×10^{19} actions but we can find a generating set of size 6.

A road map for the Rubik's Cube

There are many solution techniques for the Rubik's Cube. If you do a Google search, you'll find several methods for solving the puzzle.

These methods describe a sequence of moves to apply relative to some starting position. In many situations, there may be a shorter sequence of moves that would get you to the solution.

In fact, it was shown in July 2010 that every configuration is at most 20 moves away from the solved position!

Let's pretend for a moment that we were interested in writing a complete solutions manual for the Rubik's Cube.

Let me be more specific about what I mean.

A road map for the Rubik's Cube

We'd like our solutions manual to have the following properties:

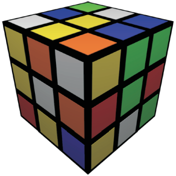
1. Given any scrambled configuration of the cube, there is a unique page in the manual corresponding to that configuration.
2. There is a method for looking up any particular configuration. (The details of how to do this are unimportant.)
3. Along with each configuration, a list of available moves is included. In each case, the page number for the outcome of each move is included, along with information about whether the corresponding move takes us closer to or farther from the solution.

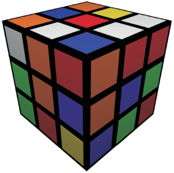
Let's call our solutions manual the *Big Book*.

Page 12,574,839,257,438,957,431 from the *Big Book*

You are **15 steps** from the solution.

Cube front





Cube back

Face	**Direction**	**Destination page**	**Progress**
Front	Clockwise	36,131,793,510,312,058,964	Closer to solved
Front	Counterclockwise	12,374,790,983,135,543,959	Farther to solved
Back	Clockwise	26,852,265,690,987,257,727	Closer to solved
Back	Counterclockwise	41,528,397,002,624,663,056	Farther to solved
Left	Clockwise	6,250,961,334,888,779,935	Closer to solved
Left	Counterclockwise	10,986,196,967,552,472,974	Farther to solved
Right	Clockwise	26,342,598,151,967,155,423	Farther to solved
Right	Counterclockwise	40,126,637,877,673,696,987	Closer to solved
Top	Clockwise	35,131,793,510,312,058,964	Closer to solved
Top	Counterclockwise	33,478,478,689,143,786,858	Farther to solved
Bottom	Clockwise	20,625,256,145,628,342,363	Farther to solved
Bottom	Counterclockwise	7,978,947,168,773,308,005	Closer to solved

A road map for the Rubik's Cube

We can think of the *Big Book* as a road map for the Rubik's Cube. Each page says, "you are here" and "if you follow this road, you'll end up over there."

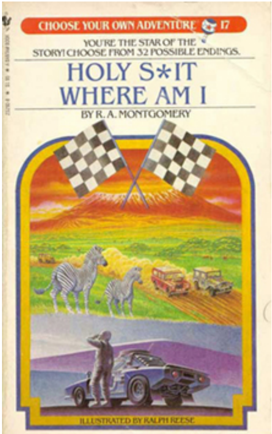


Figure: Potential cover and alternative title for the *Big Book*

A road map for the Rubik's Cube

Unlike a vintage *Choose Your Own Adventure* book, you'll additionally know whether "over there" is where you want to go or not.

Pros of the *Big Book*:

- We can solve any scrambled Rubik's Cube.
- Given any configuration, every possible sequence of moves for solving the cube is listed in the book (long sequences and short sequences).
- The *Big Book* contains complete data on the moves in the Rubik's Cube universe and how they combine.

Cons of the *Big Book*:

- We just took all the fun out of the Rubik's Cube.
- If we had such a book, using it would be fairly cumbersome.
- We can't actually make such a book. Rubik's Cube has more than 4.3×10^{19} configurations. The paper required to write the book would cover the Earth many times over. The book would require over a billion terabytes of data to store electronically, and no computer in existence can store that much data.

What have we learned?

Despite the *Big Book*'s apparent shortcomings, it made for a good thought experiment.

The most important thing to get out of this discussion is that the *Big Book* is a map of a group.

We shall not abandon the mapmaking ideas introduced by our discussion of the *Big Book* simply because the map is too large.

We can use the same ideas to map out any group. In fact, we shall frequently do exactly that.

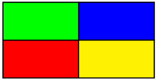
Let's try something simpler...

The Rectangle Puzzle group

- Consider a clear glass **rectangle** and label it as follows:

1	2
4	3

If you prefer, you can use colors instead of numbers:



We'll use numbers, and call the above configuration the **solved state** of our puzzle.

- The idea of the game is to scramble the puzzle and then find a way to return the rectangle to its solved state.

- Our "predefined list" consists of two actions: horizontal flip and vertical flip.

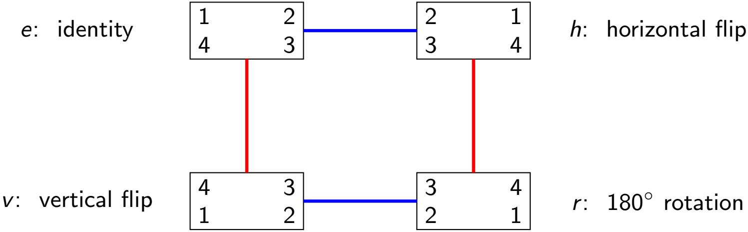
  Loosely speaking, we allow these moves because they preserve the "footprint" of the rectangle. Do any other moves preserve its footprint?

Road map for The Rectangle Puzzle

For covenience, let's say that when we flip the rectangle, the numbers automatically become "right-side-up," as they would if you rotated an iPhone.

It is not hard to see that using only sequences of horizontal and vertical flips, we can obtain only four configurations.

Unlike the Rubik's cube group, the "road map" of the rectangle puzzle is small enough that we can draw it.



Observations? What sorts of things does the map tell us about the group?

Observations

Let G denote the rectangle group. This is a **set** of four actions. We see:

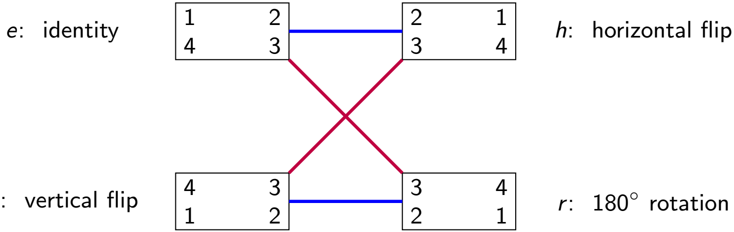
- G has 4 actions: the "*identity*" action e, a horizontal flip h, a vertical flip v, and a 180° rotation r.
$$G = \{e, h, v, r\}.$$
- We need two actions to "generate" G. In our diagram, each **generator** is represented by a different type (color) of arrow. We write:
$$G = \langle h, v \rangle .$$
- The map shows us how to get from any one configuration to any other. There is more than one way to follow the arrows! For example
$$r = hv = vh .$$
- For this particular group, the order of the actions is irrelevant! We call such a group **abelian**. Note that the Rubik's cube group is *not* abelian.
- Every action in G is its own **inverse**: That is,
$$e = e^2 = h^2 = v^2 = r^2 .$$
The Rubik's cube group does **not** have this property. Algebraically, we write:
$$e^{-1} = e, \qquad v^{-1} = v, \qquad h^{-1} = h, \qquad r^{-1} = r .$$

An alternative set of generators for the Retangle Puzzle

The rectangle puzzle can also generated by a horizontal flip and a 180° rotation:

$$G = \langle h, r \rangle .$$

Let's build a Cayley graph using this alternative set of generators.



Do you see this road map has the "same structure" as our first one? Of course, we need to "untangle it" first.

Perhaps surprisingly, this might *not* always be the case.

That is, there are (more complicated) groups for which different generating sets yield road maps that are structurally different. We'll see examples of this shortly.

Cayley diagrams

As we saw in the previous example, how we choose to layout our map is irrelevant.

What is important is that the connections between the various states are preserved.

However, we will attempt to construct our maps in a pleasing to the eye and symmetrical way.

The official name of the type of group road map that we have just created is a Cayley diagram, named after 19th century British mathematician Arthur Cayley.

In general, a Cayley diagram consists of nodes that are connected by colored (or labeled) arrows, where:

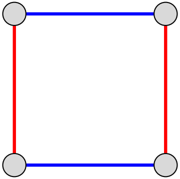
- an arrow of a particular color represents a specific generator;
- each action of the group is represented by a unique node (sometimes we will label nodes by the corresponding action).
- Equivalently, each action is represented by a (non-unique) path starting from the **solved state**.

More on arrows

- An arrow corresponding to the generator g from node x to node y means that node y is the result of applying the action $g \in G$ to node x:

- If an action $h \in G$ is its own inverse (that is, $h^2 = e$), then we have a 2-way arrow. This happens with horizontal and vertical flips. For clarity, our convention is to drop the tips on all 2-way arrows. Thus, these are exactly the same:

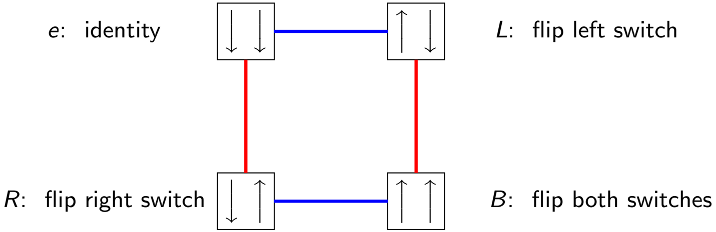
When we want to focus on a group's structure, we frequently omit the labels at the nodes. Thus, the Cayley diagram of the rectangle puzzle can be drawn as follows:



The 2-Light Switch group

Let's map out another group, which we'll call the *2-Light Switch Group*:

- Consider two light switches side by side that both start in the off position (This is our "solved state").
- We are allowed 2 actions: flip L switch and flip R switch.



Notice how the Cayley diagrams for the Rectangle Puzzle $G = \{e, v, h, r\}$ and the 2-Light Switch Group $G' = \{e, L, R, B\}$ are essentially the same.

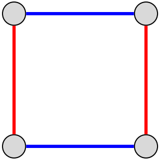
Although these groups are superficially different, the Cayley diagrams help us see that *they have the same structure*. (The fancy phrase for this property is that the "*two groups are isomorphic*"; more on this later.)

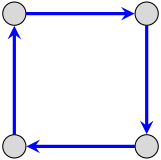
The Klein 4-group

Any group with the same Cayley diagram as the Rectangle Puzzle and the 2-Light Switch Group is called the Klein 4-group, denoted by V_4 for *vierergruppe*, "four-group" in German. It is named after the mathematician Felix Klein.

It is important to point out that the number of different types (i.e., colors) of arrows matters. For example, the Cayley diagram on the right *does not* represent V_4.

$G = V_4$





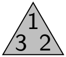
$G = ???$

Questions

- What group has a Cayley graph like the diagram on the right?
- How would you give a proof (=convincing argument) that these two groups have truly different structures? Can you find a property that one group has that the other does not?
- Can you find another group of size 4 that is different from both of these?

The Triangle Puzzle group

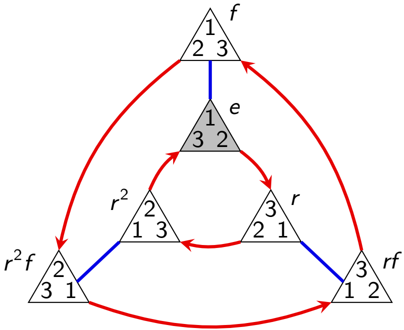
Let's play our "rectangle puzzle" game but with an equilateral triangle:



The "triangle puzzle" group, often denoted D_3, has 6 actions:

- The identity action: e
- A (clockwise) 120° rotation: r
- A (clockwise) 240° rotation: r^2
- A horizontal flip: f
- Rotate + horizontal flip: rf
- Rotate twice + horizontal flip: r^2f.

One set of generators: $D_3 = \langle r, f \rangle$.



Notice that multiple paths can lead us to the same node. These give us **relations** in our group. For example:

$$r^3 = e, \qquad r^{-1} = r^2, \qquad f^{-1} = f, \qquad rf = fr^2, \qquad r^2f = fr.$$

This group is **non-abelian**: $rf \neq fr$.

Properties of Cayley graphs

Observe that at every node of a Cayley graph, there is exactly one out-going edge of each color.

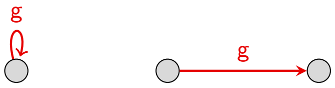
Question 1

Can an edge in a Cayley graph ever connect a node to itself?

Question 2

Suppose we have an edge corresponding to generator g that connects a node x to itself. Does that mean that the edge g connects *every* node to itself? In other words, can an action be the *identity action* when applied to some actions (or configurations) but not to others?

Visually, we're asking if the following scenerio can ever occur in a Cayley diagram:



A Theorem and Proof!

Perhaps surprisingly, the previous situation is *impossible*! Let's properly formulate and prove this.

Theorem

Suppose an action g has the property that $gx = x$ for some other action x. Then g is the *identity action*, i.e., $gh = h = hg$ for all other actions h.

Proof

The identity action (we'll denote by 1) is simply the action hh^{-1}, for any action h.

If $gx = x$, then multipling by x^{-1} *on the right* yields:

$$g = gxx^{-1} = xx^{-1} = 1.$$

Thus g is the identity action. $\square$

This was our first mathematical proof! It shows how we can deduce interesting properties about groups *from* the rules, which were not explicitly *built into* the rules.

Applications of groups

Thus far, we have introduced groups and explored a few basic examples.

At this point, we will pause to discuss a few practical (but not complicated) applications of groups.

We will see applications of group theory in 3 areas:

1. Science
2. Art
3. Mathematics

Our choice of examples is influenced by how well they illustrate the material rather than how useful they are.

Groups of symmetries

Intuitively, something is symmetrical when it looks the same from more than one point of view.

Can you think of an object that exhibits symmetry? Have we already seen some?

How does symmetry relate to groups?

The examples of groups that we've seen so far deal with arrangements of similar things.

In the next section, we will uncover the following fact (we'll be more precise later):

Cayley's Theorem

Every group can be viewed as a collection of ways to rearrange some set of things.

How to make a group out of symmetries

Groups relate to symmetry because an object's symmetries can be described using arrangements of the object's parts.

The following algorithm tells us how to construct a group that describes (or measures) a physical object's symmetry.

Algorithm

1. Identify all the parts of the object that are similar (e.g., the corners of an n-gon), and give each such part a different number.
2. Consider the actions that may rearrange the numbered parts, but leave the object it the same physical space. (This collection of actions forms a group.)
3. (Optional) If you want to visualize the group, explore and map it as we did in the previous lecture with the rectangle puzzle, etc.

Comments

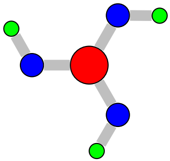
- We'll refer to the physical space that an object occupies as its footprint (this terminology does not appear in the text).
- Step 1 of our Algorithm numbers the object's parts so that we can track the manipulations permitted in Step 2. Each new state is a rearrangement of the object's similar parts and allows us to distinguish each of these rearrangements; otherwise we could not tell them apart.
- Not *every* rearrangement is valid. We are only allowed actions that maintain the object's physical integrity *and* preserve its footprint. For example, we can't rip two arms off a starfish and glue them back on in different places.
- Step 2 requires us to find *all* actions that preserve the object's footprint and physical integrity; not just the generators.
- However, if we choose to complete Step 3 (make a Cayley diagram), we must make a choice concerning generators. Different choices in generators may result in different Cayley diagrams.
- When selecting a set of generators, we would ideally like to select as small a set as possible. We can never choose too many generators, but we can choose too few. However, having "extra" generators only clutters our Cayley diagram.

Shapes of molecules

Because the shape of molecules impacts their behavior, chemists use group theory to classify their shapes. Let's look at an example.

The following figure depicts a molecule of Boric acid, $B(OH)_3$.

Note that a mirror reflection is *not* a symmetry of this molecule.

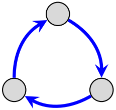


Exercise

Follow the steps of our Algorithm to find the group that describes the symmetry of the molecule and draw a possible Cayley diagram.

The group of symmetries of Boric acid has 3 actions requiring at least one generator. If we choose "120° clockwise rotation" as our generator, then the actions are:

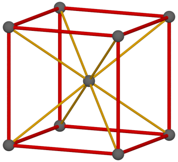
1. the *identity* (or "do nothing") action: e
2. 120° clockwise rotation: r
3. 240° clockwise rotation: r^2.

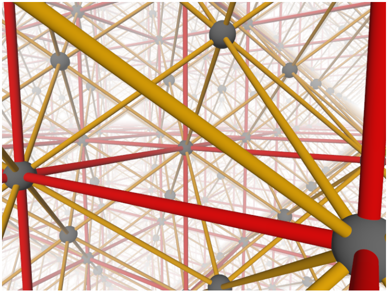


This is the cyclic group, C_3. (We'll discuss cyclic groups in a later lecture.)

Crystallography

Solids whose atoms arrange themselves in a regular, repeating pattern are called crystals. The study of crystals is called crystallography.





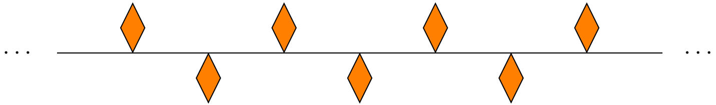
When chemists study such crystals they treat them as patterns that repeat without end. This allows a new manipulation that preserves the infinite footprint of the crystal and its physical integrity: translation.

In this case, the groups describing the symmetry of crystals are infinite. Why?

Frieze patterns

Crystals are patterns that repeat in 3 dimensions. Patterns that only repeat in 1 dimension are called frieze patterns. The groups that describe their symmetries are called frieze groups.

Frieze patterns (or at least finite sections of them) occur throughout art and architecture. Here is an example:



This frieze admits a new type of manipulation that preserves its footprint and physical integrity: a glide reflection. This action consists of a horizontal translation (by the appropriate amount) followed by a vertical flip.

Note that for this pattern, a vertical flip all by itself does not preserve the footprint, and thus is not one of the actions of the group of symmetries.

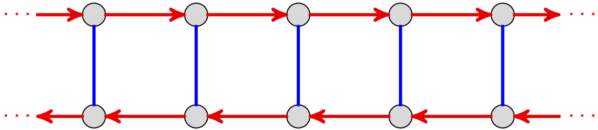
Exercise

Determine the group of symmetries of this frieze pattern and draw a possible Cayley diagram.

Frieze patterns

The group of symmetries of the frieze pattern on the previous slide turns out to be infinite, but we only needed two generators: horizontal flip and glide reflection.

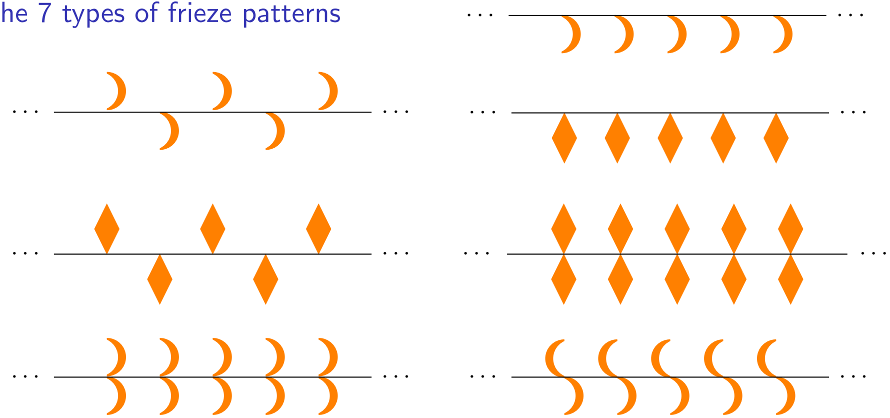
Here is a possible Cayley diagram:



Friezes, wallpapers, and crystals

- The symmetry of any frieze pattern can be described by one of 7 different infinite groups. Some frieze groups are *isomorphic* (have the same structure) even though the visual appearance of the patterns (and Cayley graphs) may differ.
- The symmetry of 2-dimensional repeating patterns, called "wallpaper patterns," has also been classified. There are 17 different wallpaper groups.
- There are 230 crystallographic groups, which describe the symmetries of 3-dimensional repeating patterns.

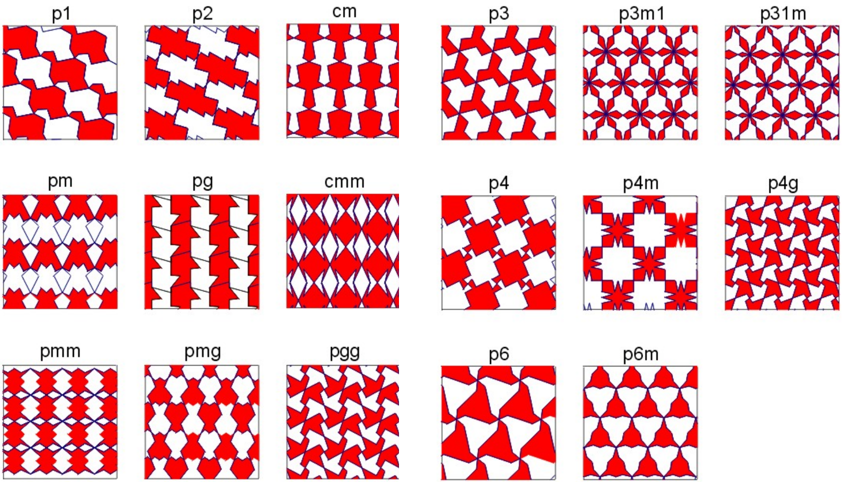
The 7 types of frieze patterns



Questions

- What basic types of symmetries (e.g., translation, reflection, rotation, glide reflection) do these frieze groups have?
- What are the (minimal) generators for the corresponding frieze groups?
- Which of these frieze patterns have isomorphic frieze groups?
- Which of these frieze groups are abelian?

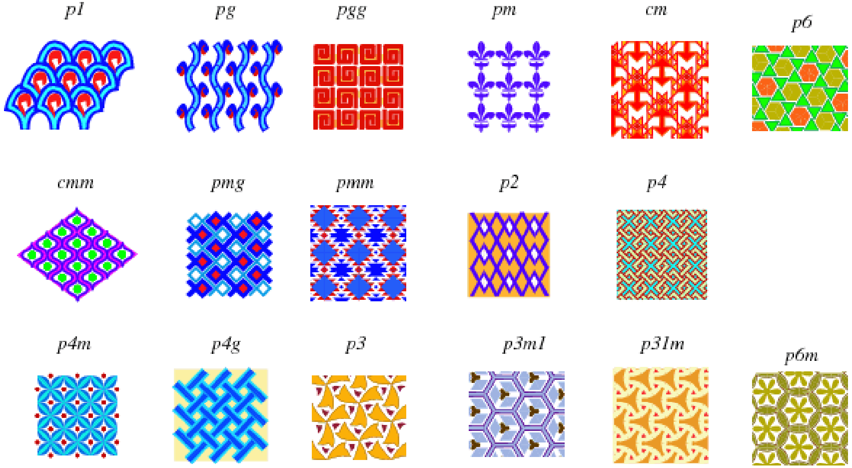
The 17 types of wallpaper patterns



Images courtesy of Patrick Morandi (New Mexico State University).

The 17 types of wallpaper patterns

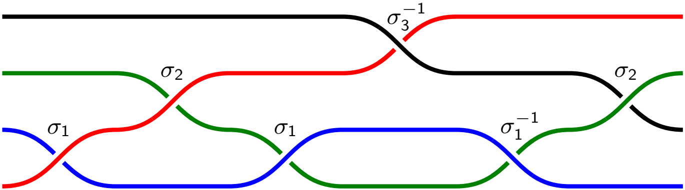
Here is another picture of all 17 wallpapers, with the official IUC notation for the symmetry group, adopted by the International Union of Crystallography in 1952.



Braid groups

Another area where groups arise in both art and mathematics is the study of braids.

This is best seen by an example. The following is a picture of an element (action) from the braid group $B_4 = \langle \sigma_1, \sigma_2, \sigma_3 \rangle$:



The braid $b = \sigma_1\sigma_2\sigma_1\sigma_3^{-1}\sigma_1^{-1}\sigma_2 = \sigma_1\sigma_2\sigma_3^{-1}\sigma_2$.

Do you see why the set of braids on n strings forms a group?

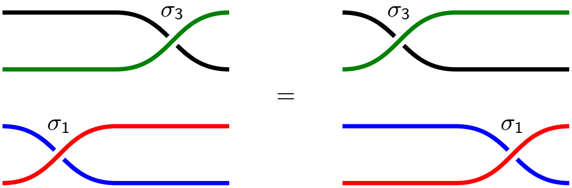
To combine two braids, just concatenate them.

Every braid is reversible – just "undo" each crossing. In the example above,

$$e = bb^{-1} = (\sigma_1\sigma_2\sigma_1\sigma_3^{-1}\sigma_1^{-1}\sigma_2)(\sigma_2^{-1}\sigma_1\sigma_3\sigma_1^{-1}\sigma_2^{-1}\sigma_1^{-1}).$$

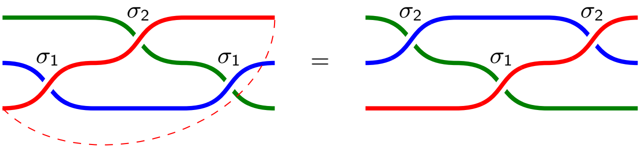
Braid groups

There are two fundamental relations in braid groups:



$$\sigma_i \sigma_j = \sigma_j \sigma_i$$

$$(\text{if } |i-j| \geq 2)$$



$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$$

We can describe the braid group B_4 by the following presentation:

$$B_4 = \langle \sigma_1, \sigma_2, \sigma_3 \mid \sigma_1\sigma_3 = \sigma_3\sigma_1, \;\; \sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2, \;\; \sigma_2\sigma_3\sigma_2 = \sigma_3\sigma_2\sigma_3 \rangle.$$

We will study presentations more in the next lecture; this is just an introduction.

Labeled Cayley diagrams

Recall that arrows in a Cayley diagram represent one choice of generators of the group. In particular, all arrows of a fixed color correspond to the same generator.

Our choice of generators influenced the resulting Cayley diagram!

When we have been drawing Cayley diagrams, we have been doing one of two things with the nodes:

1. Labeling the nodes with configurations of a thing we are acting on.
2. Leaving the nodes unlabeled (this is the "abstract Cayley diagram").

There is a 3rd thing we can do with the nodes, motivated by the fact that every path in the Cayley diagram represents an action of the group:

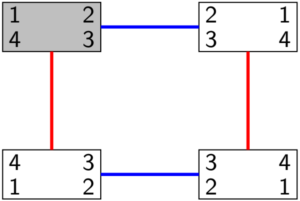
3. Label the nodes with actions (this is called a "diagram of actions").

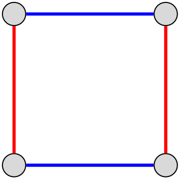
Motivating idea

If we distinguish one node as the "unscrambled" configuration and label that with the identity action, then we can label each remaining node with the action that it takes to reach it from the unscrambled state.

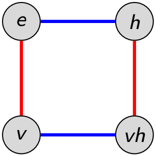
An example: The Klein 4-group

Recall the "rectangle puzzle." If we use horizontal flip (h) and vertical flip (v) as generators, then here is the Cayley diagram labeled by configurations (left), and unlabeled Cayley diagram (right):





Let's apply the steps to the abstract Cayley diagram for V_4, using the upper-left node as the "unscrambled configuration":



Note that we could also have labeled the node in the lower right corner as hv, as well.

How to label nodes with actions

Let's summarize the process that we just did.

Node labeling algorithm

The following steps transform a Cayley diagram into one that focuses on the group's actions.

(i) Choose a node as our initial reference point; label it e. (This will correspond to our "identity action.")

(ii) Relabel each remaining node in the diagram with a path that leads there from node e. (If there is more than one path, pick any one; shorter is better.)

(iii) Distinguish arrows of the same type in some way (color them, label them, dashed vs. solid, etc.)

Our convention will be to label the nodes with sequences of generators, so that reading the sequence from left to right indicates the appropriate path.

Warning!

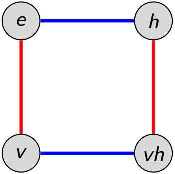
Some authors use the opposite convention, motivated by "function composition."

A "group calculator"

One neat thing about Cayley diagrams with nodes labeled by actions is that they act as a "group calculator".

For example, if we want to know what a particular sequence is equal to, we can just chase the sequence through the Cayley graph, starting at e.

Let's try one. In V_4, what is the action $hhhvhvvhv$ equal to?



We see that $hhhvhvvhv = h$. A more condensed way to write this is

$$hhhvhvvhv = h^3 vhv^2 hv = h.$$

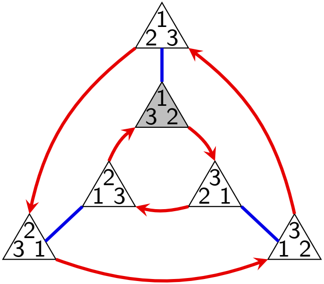
A concise way to describe V_4 is by the following group presentation (more on this later):

$$V_4 = \langle v, h \mid v^2 = e, h^2 = e, vh = hv \rangle .$$

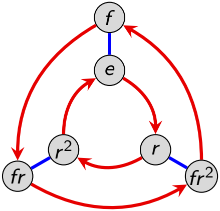
Another familiar example: D_3

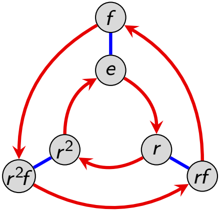
Recall the "triangle puzzle" group $G = \langle r, f \rangle$, generated by a clockwise 120° rotation r, and a horizontal flip f.

Let's take the shaded triangle to be the "unscrambled configuration."



Here are two different ways (of many!) that we can label the nodes with actions:





The following is one (of many!) presentations for this group:

$$D_3 = \langle r, f \mid r^3 = e,\ f^2 = e,\ r^2 f = fr \rangle .$$

Group presentations

Initially, we wrote $G = \langle h, v \rangle$ to say that "G is generated by the elements h and v."

All this tells us is that h and v generate G, but not how they generate G.

If we want to be more precise, we use a group presentation of the following form:

$$G = \Big\langle \text{generators} \;\Big|\; \text{relations} \Big\rangle$$

The vertical bar can be thought of as meaning "subject to".

For example, the following is a presentation for V_4:

$$V_4 = \langle a, b \mid a^2 = e,\ b^2 = e,\ ab = ba \rangle \,.$$

Caveat!

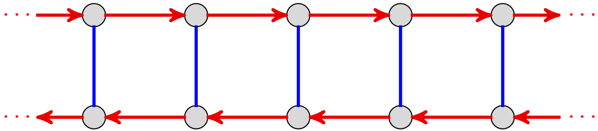
Just because there are elements in a group that "satisfy" the relations above does *not* mean that it is V_4.

For example, the trivial group $G = \{e\}$ satisfies the above presentation; just take $a = e$ and $b = e$.

Loosely speaking, the above presentation tells us that V_4 is the "largest group" that satisfies these relations. (More on this when we study quotients.)

Group presentations

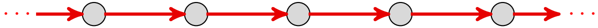
Recall the frieze group from the previous lecture that had the following Cayley diagram:



One presentation of this group is

$$G = \langle t, f \mid f^2 = e, tft = f \rangle \, .$$

Here is the Cayley diagram of another frieze group:



It has presentation

$$G = \langle a \mid \quad \rangle \, .$$

That is, "one generator subject to *no relations*."

Group presentations

Due to the aforementioned caveat, and a few other technicalities, the study of group presentations is a topic usually relegated to graduate-level algebra classes.

However, they are often introduced in an undergraduate algebra class because *they are very useful*, even if the intricate details are harmlessly swept under the rug.

The problem (called the word problem) of determining what a mystery group is from a presentation is actually computationally unsolvable! In fact, it is equivalent to the famous "halting problem" in computer science!

For (mostly) amusement, what group do you think the following presentation describes?

$$G = \langle a, b \mid ab = b^2 a,\ ba = a^2 b \rangle.$$

Surprisingly, this is the trivial group $G = \{e\}$!

Inverses

If g is a generator in a group G, then following the "g-arrow" backwards is an action that we call its inverse, and denoted by g^{-1}.

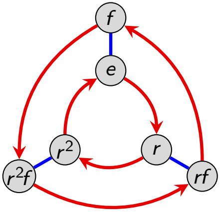
More generally, if g is represented by a path in a Cayley diagram, then g^{-1} is the action achieved by tracing out this path in reverse.

Note that by construction,

$$gg^{-1} = g^{-1}g = e\,,$$

where e is the identity (or "do nothing") action. Sometimes this is denoted by e, 1, or 0.

For example, let's use the following Cayley diagram to compute the inverses of a few actions:



$r^{-1} =$ _____ because r_____ $= e =$ _____r

$f^{-1} =$ _____ because f_____ $= e =$ _____f

$(rf)^{-1} =$ _____ because (rf)_____ $= e =$ _____(rf)

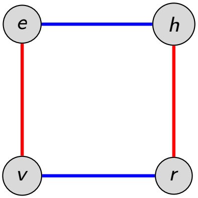
$(r^2f)^{-1} =$ _____ because (r^2f)_____ $= e =$ _____(r^2f).

Multiplication tables

Since we can use a Cayley diagram with nodes labeled by actions as a "group calculator," we can create a (group) multiplication table, that shows how every pair of group actions combine.

This is best illustrated by diving in and doing an example. Let's fill out a multiplication table for V_4.

Since order of multiplication can matter, let's use the convention that the entry in row g and column h is the element gh (rather than hg).



	e	v	h	r
e	e	v	h	r
v	v	e	r	h
h	h	r	e	v
r	r	h	v	e

Some remarks on the structure of multiplication tables

Comments

- The 1st column and 1st row repeat themselves. (Why?) Sometimes these will be omitted (*Group Explorer* does this).
- Multiplication tables can visually reveal patterns that may be difficult to see otherwise. To help make these patterns more obvious, we can color the cells of the multiplication table, assigning a unique color to each action of the group.
- A group is abelian iff its multiplication table is symmetric about the "main diagonal."
- In each row and each column, each group action occurs exactly once. (This will always happen. . . Why?)

Let's state and prove that last comment as as theorem.

A theorem and proof

Theorem

An element cannot appear twice in the same row or column of a multiplictaion table.

Proof

Suppose that in row a, the element g appears in columns b and c. Algebraically, this means

$$ab = g = ac.$$

Multiplying everything on the left by a^{-1} yields

$$a^{-1}ab = a^{-1}g = a^{-1}ac \qquad \Longrightarrow \qquad b = c.$$

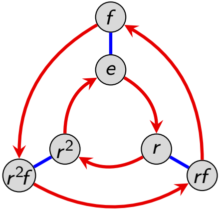
Thus, g (or any element) element cannot appear twice in the same row.

The proof that two elements cannot appear twice in the same column is similar, and will be left as a homework exercise. $\square$

Another example: D_3

Let's fill out a multiplication table for the group D_3; here are several different presentations:

$$D_3 = \langle r, f \mid r^3 = e,\ f^2 = e,\ rf = fr^2 \rangle$$
$$= \langle r, f \mid r^3 = e,\ f^2 = e,\ rfr = f \rangle .$$



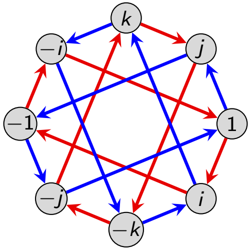
	e	r	r^2	f	rf	r^2f
e	e	r	r^2	f	rf	r^2f
r	r	r^2	e	rf	r^2f	f
r^2	r^2	e	r	r^2f	f	rf
f	f	r^2f	rf	e	r^2	r
rf	rf	f	r^2f	r	e	r^2
r^2f	r^2f	rf	f	r^2	r	e

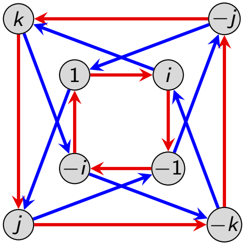
Observations? What patterns do you see?

Just for fun, what group do you get if you remove the "$r^3 = e$" relation from the presentations above? (*Hint*: We've seen it recently!)

Another example: the quaternion group

The following Cayley diagram, laid out two different ways, describes a group of size 8 called the Quaternion group, often denoted $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.





The "numbers" j and k individually act like $i = \sqrt{-1}$, because $i^2 = j^2 = k^2 = -1$.

Multiplication of $\{\pm i, \pm j, \pm k\}$ works like the cross product of unit vectors in $\mathbb{R}^3$:

$$ij = k, \quad jk = i, \quad ki = j, \qquad\qquad ji = -k, \quad kj = -i, \quad ik = -j\,.$$

Here are two possible presentations for this group:

$$\begin{aligned} Q_8 &= \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = -1 \rangle \\ &= \langle i, j \mid i^4 = j^4 = 1,\ iji = j \rangle\,. \end{aligned}$$

Moving towards the standard definition of a group

We have been calling the members that make up a group "actions" because our definition requires a group to be a collection of actions that satisfy our 4 rules.

Since the standard definition of a group is not phrased in terms of actions, we will need more general terminology.

We will call the members of a group elements. In general, a group is a set of elements satisfying some set of properties.

We will also use standard set theory notation. For example, we will write things like

$$h \in V_4$$

to mean "h is an element of the group V_4."

Binary operations

Intuitively, an operation is a method for combining objects. For example, $+$, $-$, $\cdot$, and $\div$ are all examples of operations. In fact, these are binary operations because they combine two objects into a single object.

Definition

If $*$ is a binary operation on a set S, then $s * t \in S$ for all $s, t \in S$. In this case, we say that S is closed under the operation $*$.

Combining, or "multiplying" two group elements (i.e., doing one action followed by the other) is a binary operation. We say that it is a binary operation *on* the group.

Recall that Rule 4 says that any sequence of actions is an action. This ensures that the group is closed under the binary operation of multiplication.

Multiplication tables are nice because they depict the group's binary operation in full.

However, not every table with symbols in it is going to be the multiplication table for a group.

Associativity

Recall that an operation is associative if parentheses are permitted anywhere, but required nowhere.

For example, ordinary addition and multiplication are associative. However, subtraction of integers is *not* associative:

$$4 - (1 - 2) \neq (4 - 1) - 2.$$

Is the operation of combining actions in a group associative? YES! We will not prove this fact, but rather illustrate it with an example.

Recall D_3, the group of symmetries for the equilateral triangle, generated by r (=rotate) and f (=horizontal flip).

How do the following compare?

$$rfr, \qquad (rf)r, \qquad r(fr)$$

Even though we are associating differently, the end result is that *the actions are applied left to right*.

The moral is that we never need parentheses when working with groups, though we may use them to draw our attention to a particular chunk in a sequence.

Classical definition of a group

We are now ready to state the standard definition of a group.

Definition (official)

A set G is a group if the following criteria are satisfied:

1. There is a binary operation $*$ on G.
2. $*$ is associative.
3. There is an identity element $e \in G$. That is, $e * g = g = g * e$ for all $g \in G$.
4. Every element $g \in G$ has an inverse, g^{-1}, satisfying $g * g^{-1} = e = g^{-1} * g$.

Remarks

- Depending on context, the binary operation may be denoted by $*$, $\cdot$, $+$, or $\circ$.
- As with ordinary multiplication, we frequently omit the symbol altogether and write, e.g., xy for $x * y$.
- We generally only use the $+$ symbol if the group is abelian. Thus, $g + h = h + g$ (always), but in general, $gh \neq hg$.
- Uniqueness of the identity and inverses is *not* built into the definition of a group. However, we can without much trouble, prove these properties.

Definitions of a group: Old vs. New

Do our two competing definitions agree? That is, if our informal definition says something is a group, will our official definition agree? Or vice versa?

Since our first definition of a group was informal, it is impossible to answer this question officially and absolutely. An informal definition potentially allows some technicalities and ambiguities.

This aside, our discussion leading up to our official Definition provides an informal argument for why the answer to the first question should be yes. We will answer the second question in the next chapter.

Regardless of whether the definitions agree, we always have $e^{-1} = e$. That is, the inverse of doing nothing is doing nothing.

Even though we haven't officially shown that the two definitions agree (and in some sense, we can't), we shall begin viewing groups from these two different paradigms:

- a group as a collection of actions;
- a group as a set with a binary operation.

A few simple properties

One of the first things we can prove about groups is uniqueness of the identity and inverses.

Theorem

Every element of a group has a *unique* inverse.

Proof

Let g be an element of a group G. By definition, it has at least one inverse.

Suppose that h and k are both inverses of g. This means that $gh = hg = e$ and $gk = kg = e$. It suffices to show that $h = k$. Indeed,

$$h = he = h(gk) = (hg)k = ek = k\,,$$

and the proof is complete. $\square$

The following proof is relegated to the homework; the technique is similar.

Theorem

Every group has a *unique* identity element.

Section 2: Examples of groups

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Math 4120, Modern Algebra

Overview

In this series of lectures, we will introduce 5 families of groups:

1. cyclic groups
2. abelian groups
3. dihedral groups
4. symmetric groups
5. alternating groups

Along the way, a variety of new concepts will arise, as well as some new visualization techniques.

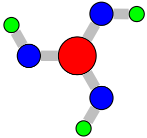
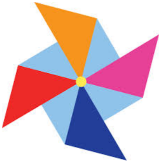
We will study permutations, how to write them concisely in cycle notation. Cayley's theorem tells us that every finite group is isomorphic to a collection of permutations (i.e., a subgroup of a symmetric group).

Cyclic groups

Definition

A group is cyclic if it can be generated by a single element.

Finite cyclic groups describe the symmetry of objects that have *only* rotational symmetry. Here are some examples of such objects.


An obvious choice of generator would be: *counterclockwise rotation by* $2\pi/n$ (called a "click"), where n is the number of "arms." This leads to the following presentation:

$$C_n = \langle r \mid r^n = e \rangle .$$

Remark

This is not the only choice of generator; but it's a natural one. Can you think of another choice of generator? Would this change the group presentation?

Cyclic groups

Definition

The order of a group G is the number of distinct elements in G, denoted by $|G|$.

The cyclic group of order n (i.e., n rotations) is denoted C_n (or sometimes by $\mathbb{Z}_n$).

For example, the group of symmetries for the objects on the previous slide are C_3 (boric acid), C_4 (pinwheel), and C_{10} (chilies).

Comment

The alternative notation $\mathbb{Z}_n$ comes from the fact that the binary operation for C_n is just modular addition. To add two numbers in $\mathbb{Z}_n$, add them as integers, divide by n, and take the remainder.

For example, in $\mathbb{Z}_6$: $3 + 5 \equiv_6 2$. "3 clicks + 5 clicks = 2 clicks". (If the context is clear, we may even write $3 + 5 = 2$.)

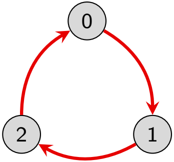
Cyclic groups, additively

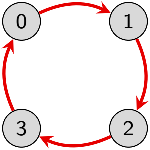
A common way to write elements in a cyclic group is with the integers $0, 1, 2, \ldots, n-1$, where

- 0 is the identity
- 1 is the single counterclockwise "click".

Observe that the set $\{0, 1, \ldots, n-1\}$ is closed under addition modulo n. That is, if we add (mod n) any two numbers in this set, the result is another member of the set.

Here are some Cayley diagrams of cyclic groups, using the canonical generator of 1.





Summary

In this setting, the cyclic group consists of the set $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ under the binary operation of $+$ (modulo n). The (additive) identity is 0.

Cyclic groups, multiplicatively

Here's another natural choice of *notation* for cyclic groups. If r is a generator (e.g., a rotation by $2\pi/n$), then we can denote the n elements by

$$1, r, r^2, \dots, r^{n-1}.$$

Think of r as the complex number $e^{2\pi i/n}$, with the group operation being *multiplication!*

Note that $r^n = 1$, $r^{n+1} = r$, $r^{n+2} = r^2$, etc. Can you see modular addition rearing its head again? Here are some Cayley diagrams, using the canonical generator of r.

Summary

In this setting, the cyclic group can be thought of as the set $C_n = \{e^{2\pi ik/n} \mid k \in \mathbb{Z}\}$ under the binary operation of $\times$. The (multiplicative) identity is 1.

More on cyclic groups

One of our notations for cyclic groups is "additive" and the other is "multiplicative." This doesn't change the actual group; only our choice of notation.

Remark

The (unique) infinite cyclic group (additively) is $(\mathbb{Z}, +)$, the integers under addition. Using multiplicative notation, the infinite cyclic group is

$$G = \langle r \mid \quad \rangle = \{r^k : k \in \mathbb{Z}\}.$$

For the infinite cyclic group $(\mathbb{Z}, +)$, only 1 or -1 can be generators. (Unless we use multiple generators, which is usually pointless.)

Proposition

Any number from $\{0, 1, \ldots, n-1\}$ that is relatively prime to n will generate $\mathbb{Z}_n$.

For example, 1 and 5 generate $\mathbb{Z}_6$, while $1, 2, 3$, and 4 all generate $\mathbb{Z}_5$. i.e.,

$$\mathbb{Z}_6 = \langle 1 \rangle = \langle 5 \rangle \,, \qquad \mathbb{Z}_5 = \langle 1 \rangle = \langle 2 \rangle = \langle 3 \rangle = \langle 4 \rangle \,.$$

Note that the above notation isn't a presentation, it just means "generated by."

More on cyclic groups

Modular addition has a nice visual appearance in the multiplication tables of cyclic groups.

	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

There are many things worth commenting on, but one of the most important properties of the multiplication tables for cyclic groups is the following:

Observation

If the headings on the multiplication table are arranged in the "natural" order $(0, 1, 2, \dots n-1)$ or $(e, r, r^2, \dots r^{n-1})$, then each row is a cyclic shift to the left of the row above it.

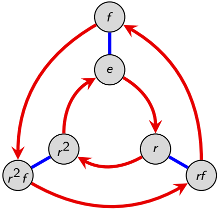
Do you see *why* this happens?

Orbits

We started our discussion with cyclic groups because of their simplicity, but also because they play a fundamental role in more complicated groups.

Before continuing our exploration into the 5 families, let's observe how cyclic groups "fit" into other groups.

Consider the Cayley diagram for D_3:



Do you see any copies of the Cayley diagram for any cyclic groups in this picture?

Starting at e, the red arrows lead in a length-3 cycle around the inside of the diagram. We refer to this cycle as the orbit of the element r.

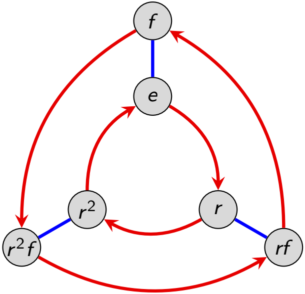
The blue arrows lead in a length-2 cycle – the orbit of f.

Orbits are usually written with braces. In this case, the orbit of r is $\{e, r, r^2\}$, and the orbit of f is $\{e, f\}$.

Orbits

Every element in a group traces out an orbit. Some of these may not be obvious from the Cayley diagram, but they are there nonetheless.

Let's work out the orbits for the remaining elements of D_3.



element	orbit
e	$\{e\}$
r	$\{e, r, r^2\}$
r^2	$\{e, r^2, r\}$
f	$\{e, f\}$
rf	$\{e, rf\}$
r^2f	$\{e, r^2f\}$

Note that there are 5 *distinct* orbits. The elements r and r^2 have the same orbit.

Orbits

Definition

The order of an element $g \in G$, denoted $|g|$, is the size of its orbit. That is, $|g| := |\langle g \rangle|$. (Recall that the order of G is defined to be $|G|$.)

Note that in any group, the orbit of e will simply be $\{e\}$.

In general, the orbit of an element g is the set

$$\langle g \rangle := \{g^k : k \in \mathbb{Z}\}.$$

This set is not necessarily infinite, as we've seen with the finite cyclic groups.

We allow negative exponents, though this only matters in infinite groups.

One way of thinking about this is that the orbit of an element g is the collection of elements that you can get to by doing g or its inverse any number of times.

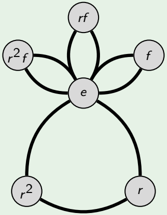
Remark

In *any* group G, the orbit of an element $g \in G$ is a cyclic group that "sits inside" G. This is an example of a subgroup, which we will study in more detail later.

Visualizing the orbits of a groups using "cycle graphs"

Example: Cycle graph of D_3

element	orbit
e	$\{e\}$
r	$\{e, r, r^2\}$
r^2	$\{e, r^2, r\}$
f	$\{e, f\}$
rf	$\{e, rf\}$
r^2f	$\{e, r^2f\}$



Comments

- In a cycle graph (also called an orbit graph), each cycle represents an orbit.
- The convention is that orbits that are subsets of larger orbits are only shown within the larger orbit.
- We don't color or put arrows on the edges of the cycles, because one orbit could have multiple generators.
- Intersections of cycles show what elements they have in common.
- What do the cycle graphs of cyclic groups look like? (*Answer*: a single cycle.)

Abelian groups

Recall that a group is abelian (named after Neils Abel) if the order of actions is irrelevant (i.e., the actions *commute*). Here is the formal mathematical definition.

Definition

A group G is abelian if $ab = ba$ for all $a, b \in G$.

Abelian groups are sometimes referred to as commutative.

Remark

To check that a group G is abeliean, it suffices to only check that $ab = ba$ for all pairs of generators of G. (*Why?*)

The pattern on the left *never* appears in the Cayley graph for an abelian group, whereas the pattern on the right illustrates the relation $ab = ba$:

Examples

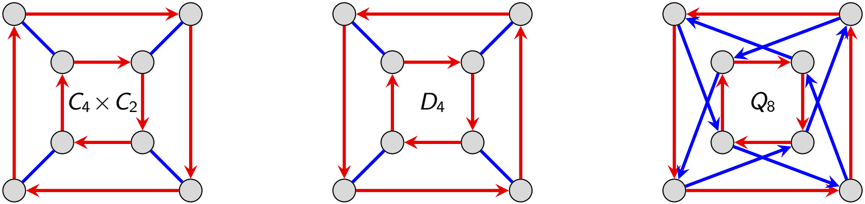
Cyclic groups are abelian.

Reason 1: The left configuration on the previous slide can never occur (since there is only one generator).

Reason 2: In the cyclic group $\langle r \rangle$, every element can be written as r^k for some k. Clearly, $r^k r^m = r^m r^k$ for all k and m.

Note that the converse fails: if a group is abelian, it need not be cyclic. (Take V_4 as an example.)

Let's explore a little further. The following are Cayley diagrams for three groups of order 8.

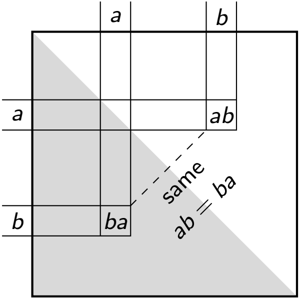


Are any of these groups abelian?

Multiplication tables of abelian groups

Abelian groups are easy to spot if you look at their multiplication tables.

The property "$ab = ba$ for all a and b" means that the table must be symmetric across the main diagonal.



	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Dihedral groups

While cyclic groups describe 2D objects that only have rotational symmetry, dihedral groups describe 2D objects that have rotational *and* reflective symmetry.

Regular polygons have rotational and reflective symmetry. The dihedral group that describes the symmetries of a regular n-gon is written D_n.

All actions in C_n are also actions of D_n, but there are more than that. The group D_n contains $2n$ actions:

- n rotations
- n reflections.

However, we only need two generators. Here is one possible choice:

1. $r =$ counterclockwise rotation by $2\pi/n$ radians. (A single "click.")
2. $f =$ flip (fix an axis of symmetry).

Here is one of (of many) ways to write the $2n$ actions of D_n:

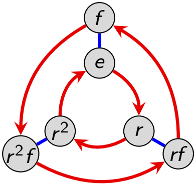
$$D_n = \{\underbrace{e, r, r^2, \dots, r^{n-1}}_{\text{rotations}}, \underbrace{f, rf, r^2 f, \dots, r^{n-1} f}_{\text{reflections}}\}.$$

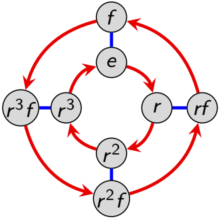
Cayley diagrams of dihedral groups

Here is one possible presentation of D_n:

$$D_n = \langle r, f \mid r^n = e,\ f^2 = e,\ rfr = f \rangle.$$

Using this generating set, the Cayley diagrams for the dihedral groups all look similar. Here they are for D_3 and D_4, respectively.

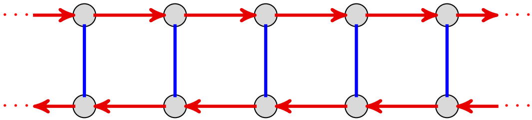




There is a related infinite dihedral group D_∞, with presentation

$$D_\infty = \langle r, f \mid f^2 = e,\ rfr = f \rangle.$$

We have already seen its Cayley diagram:



Cayley diagrams of dihedral groups

If s and t are two reflections of an n-gon across adjacent axes of symmetry (i.e., axes incident at π/n radians), then st is a rotation by $2\pi/n$.

To see an explicit example, take $s = rf$ and $t = f$ in D_n; obviously $st = (rf)f = r$.

Thus, D_n can be generated by two reflections. This has group presentation

$$\begin{aligned} D_n &= \langle s, t \mid s^2 = e,\ t^2 = e,\ (st)^n = e \rangle \\ &= \{\underbrace{e, st, ts, (st)^2, (ts)^2, \dots,}_{\text{rotations}}\ \underbrace{s, t, sts, tst, \dots}_{\text{reflections}}\}. \end{aligned}$$

What would the Cayley diagram corresponding to this generating set look like?

Remark

If $n \geq 3$, then D_n is nonabelian, because $rf \neq fr$. However, the following relations are very useful:

$$rf = fr^{n-1}, \qquad fr = r^{n-1}f.$$

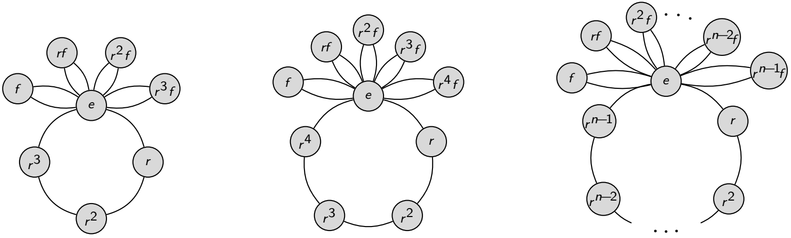
Looking at the Cayley graph should make these relations visually obvious.

Cycle graphs of dihedral groups

The (maximal) orbits of D_n consist of

- 1 orbit of size n consisting of $\{e, r, \ldots, r^{n-1}\}$;
- n orbits of size 2 consisting of $\{e, r^k f\}$ for $k = 0, 1, \ldots, n-1$.

Here is the general pattern of the cycle graphs of the dihedral groups:



Note that the size-n orbit may have smaller subsets that are orbits. For example, $\{e, r^2, r^4, \ldots, r^{n-2}\}$ and $\{e, r^{n/2}\}$ are orbits if n is even.

Multiplication tables of dihedral groups

The separation of D_n into rotations and reflections is also visible in their multiplication tables. For example, here is D_4:

	e	r	r^2	r^3	f	rf	r^2f	r^3f
e	e	r	r^2	r^3	f	rf	r^2f	r^3f
r	r	r^2	r^3	e	rf	r^2f	r^3f	f
r^2	r^2	r^3	e	r	r^2f	r^3f	f	rf
r^3	r^3	e	r	r^2	r^3f	f	rf	r^2f
f	f	r^3f	r^2f	rf	e	r^3	r^2	r
rf	rf	f	r^3f	r^2f	r	e	r^3	r^2
r^2f	r^2f	rf	f	r^3f	r^2	r	e	r^3
r^3f	r^3f	r^2f	rf	f	r^3	r^2	r	e

	e	r	r^2	r^3	f	rf	r^2f	r^3f
e	e	r	r^2	r^3	f	rf	r^2f	r^3f
r	r	r^2	r^3	e	rf	r^2f	r^3f	f
r^2	r^2	r^3	e	r	r^2f	r^3f	f	rf
r^3	r^3	e	r	r^2	r^3f	f	rf	r^2f
f	f	r^3f	r^2f	rf	e	r^3	r^2	r
rf	rf	f	r^3f	r^2f	r	e	r^3	r^2
r^2f	r^2f	rf	f	r^3f	r^2	r	e	r^3
r^3f	r^3f	r^2f	rf	f	r^3	r^2	r	e

non-flip **flip**

flip **non-flip**

As we shall see later, the partition of D_n as depicted above forms the structure of the group C_2. "Shrinking" a group in this way is called taking a quotient.

It yields a group of order 2 with the following Cayley diagram:

	e	f
e	e	f
f	f	e

Symmetric groups

Most groups we have seen have been collections of ways to rearrange things. This can be formalized.

Definition

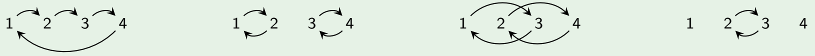
A permutation is an action that rearranges a collection of things.

For convenience, we will usually refer to permutations of positive integers (just like we did when we numbered our rectangle, etc.).

There are many ways to represent permutations, but we will start with the notation illustrated by the following example.

Example

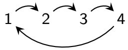
Here are some permutations of 4 objects.



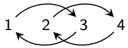
Combining permutations

In order for the set of permutations of n objects to form a group (what we want!), we need to understand how to combine permutations. Let's consider an example.

What should

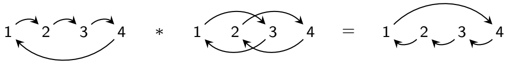


followed by



be equal to?

The first permutation rearranges the 4 objects, and then we shuffle the result according to the second permutation:



Groups of permutations

Fact

There are $n! = n(n-1)\cdots 3\cdot 2\cdot 1$ permutations of n items.

For example, there are $4! = 24$ "permutation pictures" on 4 objects.

The collection of permutations of n items forms a group!

To verify this, we just have to check that the appropriate rules of *one* of our definitions of a group hold.

How do we find the inverse of a permutation? Just reverse all of the arrows in the permutation picture. For example, the inverse of

$$\begin{tikzcd} 1 \arrow[r, bend left] & 2 \arrow[r, bend left] & 3 \arrow[r, bend left] & 4 \arrow[lll, bend left] \end{tikzcd}$$

is simply

$$\begin{tikzcd} 1 \arrow[rrr, bend left] & 2 \arrow[l, bend left] & 3 \arrow[l, bend left] & 4 \arrow[l, bend left] \end{tikzcd}$$

The symmetric group

Definition

The group of all permutations of n items is called the symmetric group (on n objects) and is denoted by S_n.

We've already seen the group S_3, which happens to be the same as the dihedral group D_3, but this is the only time the symmetric groups and dihedral groups coincide. (*Why?*)

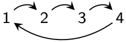
Although the set of *all* permutations of n items forms a group, creating a group does not require taking all permutations.

If we choose carefully, we can form groups by taking a subset of the permutations.

For example, the cyclic group C_n and the dihedral group D_n can both be thought of groups of certain permutations of $\{1, \ldots, n\}$. (*Why?* Do you see *which* permutations they represent?)

Cycle notation for S_n

We can concisely describe the permutation



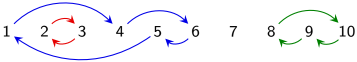
as $(1\ 2\ 3\ 4)$.

This is called cycle notation.

Observation 1

Every permutation can be decomposed into a product of disjoint cycles.

For example, in S_{10}, we can write



as $(1\ 4\ 6\ 5)\,(2\ 3)\,(8\ 10\ 9)$.

Observation 2

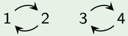
Disjoint cycles commute.

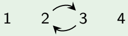
For example: $(1\ 4\ 6\ 5)\,(2\ 3)\,(8\ 10\ 9) = (2\ 3)\,(8\ 10\ 9)\,(1\ 4\ 6\ 5)$.

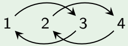
Cycle notation for S_n

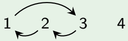
Example

Consider the following permutations in S_4:

- 

is $(1\ 2)\,(3\ 4)$
- 

is $(2\ 3)$
- 

is $(1\ 3)\ (2\ 4)$
- 

is $(1\ 3\ 2)$

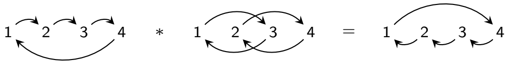
Remark

It doesn't matter "where we start" when writing the cycle. In the last example above,

$$(1\ 3\ 2) = (3\ 2\ 1) = (2\ 1\ 3) = (1\ 2)\,(2\ 3) = (1\ 2)\,(2\ 3)\,(2\ 3)\,(2\ 3)\,.$$

Composing permutations in cycle notation

Recall how we combined permutations:



In cycle notation, this is

$$(1\ 2\ 3\ 4) * (1\ 3)\,(2\ 4) = (1\ 4\ 3\ 2)\,.$$

We read left-to-right. (*Caveat*: some books use the right-to-left convention as in function composition.)

Do you see how to combine permutations in cycle notation? In the example above, we start with 1 and then read off:

- "1 goes to 2, then 2 goes to 4"; Write: (1 4
- "4 goes to 1, then 1 goes to 3"; Write: (1 4 3
- "3 goes to 4, then 4 goes to 2"; Write: (1 4 3 2
- "2 goes to 3, then 3 goes to 1"; Write: (1 4 3 2)

In this case, we've used up each number in $\{1, \ldots, n\}$. If we hadn't, we'd take the the smallest unused number and continue the process with a new (disjoint) cycle.

Transpositions

A transposition is a permutation that swaps two objects and fixes the rest, e.g.:

$$\begin{tikzcd}[column sep=small]
1 & 2 & \cdots & i-1 & i \arrow[rrrr, bend left] & i+1 & \cdots & j-1 & j \arrow[llll, bend left] & j+1 & \cdots & n-1 & n
\end{tikzcd}$$

In cycle notation, a transposition is just a 2-cycle, e.g., $(i\ \ j)$.

Theorem

The group S_n is generated by transpositions.

Intuitively, this means that every permutation can be constructed by successively exchanging pairs of objects.

In other words, if n people are standing in a row, and we want to rearrange them in some other order, we can always do this by successively having pairs of people swap places.

In fact, we only need adjacent transpositions to generate S_n:

$$S_n = \langle\, (1\ 2)\,,\ (2\ 3)\,,\ \ldots,\ (n-1\ \ n)\,\rangle\,.$$

Transpositions and the alternating groups

Remark

Even though every permutation in S_n can be written as a product of transpositions, there may be many ways to do this.

For example:

$$(1\ 3\ 2) = (1\ 2)\,(2\ 3) = (1\ 2)\,(2\ 3)\,(2\ 3)\,(2\ 3) = (1\ 2)\,(2\ 3)\,(1\ 2)\,(1\ 2).$$

Theorem

The parity of the number of transpositions of a fixed permutation is unique.

That is, a fixed permutation can either be written with an even number of transpositions, or an odd number of transpositions, but *not both*!

We thus have a notion of even permutations and odd permutations.

Theorem

Exactly half of the permutations in S_n are even, and they form a group called the alternating group, denoted A_n.

Practice

At this point, it helps to "get your hands dirty" and try a few examples. Here are some good exercises.

1. Write the following products of permutations into a product of *disjoint cycles*:
   - $(1\ 2\ 3)\,(1\ 2\ 3\ 4)$ in S_4
   - $(1\ 6)\,(1\ 2\ 4\ 5)\,(1\ 6\ 4\ 2\ 5\ 3)$ in S_6.

Let $G = S_3$, the symmetric group on three objects. This group has six elements.

2. Do the following for each element in S_3:
   - Draw its "permutation picture."
   - Write it as a product of disjoint transpositions (that is, using only $(1\ 2)$, $(2\ 3)$, and $(1\ 3)$).
   - Write it as a product of disjoint *adjacent* transpositions (that is, using only $(1\ 2)$ and $(2\ 3)$).
   - Determine whether it is even or odd.
3. Now, write down the alternating group A_3. This is the group consisting of only the *even* permutations. What familiar group is this isomorphic to?

Alternating groups

How can we verify that A_n a group?

The only major concern is it must be closed under combining permutations (all other necessary properties are inherited from S_n).

Do you see why combining two even permutations yields an even permutation?

Interesting fact

For $n \leq 5$, the group A_n consists precisely of the set of "squares" in S_n. By "square," we mean an element that can be written as an element of S_n times itself.

For example, the permutation  is a square in S_3, because:

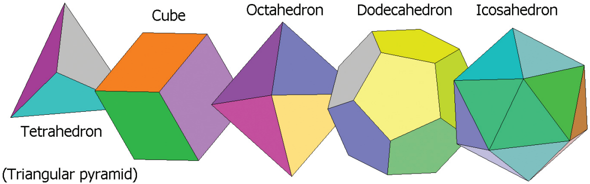
In cycle notation, this is $(1\ 3\ 2) = (1\ 2\ 3)\,(1\ 2\ 3)$.

Note that A_n has order $\dfrac{n!}{2}$.

Platonic solids

The symmetric groups and alternating groups arise throughout group theory. In particular, the groups of symmetries of the 5 Platonic solids are symmetric and alternating groups.

There are only *five* 3-dimensional shapes (polytopes) all of whose faces are regular polygons that meet at equal angles. These are called the Platonic solids:



The groups of symmetries of the Platonic solids are as follows:

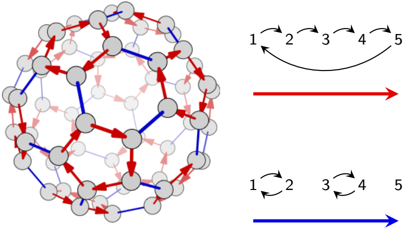
shape	group
Tetrahedron	A_4
Cube	S_4
Octahedron	S_4
Icosahedron	A_5
Dodecahedron	A_5

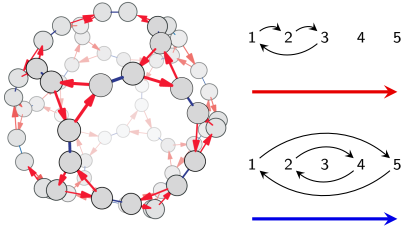
Platonic solids

The Cayley diagrams for these 3 groups can be arranged in some very interesting configurations.

In particular, the Cayley diagram for Platonic solid 'X' can be arranged on a truncated 'X', where truncated refers to cutting off some corners.

For example, here are two representations for Cayley diagrams of A_5. At left is a truncated icosahedron and at right is a truncated dodecahedron.





Cayley's theorem

Any set of permutations that forms a group is called a permutation group.

Cayley's theorem says that permutations can be used to construct any finite group.

In other words, every group has the same structure as (we say "*is isomorphic to*") some permutation group.

Warning! We are *not* saying that every group is isomorphic to some symmetric group, S_n. Rather, every group is isomorphic to a subgroup of some symmetric group S_n – i.e., a subset of S_n that is *also* a group in its own right.

Question

Given a group, how do we associate it with a set of permutations?

Cayley's theorem; how to construct permutations

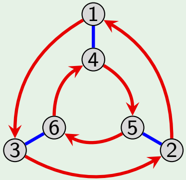
Here is an algorithm given a Cayley diagram with n nodes:

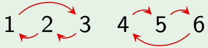
1. number the nodes 1 through n,
2. interpret each arrow type in the Cayley diagram as a permutation.

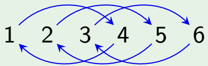
The resulting permutations are the generators of the corresponding permutation group.

Example

Let's try this with $D_3 = \langle r, f \rangle$.







We see that D_3 is isomorphic to the subgroup $\langle (132)(456), (14)(25)(36) \rangle$ of S_6.

Cayley's theorem; how to construct permutations

Here is an algorithm given a multiplication table with n elements:

1. replace the table headings with 1 through n,
2. make the appropriate replacements throughout the rest of the table,
3. interpret each column as a permutation.

This results in a 1-1 correspondence between the original group elements (not just the generators) and permutations.

Example

Let's try this with the multiplication table for $V_4 = \langle v, h \rangle$.

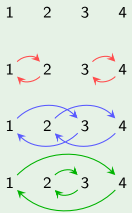
	1	2	3	4
1	1	2	3	4
2	2	1	4	3
3	3	4	1	2
4	4	3	2	1

Column 1:

Column 2:

Column 3:

Column 4:



We see that V_4 is isomorphic to the subgroup $\langle (12)(34),\ (13)(24) \rangle$ of S_4.

Cayley's theorem

Intuitively, two groups are isomorphic if they have the same structure.

Two groups are *isomorphic* if we can construct Cayley diagrams for each that look identical.

Cayley's Theorem

Every finite group is isomorphic to a collection of permutations.

Our algorithms exhibit a 1-1 correspondence between group elements and permutations.

However, we have *not* shown that the corresponding permutations form a group, or that the resulting permutation group has the same structure as the original.

What needs to be shown is that the permutation from the i^{th} row followed by the permutation from the j^{th} column, results in the permutation that corresponding to the cell in the i^{th} row and j^{th} column of the original table (Exercise.)

Section 3: The structure of groups

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Math 4120, Modern Algebra

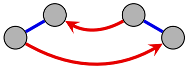
Regularity

Cayley diagrams have an important structural property call *regularity* that we've mentioned, but haven't analyzed in depth.

This is best seen with an example: Consider the group D_3. It is easy to verify that $frf = r^{-1}$.

Thus, starting at *any node* in the Cayley diagram, the path frf will *always* lead to the same node as the path r^{-1}.

That is, the following fragment permeates throughout the diagram.



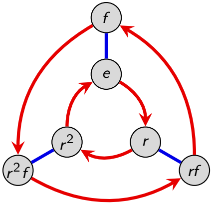
Equivalently, the path $frfr$ will always bring you back to where you started. (Because $frfr = e$).

Key observation

The algebraic relations of a group, like $frf = r^{-1}$, give Cayley diagrams a uniform symmetry – every part of the diagram is structured like every other.

Regularity

Let's look at the Cayley diagram for D_3:



Check that indeed, $frf = r^{-1}$ holds by following the corresponding paths starting at any of the six nodes.

There are other patterns that permeate this diagram, as well. Do you see any?

Here are a couple: $f^2 = e$, $r^3 = e$.

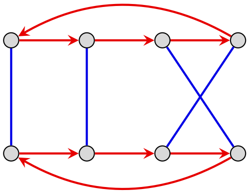
Definition

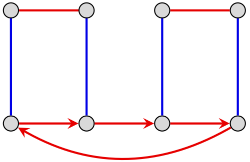
A diagram is called regular if it repeats every one of its interval patterns throughout the whole diagram, in the sense described above.

Regularity

Every Cayley diagram is regular. In particular, diagrams lacking regularity do *not* represent groups (and so they are not called Cayley diagrams).

Here are two diagrams that *cannot* be the Cayley diagram for a group because they are not regular.





Recall that our original definition of a group was informal and "unofficial."

One reason for this is that technically, regularity needs to be incorporated in the rules. Otherwise, the previous diagram would describe a group of actions.

We've indirectly discussed the regularity property of Cayley diagrams, and it was implied, but we haven't spelled out the details until now.

Subgroups

Definition

When one group is contained in another, the smaller group is called a subgroup of the larger group. If H is a subgroup of G, we write $H < G$ or $H \leq G$.

All of the orbits that we saw in previous lectures are subgroups. Moreover, they are *cyclic* subgroups. (Why?)

For example, the orbit of r in D_3 is a subgroup of order 3 living inside D_3. We can write

$$\langle r \rangle = \{e, r, r^2\} < D_3.$$

In fact, since $\langle r \rangle$ is really just a copy of C_3, we may be less formal and write

$$C_3 < D_3.$$

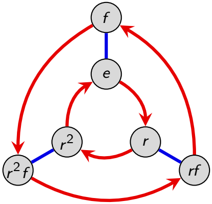
An example: D_3

Recall that the orbits of D_3 are

$$\langle e \rangle = \{e\}, \qquad \langle r \rangle = \langle r^2 \rangle = \{e, r, r^2\}, \qquad \langle f \rangle = \{e, f\}$$

$$\langle rf \rangle = \{e, rf\}, \qquad \langle r^2 f \rangle = \{e, r^2 f\}\,.$$

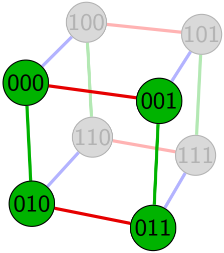
The orbits corresponding to the generators are staring at us in the Cayley diagram. The others are more hidden.



It turns out that all of the subgroups of D_3 are just (cyclic) orbits.

However, there are groups that have subgroups that are *not* cyclic.

Another example: $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$



Here is the Cayley diagram for the group $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ (the "three-light switch group").

A copy of the subgroup V_4 is highlighted.

The group V_4 requires at least two generators and hence is *not* a cyclic subgroup of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. In this case, we can write

$$\langle 001, 010 \rangle = \{000, 001, 010, 011\} < \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Every (nontrivial) group G has *at least* two subgroups:

1. the trivial subgroup: $\{e\}$
2. the non-proper subgroup: G. (Every group is a subgroup of itself.)

Question

Which groups have *only* these two subgroups?

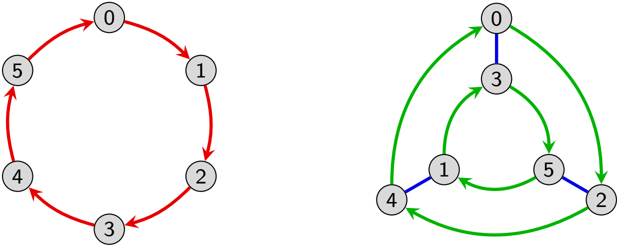
Yet one more example: $\mathbb{Z}_6$

It is not difficult to see that the subgroups of $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ are

$$\langle 0 \rangle = \{0\}, \qquad \langle 2 \rangle = \langle 4 \rangle = \{0, 2, 4\}, \qquad \langle 3 \rangle = \{0, 3\}, \qquad \langle 1 \rangle = \langle 5 \rangle = \mathbb{Z}_6.$$

Depending on our choice of generators and layout of the Cayley diagram, not all of these subgroups may be "visually obvious."

Here are two Cayley diagrams for $\mathbb{Z}_6$, one generated by $\langle 1 \rangle$ and the other by $\langle 2, 3 \rangle$:



One last example: D_4

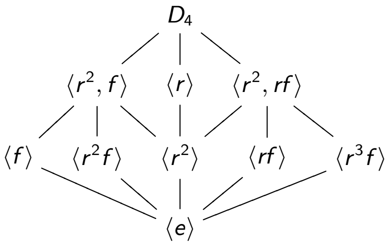
The dihedral group D_4 has 10 subgroups, though some of these are isomorphic to each other:

$$\{e\}, \underbrace{\langle r^2\rangle, \langle f\rangle, \langle rf\rangle, \langle r^2 f\rangle, \langle r^3 f\rangle}_{\text{order 2}}, \underbrace{\langle r\rangle, \langle r^2, f\rangle, \langle r^2, rf\rangle}_{\text{order 4}}, D_4.$$

Remark

We can arrange the subgroups in a diagram called a subgroup lattice that shows which subgroups contain other subgroups. This is best seen by an example.

The subgroup lattice of D_4:



A (terrible) way to find all subgroups

Here is a brute-force method for finding all subgroups of a given group G of order n.

Though this algorithm is horribly inefficient, it makes a good thought exercise.

0. we always have $\{e\}$ and G as subgroups
1. find all subgroups generated by a single element ("cyclic subgroups")
2. find all subgroups generated by 2 elements

$$\vdots$$

n-1. find all subgroups generated by $n-1$ elements

Along the way, we will certainly duplicate subgroups; one reason why this is so inefficient and impractible.

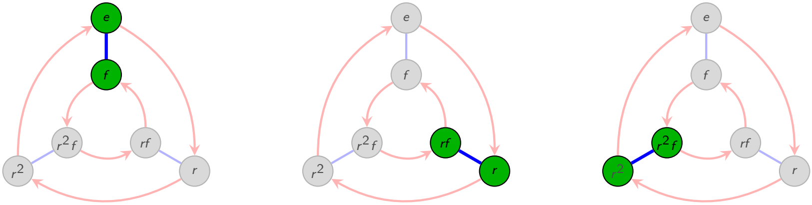
This algorithm works because every group (and subgroup) has a set of generators.

Soon, we will see how a result known as Lagrange's theorem greatly narrows down the possibilities for subgroups.

Cosets

The regularity property of Cayley diagrams implies that identical copies of the fragment of the diagram that correspond to a subgroup appear throughout the rest of the diagram.

For example, the following figures highlight the repeated copies of $\langle f \rangle = \{e, f\}$ in D_3:



However, only one of these copies is actually a group! Since the other two copies do *not* contain the identity, they cannot be groups.

Key concept

The elements that form these repeated copies of the subgroup fragment in the Cayley diagram are called cosets.

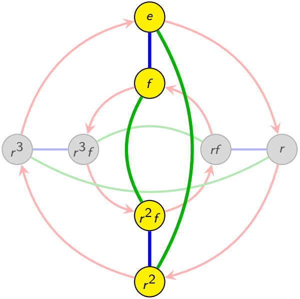
An example: D_4

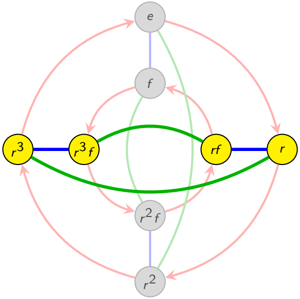
Let's find all of the cosets of the subgroup $H = \langle f, r^2 \rangle = \{e, f, r^2, r^2 f\}$ of D_4.

If we use r^2 as a generator in the Cayley diagram of D_4, then it will be easier to "see" the cosets.

Note that $D_4 = \langle r, f \rangle = \langle r, f, r^2 \rangle$. The cosets of $H = \langle f, r^2 \rangle$ are:

$$H = \langle f, r^2 \rangle = \underbrace{\{e, f, r^2, r^2 f\}}_{\text{original}}, \qquad rH = r\langle f, r^2 \rangle = \underbrace{\{r, r^3, rf, r^3 f\}}_{\text{copy}}.$$





More on cosets

Definition

If H is a subgroup of G, then a (left) coset is a set

$$aH = \{ah : h \in H\},$$

where $a \in G$ is some fixed element. The distinguished element (in this case, a) that we choose to use to name the coset is called the representative.

Remark

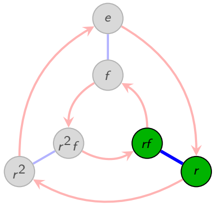
In a Cayley diagram, the (left) coset aH can be found as follows: start from node a and follow *all* paths in H.

For example, let $H = \langle f \rangle$ in D_3. The coset $\{r, rf\}$ of H is the set

$$rH = r\langle f \rangle = r\{e, f\} = \{r, rf\}.$$

Alternatively, we could have written $(rf)H$ to denote the same coset, because

$$rfH = rf\{e, f\} = \{rf, rf^2\} = \{rf, r\}.$$



More on cosets

The following results should be "visually clear" from the Cayley diagrams and the regularity property. Formal algebraic proofs that are not done here will be assigned as homework.

Proposition

For any subgroup $H \leq G$, the union of the (left) cosets of H is the whole group G.

Proof

The element $g \in G$ lies in the coset gH, because $g = ge \in gH = \{gh \mid h \in H\}$. $\square$

Proposition

Each (left) coset can have multiple representatives. Specifically, if $b \in aH$, then $aH = bH$. $\square$

Proposition

All (left) cosets of $H \leq G$ have the same size. $\square$

More on cosets

Proposition

For any subgroup $H \leq G$, the (left) cosets of H partition the group G.

Proof

We know that the element $g \in G$ lies in a (left) coset of H, namely gH. Uniqueness follows because if $g \in kH$, then $gH = kH$. $\square$

Subgroups also have right cosets:

$$Ha = \{ha \colon h \in H\}.$$

For example, the right cosets of $H = \langle f \rangle$ in D_3 are

$$Hr = \langle f \rangle r = \{e, f\}r = \{r, fr\} = \{r, r^2 f\}$$

(recall that $fr = r^2 f$) and

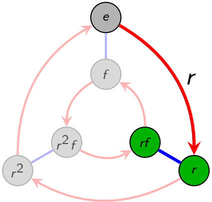
$$\langle f \rangle r^2 = \{e, f\}r^2 = \{r^2, fr^2\} = \{r^2, rf\}.$$

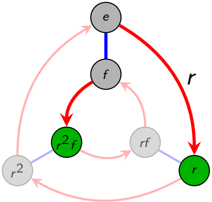
In this example, the left cosets for $\langle f \rangle$ are different than the right cosets. Thus, they must look different in the Cayley diagram.

Left vs. right cosets

The left diagram below shows the left coset $r\langle f\rangle$ in D_3: the nodes that f arrows can reach after the path to r has been followed.

The right diagram shows the right coset $\langle f\rangle r$ in D_3: the nodes that r arrows can reach from the elements in $\langle f\rangle$.





Thus, left cosets look like copies of the subgroup, while the elements of right cosets are usually scattered, because we adopted the convention that arrows in a Cayley diagram represent right multiplication.

Key point

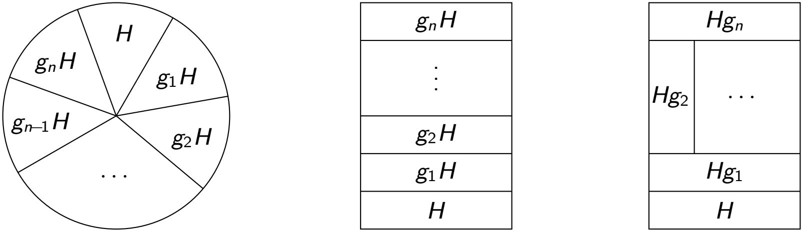
Left and right cosets are generally different.

Left vs. right cosets

For any subgroup $H \leq G$, we can think of G as the union of non-overlapping and equal size copies of *any* subgroup, namely that subgroup's left cosets.

Though the right cosets also partition G, the corresponding partitions could be different!

Here are a few visualizations of this idea:



Definition

If $H < G$, then the index of H in G, written $[G : H]$, is the number of distinct left (or equivalently, right) cosets of H in G.

Left vs. right cosets

The left and right cosets of the subgroup $H = \langle f \rangle \leq D_3$ are *different*:

r^2H	r^2f	r^2
rH	r	rf
H	e	f

	Hr	Hr^2
	r^2f	r^2
	r	rf
H	e	f

The left and right cosets of the subgroup $N = \langle r \rangle \leq D_3$ are *the same*:

fN	f	rf	r^2f
N	e	r	r^2

Nf	f	rf	r^2f
N	e	r	r^2

Proposition

If $H \leq G$ has index $[G : H] = 2$, then the left and right cosets of H are the same.

Cosets of abelian groups

Recall that in some abelian groups, we use the symbol $+$ for the binary operation.

In this case, left cosets have the form $a + H$ (instead of aH).

For example, let $G = (\mathbb{Z}, +)$, and consider the subgroup $H = 4\mathbb{Z} = \{4k \mid k \in \mathbb{Z}\}$ consisting of multiples of 4.

The left cosets of H are

$$\begin{aligned} H &= \{\dots, -12, -8, -4, 0, 4, 8, 12, \dots\} \\ 1 + H &= \{\dots, -11, -7, -3, 1, 5, 9, 13, \dots\} \\ 2 + H &= \{\dots, -10, -6, -2, 2, 6, 10, 14, \dots\} \\ 3 + H &= \{\dots, -9, -5, -1, 3, 7, 11, 15, \dots\}\,. \end{aligned}$$

Notice that these are the same as the right cosets of H:

$$H\,, \qquad H + 1\,, \qquad H + 2\,, \qquad H + 3\,.$$

Do you see why the left and right cosets of an abelian group will *always* be the same?

Also, note why it would be incorrect to write $3H$ for the coset $3 + H$. In fact, $3H$ would usually be interpreted to mean the subgroup $3(4\mathbb{Z}) = 12\mathbb{Z}$.

A theorem of Joseph Lagrange

The following result is named after the prolific 18th century Italian/French mathematician Joseph Lagrange.

Lagrange's Theorem

Assume G is finite. If $H < G$, then $|H|$ divides $|G|$.

Proof

Suppose there are n left cosets of the subgroup H. Since they are all the same size, and they partition G, we must have

$$|G| = \underbrace{|H| + \cdots + |H|}_{n \text{ copies}} = n\,|H|.$$

Therefore, $|H|$ divides $|G|$. $\square$

Corollary

If $|G| < \infty$ and $H \leq G$, then

$$[G : H] = \frac{|G|}{|H|}.$$

Normal subgroups

Definition

A subgroup H of G is a normal subgroup of G if $gH = Hg$ for *all* $g \in G$. We denote this as $H \lhd G$, or $H \unlhd G$.

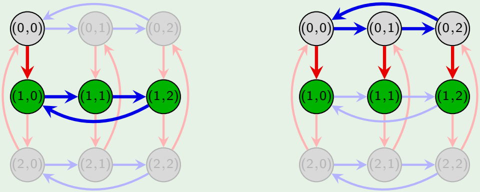
Observation

Subgroups of abelian groups are always normal, because for any $H < G$,

$$aH = \{ah \colon h \in H\} = \{ha \colon h \in H\} = Ha\,.$$

Example

Consider the subgroup $H = \langle (0,1) \rangle = \{(0,0),(0,1),(0,2)\}$ in the group $\mathbb{Z}_3 \times \mathbb{Z}_3$ and take $g = (1,0)$. Addition is done modulo 3, componentwise. The following depicts the equality $g + H = H + g$:



Normal subgroups of nonabelian groups

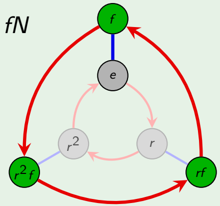
Subgroups whose left and right cosets agree are normal and they have very special properties.

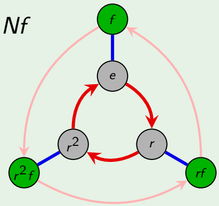
Since subgroups of abelian groups are always normal, we will be particularly interested in normal subgroups of non-abelian groups.

Example

Consider the subgroup $N = \{e, r, r^2\} \leq D_3$.

The cosets (left or right) of N are $N = \{e, r, r^2\}$ and $Nf = \{f, rf, r^2f\} = fN$. The following depicts this equality; the coset $fN = Nf$ are the green nodes.





Conjugate subgroups

For a fixed element $g \in G$, the set

$$gHg^{-1} = \{ghg^{-1} \mid h \in H\}$$

is called the conjugate of H by g.

Observation 1

For any $g \in G$, the conjugate gHg^{-1} is a subgroup of G.

Proof

1. Identity: $e = geg^{-1}$. ✓
2. Closure: $(gh_1g^{-1})(gh_2g^{-1}) = gh_1h_2g^{-1}$. ✓
3. Inverses: $(ghg^{-1})^{-1} = gh^{-1}g^{-1}$. ✓ □

Observation 2

$gh_1g^{-1} = gh_2g^{-1}$ if and only if $h_1 = h_2$. □

On the homework, you will show that H and gHg^{-1} are isomorphic subgroups. (Though we don't yet know how to do this, or precisely what it means.)

How to check if a subgroup is normal

If $gH = Hg$, then right-multiplying both sides by g^{-1} yields $gHg^{-1} = H$.

This gives us a new way to check whether a subgroup H is normal in G.

Useful remark

The following conditions are all equivalent to a subgroup $H \leq G$ being normal:

(i) $gH = Hg$ for all $g \in G$; ("left cosets are right cosets");

(ii) $gHg^{-1} = H$ for all $g \in G$; ("only one conjugate subgroup")

(iii) $ghg^{-1} \in H$ for all $g \in G$; ("closed under conjugation").

Sometimes, one of these methods is *much* easier than the others!

For example, all it takes to show that H is not normal is finding *one element* $h \in H$ for which $ghg^{-1} \notin H$ for some $g \in G$.

As another example, if we happen to know that G has a unique subgroup of size $|H|$, then H *must* be normal. (Why?)

Products and quotients of groups

Previously, we looked for smaller groups lurking inside a group.

Exploring the subgroups of a group gives us insight into the its internal structure.

Next, we will introduce the following topics:

1. direct products: a method for making *larger* groups from smaller groups.
2. quotients: a method for making *smaller* groups from larger groups.

Before we begin, we'll note that we can *always* form a direct product of two groups.

In constrast, we cannot always take the quotient of two groups. In fact, quotients are restricted to some pretty specific circumstances, as we shall see.

Direct products, algebraically

It is easiest to think of direct products of groups algebraically, rather than visually.

If A and B are groups, there is a natural group structure on the set

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

Definition

The **direct product** of groups A and B consists of the set $A \times B$, and the group operation is done componentwise: if $(a, b), (c, d) \in A \times B$, then

$$(a, b) * (c, d) = (ac, bd).$$

We call A and B the factors of the direct product.

Note that the binary operations on A and B could be different. One might be $*$ and the other $+$.

For example, in $D_3 \times \mathbb{Z}_4$:

$$(r^2, 1) * (fr, 3) = (r^2 fr, 1 + 3) = (rf, 0).$$

These elements do *not* commute:

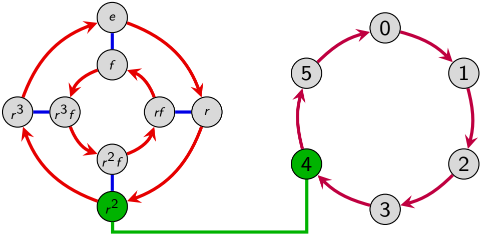
$$(fr, 3) * (r^2, 1) = (fr^3, 3 + 1) = (f, 0).$$

Direct products, visually

Here's one way to think of the direct product of two cyclic groups, say $\mathbb{Z}_n \times \mathbb{Z}_m$: Imagine a slot machine with two wheels, one with n spaces (numbered 0 through $n-1$) and the other with m spaces (numbered 0 through $m-1$).

The actions are: spin one or both of the wheels. Each action can be labeled by where we end up on each wheel, say (i, j).

Here is an example for a more general case: the element $(r^2, 4)$ in $D_4 \times \mathbb{Z}_6$.



Key idea

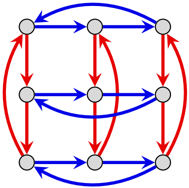
The direct product of two groups joins them so they act independently of each other.

Cayley diagrams of direct products

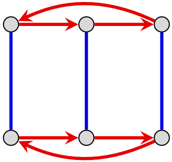
Remark

Just because a group is not written with $\times$ doesn't mean that there isn't some hidden direct product structure lurking. For example, V_4 is really just $C_2 \times C_2$.

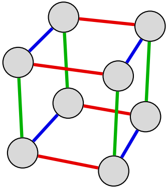
Here are some examples of direct products:



$C_3 \times C_3$



$C_3 \times C_2$



$C_2 \times C_2 \times C_2$

Even more surprising, the group $C_3 \times C_2$ is actually isomorphic to the cyclic group C_6!

Indeed, the Cayley diagram for C_6 using generators r^2 and r^3 is the same as the Cayley diagram for $C_3 \times C_2$ above.

We'll understand this better later in the class when we study homomorphisms. For now, we will focus our attention on direct products.

Cayley diagrams of direct products

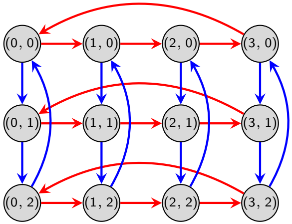
Let e_A be the identity of A and e_B the identity of B.

Given a Cayley diagram of A with generators $a_1, \dots, a_k$, and a Cayley diagram of B with generators $b_1, \dots, b_\ell$, we can create a Cayley diagram for $A \times B$ as follows:

- Vertex set: $\{(a, b) \mid a \in A,\ b \in B\}$.
- Generators: $(a_1, e_b), \dots, (a_k, e_b)$ and $(e_a, b_1), \dots, (e_a, b_\ell)$.

Frequently it is helpful to arrange the vertices in a rectangular grid.

For example, here is a Cayley diagram for the group $\mathbb{Z}_4 \times \mathbb{Z}_3$:



What are the subgroups of $\mathbb{Z}_4 \times \mathbb{Z}_3$? There are six (did you find them all?), they are:

$$\mathbb{Z}_4 \times \mathbb{Z}_3, \qquad \{0\} \times \{0\}, \qquad \{0\} \times \mathbb{Z}_3, \qquad \mathbb{Z}_4 \times \{0\}, \qquad \mathbb{Z}_2 \times \mathbb{Z}_3, \qquad \mathbb{Z}_2 \times \{0\}.$$

Subgroups of direct products

Remark

If $H \leq A$, and $K \leq B$, then $H \times K$ is a subgroup of $A \times B$.

For $\mathbb{Z}_4 \times \mathbb{Z}_3$, *all* subgroups had this form. However, this is not always true.

For example, consider the group $\mathbb{Z}_2 \times \mathbb{Z}_2$, which is really just V_4. Since $\mathbb{Z}_2$ has two subgroups, the following four sets are subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2$:

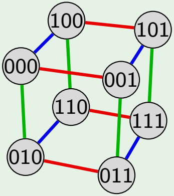
$$\mathbb{Z}_2 \times \mathbb{Z}_2, \qquad \{0\} \times \{0\}, \qquad \mathbb{Z}_2 \times \{0\} = \langle (1,0) \rangle, \qquad \{0\} \times \mathbb{Z}_2 = \langle (0,1) \rangle.$$

However, one subgroup of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is missing from this list: $\langle (1,1) \rangle = \{(0,0), (1,1)\}$.

Exercise

What are the subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$?

Here is a Cayley diagram, writing the elements of the product as abc rather than (a, b, c).



Hint: There are 16 subgroups!

Direct products, visually

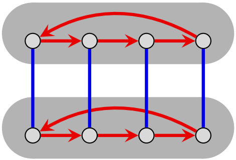
It's not needed, but one can construct the Cayley diagram of a direct product using the following "inflation" method.

Inflation algorithm

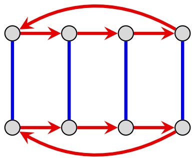
To make a Cayley diagram of $A \times B$ from the Cayley diagrams of A and B:

1. Begin with the Cayley diagram for A.
2. Inflate each node, and place in it a copy of the Cayley diagram for B. (Use different colors for the two Cayley diagrams.)
3. Remove the (inflated) nodes of A while using the arrows of A to connect corresponding nodes from each copy of B. That is, remove the A diagram but treat its arrows as a blueprint for how to connect corresponding nodes in the copies of B.

Cyclic group $\mathbb{Z}_2$



each node contains a copy of $\mathbb{Z}_4$



direct product group $\mathbb{Z}_4 \times \mathbb{Z}_2$

Properties of direct products

Recall the following important definition.

Definition

A subgroup $H < G$ is normal if $xH = Hx$ for all $x \in G$. We denote this by $H \lhd G$.

Assuming A and B are not trivial, the direct product $A \times B$ has *at least* four normal subgroups:

$$\{e_A\} \times \{e_B\}\,, \qquad A \times \{e_B\}\,, \qquad \{e_A\} \times B\,, \qquad A \times B\,.$$

Sometimes we "abuse notation" and write $A \lhd A \times B$ and $B \lhd A \times B$ for the middle two. (Technically, A and B are not even subsets of $A \times B$.)

Here's another observation: "A-arrows" are independent of "B-arrows."

Observation

In a Cayley diagram for $A \times B$, following "A-arrows" neither impacts or is impacted by the location in group B.

Algebraically, this is just saying that $(a, e_b) * (e_a, b) = (a, b) = (e_a, b) * (a, e_b)$.

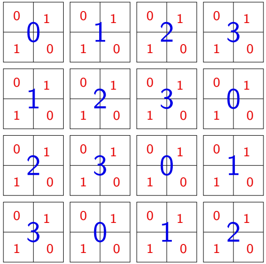
Multiplication tables of direct products

Direct products can also be visualized using multiplication tables.

The general process should be clear after seeing the following example; constructing the table for the group $\mathbb{Z}_4 \times \mathbb{Z}_2$:

0	1	2	3
1	2	3	0
2	3	0	1
3	0	1	2

multiplication table
for the group $\mathbb{Z}_4$



inflate each cell to contain a copy
of the multiplication table of $\mathbb{Z}_2$

(0,0)	(0,1)	(1,0)	(1,1)	(2,0)	(2,1)	(3,0)	(3,1)
(0,1)	(0,0)	(1,1)	(1,0)	(2,1)	(2,0)	(3,1)	(3,0)
(1,0)	(1,1)	(2,0)	(2,1)	(3,0)	(3,1)	(0,0)	(0,1)
(1,1)	(1,0)	(2,1)	(2,0)	(3,1)	(3,0)	(0,1)	(0,0)
(2,0)	(2,1)	(3,0)	(3,1)	(0,0)	(0,1)	(1,0)	(1,1)
(2,1)	(2,0)	(3,1)	(3,0)	(0,1)	(0,0)	(1,1)	(1,0)
(3,0)	(3,1)	(0,0)	(0,1)	(1,0)	(1,1)	(2,0)	(2,1)
(3,1)	(3,0)	(0,1)	(0,0)	(1,1)	(1,0)	(2,1)	(2,0)

join the little tables and element names
to form the direct product table for $\mathbb{Z}_4 \times \mathbb{Z}_2$

Quotients

Direct products make larger groups from smaller groups. It is a way to *multiply* groups.

The opposite procedure is called taking a quotient. It is a way to *divide* groups.

Unlike what we did with direct products, we will first describe the quotient operation using Cayley diagrams, and then formalize it algebraically and explore properties of the resulting group.

Definition

To divide a group G by one of its subgroups H, follow these steps:

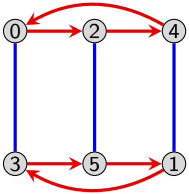
1. Organize a Cayley diagram of G by H (so that we can "see" the subgroup H in the diagram for G).
2. Collapse each left coset of H into one large node. Unite those arrows that now have the same start and end nodes. This forms a new diagram with fewer nodes and arrows.
3. IF (and *only* if) the resulting diagram is a Cayley diagram of a group, you have obtained the quotient group of G by H, denoted G/H (say: "G mod H".) If not, then G cannot be divided by H.

An example: $\mathbb{Z}_3 < \mathbb{Z}_6$

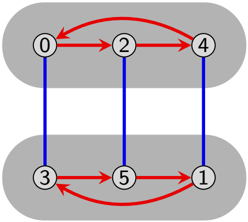
Consider the group $G = \mathbb{Z}_6$ and its normal subgroup $H = \langle 2 \rangle = \{0, 2, 4\}$.

There are two (left) cosets: $H = \{0, 2, 4\}$ and $1 + H = \{1, 3, 5\}$.

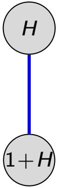
The following diagram shows how to take a quotient of $\mathbb{Z}_6$ by H.



$\mathbb{Z}_6$ organized by the subgroup $H = \langle 2 \rangle$



Left cosets of H are near each other



Collapse cosets into single nodes

In this example, the resulting diagram *is* a Cayley diagram. So, we *can* divide $\mathbb{Z}_6$ by $\langle 2 \rangle$, and we see that $\mathbb{Z}_6/H$ is isomorphic to $\mathbb{Z}_2$.

We write this as $\mathbb{Z}_6/H \cong \mathbb{Z}_2$.

A few remarks

- Step 3 of the Definition says "IF the new diagram is a Cayley diagram …" Sometimes it won't be, in which case there is no quotient.
- The elements of G/H are the cosets of H. Asking if G/H exists amounts to asking if the set of left (or right) cosets of H forms a group. (More on this later.)
- In light of this, given *any* subgroup $H < G$ (normal or not), we will let

  $$G/H := \{gH \mid g \in G\}$$

  denote the set of left cosets of H in G.
- Not surprisingly, if $G = A \times B$ and we divide G by A (technically $A \times \{e\}$), the quotient group is B. (We'll see why shortly).

Caveat!

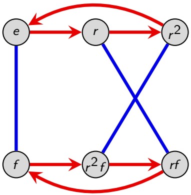
The converse of the previous statement is generally *not* true. That is, if G/H is a group, then G is in general *not* a direct product of H and G/H.

An example: $C_3 < D_3$

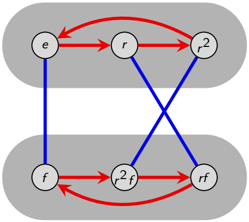
Consider the group $G = D_3$ and its normal subgroup $H = \langle r \rangle \cong C_3$.

There are two (left) cosets: $H = \{e, r, r^2\}$ and $fH = \{f, rf, r^2 f\}$.

The following diagram shows how to take a quotient of D_3 by H.



D_3 organized by the subgroup $H = \langle r \rangle$



Left cosets of H are near each other

Collapse cosets into single nodes

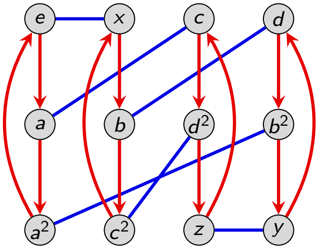
The result is a Cayley diagram for C_2, thus

$$D_3/H \cong C_2\,. \qquad \text{However...} \quad C_3 \times C_2 \not\cong D_3\,.$$

Note that $C_3 \times C_2$ is abelian, but D_3 is not.

Example: $G = A_4$ and $H = \langle x, z \rangle \cong V_4$

Consider the following Cayley diagram for $G = A_4$ using generators $\langle a, x \rangle$.

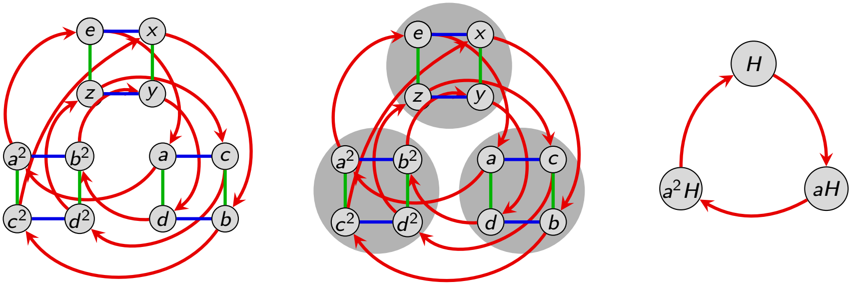


Consider $H = \langle x, z \rangle = \{e, x, y, z\} \cong V_4$. This subgroup is not "visually obvious" in this Cayley diagram.

Let's add z to the generating set, and consider the resulting Cayley diagram.

Example: $G = A_4$ and $H = \langle x, z \rangle \cong V_4$

Here is a Cayley diagram for A_4 (with generators x, z, and a), organized by the subgroup $H = \langle x, z \rangle$ which allows us to see the left cosets of H clearly.



A_4 organized by the subgroup $H = \langle x, z \rangle$

Left cosets of H are near each other

Collapse cosets into single nodes

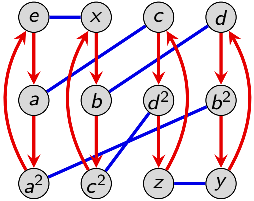
The resulting diagram is a Cayley diagram! Therefore, $A_4/H \cong C_3$. However, A_4 is *not* isomorphic to the (abelian) group $V_4 \times C_3$.

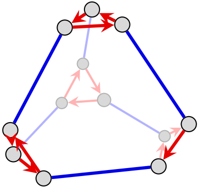
Example: $G = A_4$ and $H = \langle a \rangle \cong C_3$

Let's see an example where we cannot divide G by a particular subgroup H.

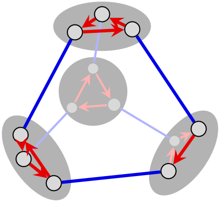
Consider the subgroup $H = \langle a \rangle \cong C_3$ of A_4.

Do you see what will go wrong if we try to divide A_4 by $H = \langle a \rangle$?

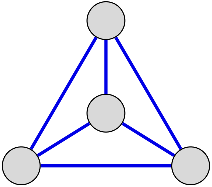




A_4 organized by the subgroup $H = \langle a \rangle$



Left cosets of H are near each other



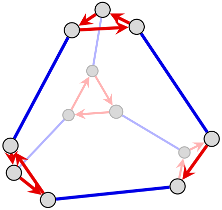
Collapse cosets into single nodes

This resulting diagram is *not* a Cayley diagram! There are multiple outgoing blue arrows from each node.

When can we divide G by a subgroup H?

Consider $H = \langle a \rangle \leq A_4$ again.

The left cosets are easy to spot.



Remark

The right cosets are *not* the same as the left cosets! The blue arrows out of any single coset scatter the nodes.

Thus, $H = \langle a \rangle$ is *not* normal in A_4.

If we took the effort to check our first 3 examples, we would find that in each case, the left cosets and right cosets coincide. In those examples, G/H existed, and H was normal in G.

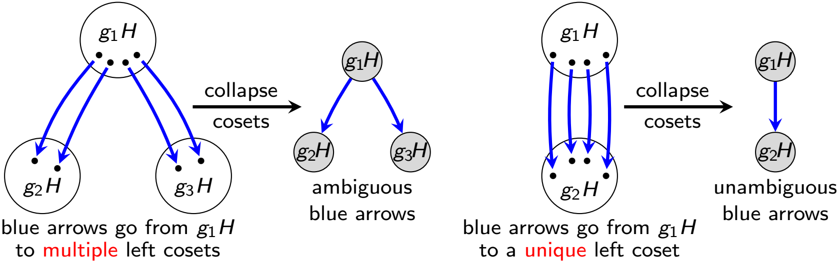
However, these 4 examples do not constitute a proof; they only provide evidence that the claim is true.

When can we divide G by a subgroup H?

Let's try to gain more insight. Consider a group G with subgroup H. Recall that:

- each **left coset** gH is the set of nodes that the H-arrows can reach from g (which looks like a copy of H at g);
- each **right coset** Hg is the set of nodes that the g-arrows can reach from H.

The following figure depicts the potential ambiguity that may arise when cosets are collapsed in the sense of our quotient definition.



The action of the blue arrows above illustrates multiplication of a left coset on the right by some element. That is, the picture shows how left and right cosets *interact*.

When can we divide G by a subgroup H?

When H is normal, $gH = Hg$ for all $g \in G$.

In this case, to whichever coset one g arrow leads from H (the left coset), *all g arrows lead unanimously and unambiguously* (because it is also a right coset Hg).

Thus, in this case, collapsing the cosets is a *well-defined* operation.

Finally, we have an answer to our original question of when we can take a quotient.

Quotient theorem

If $H < G$, then the quotient group G/H can be constructed *if and only if* $H \lhd G$.

To summarize our "visual argument": The quotient process succeeds iff the resulting diagram is a valid Cayley diagram.

Nearly all aspects of valid Cayley diagrams are guaranteed by the quotient process: Every node has exactly one incoming and outgoing edge of each color, because $H \lhd G$. The diagram is regular too.

Though it's convincing, this argument isn't quite a formal proof; we'll do a rigorous algebraic proof next.

Quotient groups, algebraically

To prove the Quotient Theorem, we need to describe the quotient process algebraically.

Recall that even if H is not normal in G, we will still denote the set of left cosets of H in G by G/H.

Quotient theorem (restated)

When $H \lhd G$, the set of cosets G/H forms a group.

This means there is a well-defined binary operation on the set of cosets. But how do we "multipy" two cosets?

If aH and bH are left cosets, define

$$aH \cdot bH := abH\,.$$

Clearly, G/H is closed under this operation. But we also need to verify that this definition is well-defined.

By this, we mean that it *does not depend* on our choice of coset representative.

Quotient groups, algebraically

Lemma

Let $H \lhd G$. Multiplication of cosets is well-defined:

$$\text{if } a_1H = a_2H \text{ and } b_1H = b_2H, \text{ then } a_1H \cdot b_1H = a_2H \cdot b_2H.$$

Proof

Suppose that $H \lhd G$, $a_1H = a_2H$ and $b_1H = b_2H$. Then

$$\begin{array}{rcll}
a_1H \cdot b_1H & = & a_1b_1H & \text{(by definition)} \\
 & = & a_1(b_2H) & (b_1H = b_2H \text{ by assumption}) \\
 & = & (a_1H)b_2 & (b_2H = Hb_2 \text{ since } H \lhd G) \\
 & = & (a_2H)b_2 & (a_1H = a_2H \text{ by assumption}) \\
 & = & a_2b_2H & (b_2H = Hb_2 \text{ since } H \lhd G) \\
 & = & a_2H \cdot b_2H & \text{(by definition)}
\end{array}$$

Thus, the binary operation on G/H is well-defined. $\square$

Quotient groups, algebraically

Quotient theorem (restated)

When $H \lhd G$, the set of cosets G/H forms a group.

Proof

There is a well-defined binary operation on the set of left (equivalently, right) cosets: $aH \cdot bH = abH$. We need to verify the three remaining properties of a group:

Identity. The coset $H = eH$ is the identity because for any coset $aH \in G/H$,

$$aH \cdot H = aeH = aH = eaH = H \cdot aH\,.$$

Inverses. Given a coset aH, its inverse is $a^{-1}H$, because

$$aH \cdot a^{-1}H = eH = a^{-1}H \cdot aH\,.$$

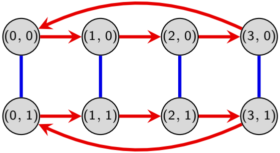
Closure. This is immediate, because $aH \cdot bH = abH$ is another coset in G/H. $\square$

Properties of quotients

Question

If H and K are subgroups and $H \cong K$, then are G/H and G/K isomorphic?

For example, here is a Cayley diagram for the group $\mathbb{Z}_4 \times \mathbb{Z}_2$:



It is visually obvious that the quotient of $\mathbb{Z}_4 \times \mathbb{Z}_2$ by the subgroup $\langle (0,1) \rangle \cong \mathbb{Z}_2$ is the group $\mathbb{Z}_4$.

The quotient of $\mathbb{Z}_4 \times \mathbb{Z}_2$ by the subgroup $\langle (2,0) \rangle \cong \mathbb{Z}_2$ is a bit harder to see. Algebraically, it consists of the cosets

$$\langle (2,0) \rangle\,, \qquad (1,0) + \langle (2,0) \rangle\,, \qquad (0,1) + \langle (2,0) \rangle\,, \qquad (1,1) + \langle (2,0) \rangle\,.$$

It is now apparent that this group is isomorphic to V_4.

Thus, the answer to the question above is "no." Surprised?

Normalizers

Question

If $H < G$ but H is *not* normal, can we measure "how far" H is from being normal?

Recall that $H \lhd G$ iff $gH = Hg$ for all $g \in G$. So, one way to answer our question is to check how many $g \in G$ satisfy this requirement. Imagine that each $g \in G$ is voting as to whether H is normal:

$$gH = Hg \quad \text{"yea"} \qquad\qquad gH \neq Hg \quad \text{"nay"}$$

At a *minimum*, every $g \in H$ votes "yea." (Why?)

At a *maximum*, every $g \in G$ could vote "yea," but this only happens when H really is normal.

There can be levels between these 2 extremes as well.

Definition

The set of elements in G that vote in favor of H's normality is called the normalizer of H in G, denoted $N_G(H)$. That is,

$$N_G(H) = \{g \in G : gH = Hg\} = \{g \in G : gHg^{-1} = H\}.$$

Normalizers

Let's explore some possibilities for what the normalizer of a subgroup can be. In particular, is it a subgroup?

Observation 1

If $g \in N_G(H)$, then $gH \subseteq N_G(H)$.

Proof

If $gH = Hg$, then $gH = bH$ for all $b \in gH$. Therefore, $bH = gH = Hg = Hb$. $\square$

The deciding factor in how a left coset votes is whether it is a right coset (members of gH vote as a block – exactly when $gH = Hg$).

Observation 2

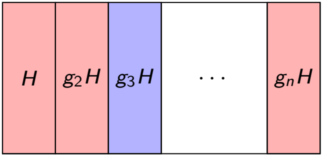
$|N_G(H)|$ is a multiple of $|H|$.

Proof

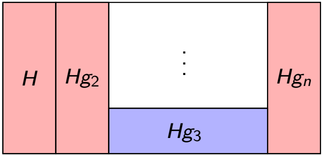
By Observation 1, $N_G(H)$ is made up of whole (left) cosets of H, and all (left) cosets are the same size and disjoint. $\square$

Normalizers

Consider a subgroup $H \leq G$ of index n. Suppose that the left and right cosets partition G as shown below:



Partition of G by the left cosets of H



Partition of G by the right cosets of H

The cosets H, and $g_2H = Hg_2$, and $g_nH = Hg_n$ all vote "yea".

The left coset g_3H votes "nay" because $g_3H \neq Hg_3$.

Assuming all other cosets vote "nay", the normalizer of H is

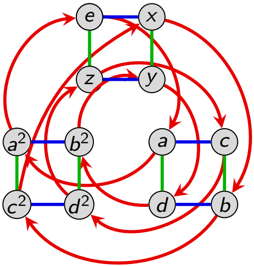
$$N_G(H) = H \cup g_2H \cup g_nH\,.$$

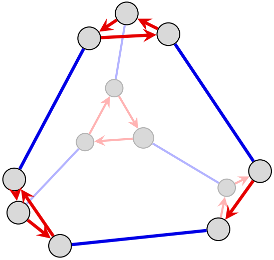
In summary, the two "extreme cases" for $N_G(H)$ are:

- $N_G(H) = G$: iff H is a normal subgroup
- $N_G(H) = H$: H is as "unnormal as possible"

An example: A_4

We saw earlier that $H = \langle x, z \rangle \lhd A_4$. Therefore, $N_{A_4}(H) = A_4$.





At the other extreme, consider $\langle a \rangle < A_4$ again, which is as far from normal as it can possibly be: $\langle a \rangle \ntriangleleft A_4$.

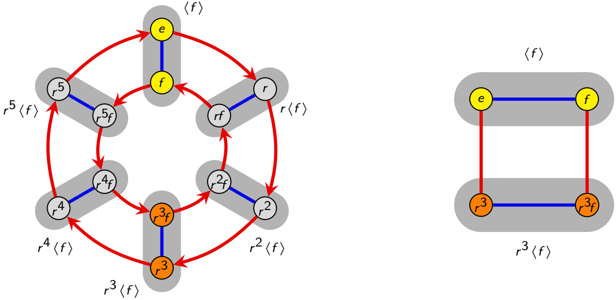
No right coset of $\langle a \rangle$ coincides with a left coset, other than $\langle a \rangle$ itself. Thus, $N_{A_4}(\langle a \rangle) = \langle a \rangle$.

Observation 3

In the Cayley diagram of G, the normalizer of H consists of the copies of H that are connected to H by unanimous arrows.

How to spot the normalizer in the Cayley diagram

The following figure depicts the six left cosets of $H = \langle f \rangle = \{e, f\}$ in D_6.



Note that r^3H is the *only* coset of H (besides H, obviously) that cannot be reached from H by more than one element of D_6.

Thus, $N_{D_6}(\langle f \rangle) = \langle f \rangle \cup r^3\langle f \rangle = \{e, f, r^3, r^3f\} \cong V_4$.

Observe that the normalizer is also a subgroup satisfying: $\langle f \rangle \lneq N_{D_6}(\langle f \rangle) \lneq D_6$.

Do you see the pattern for $N_{D_n}(\langle f \rangle)$? (It depends on whether n is even or odd.)

Normalizers are subgroups!

Theorem

For any $H < G$, we have $N_G(H) < G$.

Proof

Recall that $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$; "*the set of elements that normalize H.*" We need to verify three properties of $N_G(H)$:

(i) Contains the identity;

(ii) Inverses exist;

(iii) Closed under the binary operation.

Identity. Naturally, $eHe^{-1} = \{ehe^{-1} \mid h \in H\} = H$.

Inverses. Suppose $g \in N_G(H)$, which means $gHg^{-1} = H$. We need to show that $g^{-1} \in N_G(H)$. That is, $g^{-1}H(g^{-1})^{-1} = g^{-1}Hg = H$. Indeed,

$$g^{-1}Hg = g^{-1}(gHg^{-1})g = eHe = H\,.$$

Normalizers are subgroups!

Proof (cont.)

Closure. Suppose $g_1, g_2 \in N_G(H)$, which means that $g_1 H g_1^{-1} = H$ and $g_2 H g_2^{-1} = H$. We need to show that $g_1 g_2 \in N_G(H)$.

$$(g_1 g_2) H (g_1 g_2)^{-1} = g_1 g_2 H g_2^{-1} g_1^{-1} = g_1 (g_2 H g_2^{-1}) g_1^{-1} = g_1 H g_1^{-1} = H\,.$$

Since $N_G(H)$ contains the identity, every element has an inverse, and is closed under the binary operation, it is a (sub)group! $\square$

Corollary

Every subgroup is normal in its normalizer:

$$H \lhd N_G(H) \leq G\,.$$

Proof

By definition, $gH = Hg$ for all $g \in N_G(H)$. Therefore, $H \lhd N_G(H)$. $\square$

Conjugation

Recall that for $H \leq G$, the conjugate subgroup of H by a fixed $g \in G$ is

$$gHg^{-1} = \{ghg^{-1} \mid h \in H\}.$$

Additionally, H is normal iff $gHg^{-1} = H$ for all $g \in G$.

We can also fix the element we are conjugating. Given $x \in G$, we may ask:

"*which elements can be written as gxg^{-1} for some $g \in G$?*"

The set of all such elements in G is called the conjugacy class of x, denoted $\mathrm{cl}_G(x)$. Formally, this is the set

$$\mathrm{cl}_G(x) = \{gxg^{-1} \mid g \in G\}.$$

Remarks

- In any group, $\mathrm{cl}_G(e) = \{e\}$, because $geg^{-1} = e$ for any $g \in G$.
- If x and g commute, then $gxg^{-1} = x$. Thus, when computing $\mathrm{cl}_G(x)$, we only need to check gxg^{-1} for those $g \in G$ that *do not commute* with x.
- Moreover, $\mathrm{cl}_G(x) = \{x\}$ iff x commutes with everything in G. (Why?)

Conjugacy classes

Lemma

Conjugacy is an equivalence relation.

Proof

- *Reflexive*: $x = exe^{-1}$.
- *Symmetric*: $x = gyg^{-1} \Rightarrow y = g^{-1}xg$.
- *Transitive*: $x = gyg^{-1}$ and $y = hzh^{-1} \Rightarrow x = (gh)z(gh)^{-1}$. $\square$

Since conjugacy is an equivalence relation, it partitions the group G into equivalence classes (conjugacy classes).

Let's compute the conjugacy classes in D_4. We'll start by finding $\mathrm{cl}_{D_4}(r)$. Note that we only need to compute grg^{-1} for those g that *do not* commute with r:

$$frf^{-1} = r^3, \qquad (rf)r(rf)^{-1} = r^3, \qquad (r^2f)r(r^2f)^{-1} = r^3, \qquad (r^3f)r(r^3f)^{-1} = r^3.$$

Therefore, the conjugacy class of r is $\mathrm{cl}_{D_4}(r) = \{r, r^3\}$.

Since conjugacy is an equivalence relation, $\mathrm{cl}_{D_4}(r^3) = \mathrm{cl}_{D_4}(r) = \{r, r^3\}$.

Conjugacy classes in D_4

To compute $\mathrm{cl}_{D_4}(f)$, we don't need to check e, r^2, f, or r^2f, since these all commute with f:

$$rfr^{-1} = r^2f, \qquad r^3f(r^3)^{-1} = r^2f, \qquad (rf)f(rf)^{-1} = r^2f, \qquad (r^3f)f(r^3f)^{-1} = r^2f\,.$$

Therefore, $\mathrm{cl}_{D_4}(f) = \{f, r^2f\}$.

What is $\mathrm{cl}_{D_4}(rf)$? Note that it has size greater than 1 because rf does not commute with everything in D_4.

It also *cannot* contain elements from the other conjugacy classes. The only element left is r^3f, so $\mathrm{cl}_{D_4}(rf) = \{rf, r^3f\}$.

The "Class Equation", visually:
Partition of D_4 by its
conjugacy classes

e	r	f	r^2f
r^2	r^3	rf	r^3f

We can write $D_4 = \underbrace{\{e\} \cup \{r^2\}}_{\text{these commute with everything in } D_4} \cup \{r, r^3\} \cup \{f, r^2f\} \cup \{r, r^3f\}$.

The class equation

Definition

The center of G is the set $Z(G) = \{z \in G \mid gz = zg, \ \forall g \in G\}$.

Observation

$\mathrm{cl}_G(x) = \{x\}$ if and only if $x \in Z(G)$.

Proof

Suppose x is in its own conjugacy class. This means that

$$\mathrm{cl}_G(x) = \{x\} \iff gxg^{-1} = x, \ \forall g \in G \iff gx = xg, \ \forall g \in G \iff x \in Z(G).$$

$\square$

The Class Equation

For any finite group G,

$$|G| = |Z(G)| + \sum |\,\mathrm{cl}_G(x_i)|$$

where the sum is taken over distinct conjugacy classes of size greater than 1.

More on conjugacy classes

Proposition

Every normal subgroup is the union of conjugacy classes.

Proof

Suppose $n \in N \lhd G$. Then $gng^{-1} \in gNg^{-1} = N$, thus if $n \in N$, its entire conjugacy class $\mathrm{cl}_G(n)$ is contained in N as well. $\square$

Proposition

Conjugate elements have the same order.

Proof

Consider x and $y = gxg^{-1}$.

If $x^n = e$, then $(gxg^{-1})^n = (gxg^{-1})(gxg^{-1}) \cdots (gxg^{-1}) = gx^n g^{-1} = geg^{-1} = e$. Therefore, $|x| \geq |gxg^{-1}|$.

Conversely, if $(gxg^{-1})^n = e$, then $gx^n g^{-1} = e$, and it must follow that $x^n = e$. Therefore, $|x| \leq |gxg^{-1}|$. $\square$

Conjugacy classes in D_6

Let's determine the conjugacy classes of $D_6 = \langle r, f \mid r^6 = e,\ f^2 = e,\ r^i f = fr^{-i} \rangle$.

The center of D_6 is $Z(D_6) = \{e, r^3\}$; these are the *only* elements in size-1 conjugacy classes.

The only two elements of order 6 are r and r^5; so we must have $\mathrm{cl}_{D_6}(r) = \{r, r^5\}$.

The only two elements of order 3 are r^2 and r^4; so we must have $\mathrm{cl}_{D_6}(r^2) = \{r^2, r^4\}$.

Let's compute the conjugacy class of a reflection $r^i f$. We need to consider two cases; conjugating by r^j and by $r^j f$:

- $r^j(r^i f)r^{-j} = r^j r^i r^j f = r^{i+2j} f$
- $(r^j f)(r^i f)(r^j f)^{-1} = (r^j f)(r^i f) f r^{-j} = r^j f r^{i-j} = r^j r^{j-i} f = r^{2j-i} f.$

Thus, $r^i f$ and $r^k f$ are conjugate iff i and k are both even, or both odd.

The Class Equation, visually:
Partition of D_6 by its
conjugacy classes

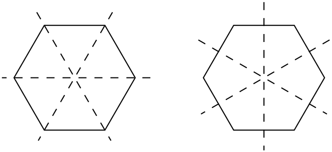
e	r	r^2	f	$r^2 f$	$r^4 f$
r^3	r^5	r^4	rf	$r^3 f$	$r^5 f$

Conjugacy "preserves structure"

Think back to linear algebra. Two matrices A and B are *similar* (=conjugate) if $A = PBP^{-1}$.

Conjugate matrices have the same eigenvalues, eigenvectors, and determinant. In fact, they represent the *same linear map*, but under a change of basis.

If n is even, then there are two "types" of reflections of an n-gon: the axis goes through two corners, or it bisects a pair of sides.



Notice how in D_n, conjugate reflections have the same "type." Do you have a guess of what the conjugacy classes of reflections are in D_n when n is odd?

Also, conjugate rotations in D_n had the same rotating angle, but in the opposite direction (e.g., r^k and r^{n-k}).

Next, we will look at conjugacy classes in the symmetric group S_n. We will see that conjugate permutations have "the same structure."

Cycle type and conjugacy

Definition

Two elements in S_n have the same cycle type if when written as a product of disjoint cycles, there are the same number of length-k cycles for each k.

We can write the cycle type of a permutation $\sigma \in S_n$ as a list $c_1, c_2, \ldots, c_n$, where c_i is the number of cycles of length i in σ.

Here is an example of some elements in S_9 and their cycle types.

- (1 8) (5) (2 3) (4 9 6 7) has cycle type 1,2,0,1.
- (1 8 4 2 3 4 9 6 7) has cycle type 0,0,0,0,0,0,0,0,1.
- $e = (1)(2)(3)(4)(5)(6)(7)(8)(9)$ has cycle type 9.

Theorem

Two elements $g, h \in S_n$ are conjugate if and only if they have the same cycle type.

Big idea

Conjugate permutations have the same structure. Such permutations are *the same up to renumbering*.

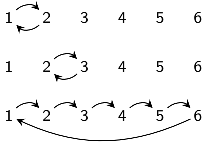
An example

Consider the following permutations in $G = S_6$:

$$g = (1\ 2)$$

$$h = (2\ 3)$$

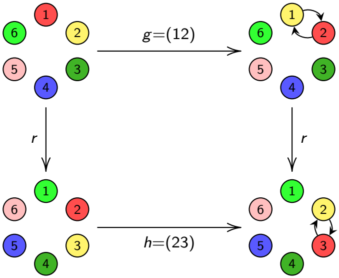
$$r = (1\ 2\ 3\ 4\ 5\ 6)$$



Since g and h have the same cycle type, they are conjugate:

$$(1\ 2\ 3\ 4\ 5\ 6)\,(2\ 3)\,(1\ 6\ 5\ 4\ 3\ 2) = (1\ 2)\,.$$

Here is a visual interpretation of $g = rhr^{-1}$:



Section 4: Maps between groups

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Math 4120, Modern Algebra

Homomorphisms

Throughout the course, we've said things like:

- "This group has the same structure as that group."
- "This group is isomorphic to that group."

However, we've never really spelled out the details about what this means.

We will study a special type of function between groups, called a *homomorphism*. An *isomorphism* is a special type of homomorphism. The Greek roots "homo" and "morph" together mean "same shape."

There are two situations where homomorphisms arise:

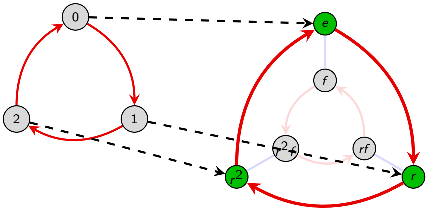
- when one group is a subgroup of another;
- when one group is a quotient of another.

The corresponding homomorphisms are called embeddings and quotient maps.

Also in this chapter, we will completely classify all finite abelian groups, and get a taste of a few more advanced topics, such as the the four "isomorphism theorems," commutator subgroups, and automorphisms.

A motivating example

Consider the statement: $\mathbb{Z}_3 < D_3$. Here is a visual:



$0 \mapsto e$

$1 \mapsto r$

$2 \mapsto r^2$

The group D_3 contains a size-3 cyclic subgroup $\langle r \rangle$, which is identical to $\mathbb{Z}_3$ in structure only. None of the elements of $\mathbb{Z}_3$ (namely 0, 1, 2) are actually in D_3.

When we say $\mathbb{Z}_3 < D_3$, we really mean that the structure of $\mathbb{Z}_3$ shows up in D_3.

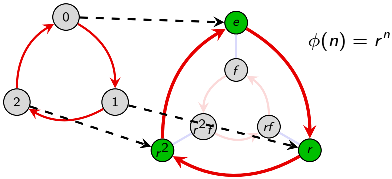
In particular, there is a bijective correspondence between the elements in $\mathbb{Z}_3$ and those in the subgroup $\langle r \rangle$ in D_3. Furthermore, the *relationship* between the corresponding nodes is the same.

A homomorphism is the mathematical tool for succinctly expressing precise structural correspondences. It is a *function* between groups satisfying a few "natural" properties.

Homomorphisms

Using our previous example, we say that this function maps elements of $\mathbb{Z}_3$ to elements of D_3. We may write this as

$$\phi \colon \mathbb{Z}_3 \longrightarrow D_3 \,.$$



The group *from* which a function originates is the domain ($\mathbb{Z}_3$ in our example). The group *into* which the function maps is the codomain (D_3 in our example).

The elements in the codomain that the function maps to are called the image of the function ($\{e, r, r^2\}$ in our example), denoted $\mathrm{Im}(\phi)$. That is,

$$\mathrm{Im}(\phi) = \phi(G) = \{\phi(g) \mid g \in G\} \,.$$

Definition

A homomorphism is a function $\phi \colon G \to H$ between two groups satisfying

$$\phi(ab) = \phi(a)\phi(b), \qquad \text{for all } a, b \in G \,.$$

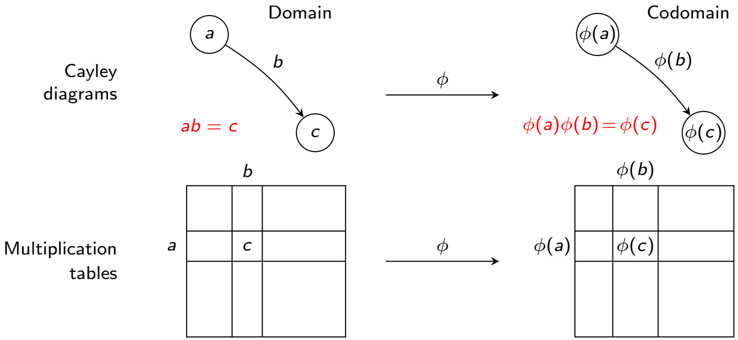
Note that the operation $a \cdot b$ is occurring in the domain while $\phi(a) \cdot \phi(b)$ occurs in the codomain.

Homomorphisms

Remark

Not every function from one group to another is a homomorphism! The condition $\phi(ab) = \phi(a)\phi(b)$ means that the map ϕ preserves the structure of G.

The $\phi(ab) = \phi(a)\phi(b)$ condition has visual interpretations on the level of Cayley diagrams and multiplication tables.



Note that in the Cayley diagrams, b and $\phi(b)$ are paths; they need not just be edges.

An example

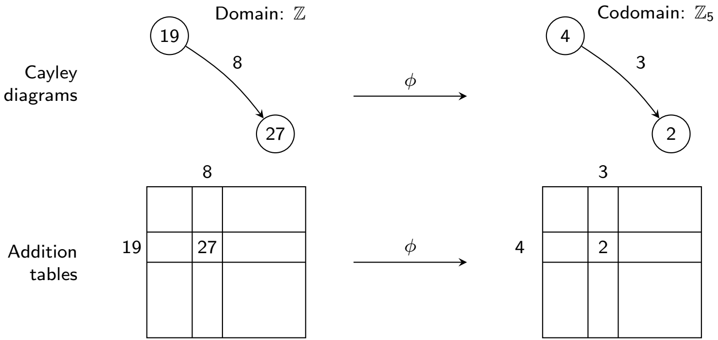
Consider the function ϕ that reduces an integer modulo 5:

$$\phi \colon \mathbb{Z} \longrightarrow \mathbb{Z}_5\,, \qquad \phi(n) = n \pmod 5.$$

Since the group operation is additive, the "homomorphism property" becomes

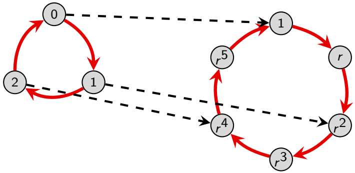
$$\phi(a+b) = \phi(a) + \phi(b)\,.$$

In plain English, this just says that one can "first add and then reduce modulo 5," OR "first reduce modulo 5 and then add."



Types of homomorphisms

Consider the following homomorphism $\theta\colon \mathbb{Z}_3 \to C_6$, defined by $\theta(n) = r^{2n}$:



$0 \mapsto 1$

$1 \mapsto r^2$

$2 \mapsto r^4$

It is easy to check that $\theta(a + b) = \theta(a)\theta(b)$: The red-arrow in $\mathbb{Z}_3$ (representing 1) gets mapped to the 2-step path representing r^2 in C_6.

A homomorphism $\phi\colon G \to H$ that is one-to-one or "injective" is called an embedding: the group G "embeds" into H as a subgroup. If θ is not one-to-one, then it is a quotient.

If $\phi(G) = H$, then ϕ is onto, or surjective.

Definition

A homomorphism that is both injective and surjective is an **isomorphism**.

An **automorphism** is an isomorphism from a group *to itself*.

Homomorphisms and generators

Remark

If we know where a homomorphism maps the generators of G, we can determine where it maps *all* elements of G.

For example, suppose $\phi : \mathbb{Z}_3 \to \mathbb{Z}_6$ was a homomorphism, with $\phi(1) = 4$. Using this information, we can construct the rest of ϕ:

$$\phi(2) = \phi(1+1) = \phi(1) + \phi(1) = 4 + 4 = 2$$
$$\phi(0) = \phi(1+2) = \phi(1) + \phi(2) = 4 + 2 = 0.$$

Example

Suppose that $G = \langle a, b \rangle$, and $\phi \colon G \to H$, and we know $\phi(a)$ and $\phi(b)$. Using this information we can determine the image of any element in G. For example, for $g = a^3 b^2 ab$, we have

$$\phi(g) = \phi(aaabbab) = \phi(a)\,\phi(a)\,\phi(a)\,\phi(b)\,\phi(b)\,\phi(a)\,\phi(b).$$

What do you think $\phi(a^{-1})$ is?

Two basic properties of homomorphisms

Proposition

Let $\phi\colon G \to H$ be a homomorphism. Denote the identity of G by 1_G, and the identity of H by 1_H.

(i) $\phi(1_G) = 1_H$ "ϕ sends the identity to the identity"

(ii) $\phi(g^{-1}) = \phi(g)^{-1}$ "ϕ sends inverses to inverses"

Proof

(i) Pick any $g \in G$. Now, $\phi(g) \in H$; observe that

$$\phi(1_G)\,\phi(g) = \phi(1_G \cdot g) = \phi(g) = 1_H \cdot \phi(g)\,.$$

Therefore, $\phi(1_G) = 1_H$. ✓

(ii) Take any $g \in G$. Observe that

$$\phi(g)\,\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1_G) = 1_H\,.$$

Since $\phi(g)\phi(g^{-1}) = 1_H$, it follows immediately that $\phi(g^{-1}) = \phi(g)^{-1}$. ✓ □

A word of caution

Just because a homomorphism $\phi\colon G \to H$ is determined by the image of its generators does *not* mean that every such image will work.

For example, suppose we try to define a homomorphism $\phi\colon \mathbb{Z}_3 \to \mathbb{Z}_4$ by $\phi(1) = 1$. Then we get

$$\phi(2) = \phi(1+1) = \phi(1) + \phi(1) = 2,$$
$$\phi(0) = \phi(1+1+1) = \phi(1) + \phi(1) + \phi(1) = 3\,.$$

This is *impossible*, because $\phi(0) = 0$. (Identity is mapped to the identity.)

That's not to say that there isn't a homomorphism $\phi\colon \mathbb{Z}_3 \to \mathbb{Z}_4$; note that there is always the trivial homomorphism between two groups:

$$\phi\colon G \longrightarrow H\,, \qquad \phi(g) = 1_H \quad \text{for all } g \in G\,.$$

Exercise

Show that there is no embedding $\phi\colon \mathbb{Z}_n \hookrightarrow \mathbb{Z}$, for $n \geq 2$. That is, *any* such homomorphism must satisfy $\phi(1) = 0$.

Isomorphisms

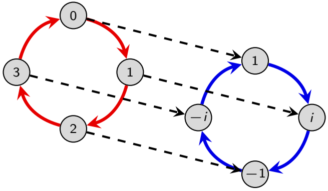
Two isomorphic groups may name their elements differently and may look different based on the layouts or choice of generators for their Cayley diagrams, but the isomorphism between them guarantees that they have the same structure.

When two groups G and H have an isomorphism between them, we say that G and H are isomorphic, and write $G \cong H$.

The roots of the polynomial $f(x) = x^4 - 1$ are called the 4th roots of unity, and denoted $R(4) := \{1, i, -1, -i\}$. They are a subgroup of $\mathbb{C}^* := \mathbb{C} \setminus \{0\}$, the nonzero complex numbers under multiplication.

The following map is an isomorphism between $\mathbb{Z}_4$ and $R(4)$.

$$\phi\colon \mathbb{Z}_4 \longrightarrow R(4)\,, \qquad \phi(k) = i^k\,.$$

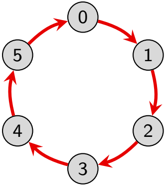


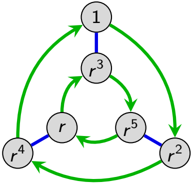
Isomorphisms

Sometimes, the isomorphism is less visually obvious because the Cayley graphs have different structure.

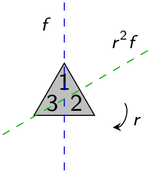
For example, the following is an isomorphism:

$$\phi\colon \mathbb{Z}_6 \longrightarrow C_6$$
$$\phi(k) = r^k$$

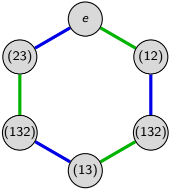


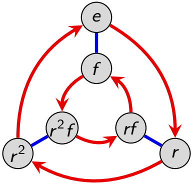


Here is another non-obvious isomorphism between $S_3 = \langle (12), (23) \rangle$ and $D_3 = \langle r, f \rangle$.



$$\phi\colon S_3 \longrightarrow D_3$$
$$\phi\colon (12) \longmapsto r^2 f$$
$$\phi\colon (23) \longmapsto f$$





Another example: the quaternions

Let $\mathrm{GL}_n(\mathbb{R})$ be the set of invertible $n \times n$ matrices with real-valued entries. It is easy to see that this is a group under multiplication.

Recall the quaternion group $Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = -1,\ ij = k \rangle$.

The following set of 8 matrices forms an isomorphic group under multiplication, where I is the 4×4 identity matrix:

$$\left\{ \pm I, \quad \pm \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \pm \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \quad \pm \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \right\}.$$

Formally, we have an embedding $\phi\colon Q_8 \to \mathrm{GL}_4(\mathbb{R})$ where

$$\phi(i) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \phi(j) = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \quad \phi(k) = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

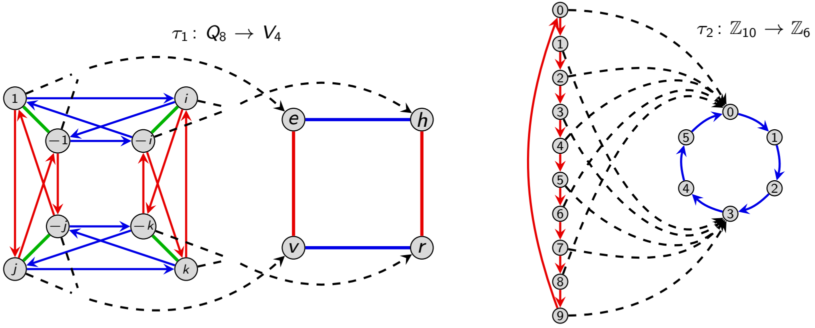
We say that Q_8 is represented by a set of matrices.

Many other groups can be represented by matrices. Can you think of how to represent V_4, C_n, or S_n, using matrices?

Quotient maps

Consider a homomorphism where more than one element of the domain maps to the same element of the codomain (i.e., non-embeddings).

Here are some examples.



Non-embedding homomorphisms are called quotient maps (as we'll see, they correspond to our quotient process).

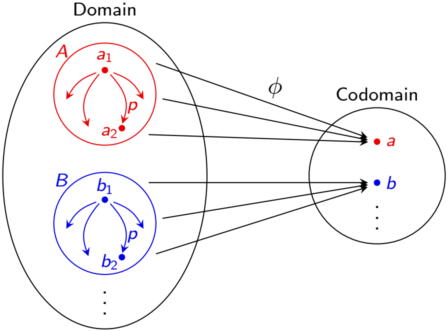
Preimages

Definition

If $\phi\colon G \to H$ is a homomorphism and $h \in \text{Im}(\phi) < H$, define the preimage of h to be the set

$$\phi^{-1}(h) := \{g \in G : \phi(g) = h\}.$$

Observe in the previous examples that the preimages all had the same structure. This always happens.



The preimage of $1_H \in H$ is called the kernel of ϕ, denoted $\text{Ker}\,\phi$.

Preimages

Observation 1

All preimages of ϕ have the same structure.

Proof (sketch)

Pick two elements $a, b \in \phi(G)$, and let $A = \phi^{-1}(a)$ and $B = \phi^{-1}(b)$ be their preimages.

Consider any path $a_1 \xrightarrow{p} a_2$ between elements in A. For any $b_1 \in B$, there is a corresponding path $b_1 \xrightarrow{p} b_2$. *We need to show that $b_2 \in B$.*

Since homomorphisms preserve structure, $\phi(a_1) \xrightarrow{\phi(p)} \phi(a_2)$. Since $\phi(a_1) = \phi(a_2)$, $\phi(p)$ is the *empty path*.

Therefore, $\phi(b_1) \xrightarrow{\phi(p)} \phi(b_2)$, i.e., $\phi(b_1) = \phi(b_2)$, and so by definition, $b_2 \in B$. $\square$

Clearly, G is partitioned by preimages of ϕ. Additionally, we just showed that they all have the same structure. (Sound familiar?)

Preimages and kernels

Definition

The kernel of a homomorphism $\phi\colon G \to H$ is the set

$$\mathrm{Ker}(\phi) := \phi^{-1}(e) = \{k \in G : \phi(k) = e\}\,.$$

Observation 2

(i) The preimage of the identity (i.e., $K = \mathrm{Ker}(\phi)$) is a subgroup of G.

(ii) All other preimages are left cosets of K.

Proof (of (i))

Let $K = \mathrm{Ker}(\phi)$, and take $a, b \in K$. We must show that K satisfies 3 properties:

Identity: $\phi(e) = e$. $\checkmark$

Closure: $\phi(ab) = \phi(a)\,\phi(b) = e \cdot e = e$. $\checkmark$

Inverses: $\phi(a^{-1}) = \phi(a)^{-1} = e^{-1} = e$. $\checkmark$

Thus, K is a subgroup of G. $\square$

Kernels

Observation 3

Ker(ϕ) is a normal subgroup of G.

Proof

Let $K = \text{Ker}(\phi)$. We will show that if $k \in K$, then $gkg^{-1} \in K$. Take any $g \in G$, and observe that

$$\phi(gkg^{-1}) = \phi(g)\,\phi(k)\,\phi(g^{-1}) = \phi(g) \cdot e \cdot \phi(g^{-1}) = \phi(g)\phi(g)^{-1} = e\,.$$

Therefore, $gkg^{-1} \in \text{Ker}(\phi)$, so $K \lhd G$. $\square$

Key observation

Given any homomorphism $\phi\colon G \to H$, we can *always* form the quotient group $G/\,\text{Ker}(\phi)$.

Quotients: via multiplication tables

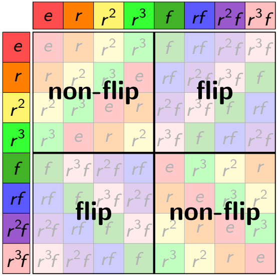
Recall that $C_2 = \{e^{0\pi i}, e^{1\pi i}\} = \{1, -1\}$. Consider the following (quotient) homomorphism:

$$\phi \colon D_4 \longrightarrow C_2\,, \qquad \text{defined by} \quad \phi(r) = 1 \quad \text{and} \quad \phi(f) = -1\,.$$

Note that $\phi(\text{rotation}) = 1$ and $\phi(\text{reflection}) = -1$.

The quotient process of "shrinking D_4 down to C_2" can be clearly seen from the multiplication tables.

	e	r	r^2	r^3	f	rf	r^2f	r^3f
e	e	r	r^2	r^3	f	rf	r^2f	r^3f
r	r	r^2	r^3	e	rf	r^2f	r^3f	f
r^2	r^2	r^3	e	r	r^2f	r^3f	f	rf
r^3	r^3	e	r	r^2	r^3f	f	rf	r^2f
f	f	r^3f	r^2f	rf	e	r^3	r^2	r
rf	rf	f	r^3f	r^2f	r	e	r^3	r^2
r^2f	r^2f	rf	f	r^3f	r^2	r	e	r^3
r^3f	r^3f	r^2f	rf	f	r^3	r^2	r	e



	1	-1
1	1	-1
-1	-1	1

Quotients: via Cayley diagrams

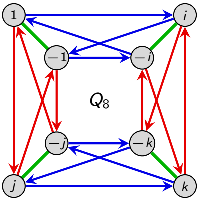
Define the homomorphism $\phi : Q_8 \to V_4$ via $\phi(i) = v$ and $\phi(j) = h$. Since $Q_8 = \langle i, j \rangle$, we can determine where ϕ sends the remaining elements:

$$\phi(1) = e\,, \qquad \phi(-1) = \phi(i^2) = \phi(i)^2 = v^2 = e\,,$$

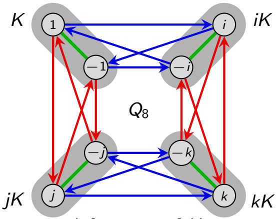
$$\phi(k) = \phi(ij) = \phi(i)\phi(j) = vh = r\,, \qquad \phi(-k) = \phi(ji) = \phi(j)\phi(i) = hv = r\,,$$

$$\phi(-i) = \phi(-1)\phi(i) = ev = v\,, \qquad \phi(-j) = \phi(-1)\phi(j) = eh = h\,.$$

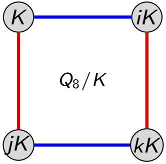
Note that $\operatorname{Ker}\phi = \{-1, 1\}$. Let's see what happens when we quotient out by $\operatorname{Ker}\phi$:



Q_8 organized by the subgroup $K = \langle -1 \rangle$



left cosets of K are near each other



collapse cosets into single nodes

Do you notice any relationship between $Q_8/\operatorname{Ker}(\phi)$ and $\operatorname{Im}(\phi)$?

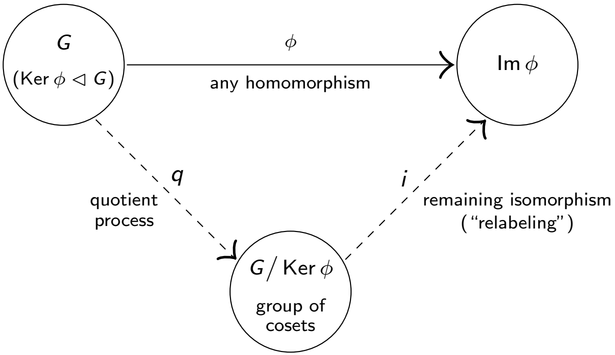
The Fundamental Homomorphism Theorem

The following is one of the central results in group theory.

Fundamental homomorphism theorem (FHT)

If $\phi \colon G \to H$ is a homomorphism, then $\mathrm{Im}(\phi) \cong G/\operatorname{Ker}(\phi)$.

The FHT says that every homomorphism can be decomposed into two steps: (i) quotient out by the kernel, and then (ii) relabel the nodes via ϕ.



Proof of the FHT

Fundamental homomorphism theorem

If $\phi\colon G \to H$ is a homomorphism, then $\mathrm{Im}(\phi) \cong G/\operatorname{Ker}(\phi)$.

Proof

We will construct an explicit map $i\colon G/\operatorname{Ker}(\phi) \longrightarrow \mathrm{Im}(\phi)$ and prove that it is an isomorphism.

Let $K = \operatorname{Ker}(\phi)$, and recall that $G/K = \{aK : a \in G\}$. Define

$$i\colon G/K \longrightarrow \mathrm{Im}(\phi)\,, \qquad i\colon gK \longmapsto \phi(g)\,.$$

- *<u>Show i is well-defined</u>*: We must show that if $aK = bK$, then $i(aK) = i(bK)$.

Suppose $aK = bK$. We have

$$aK = bK \quad \Longrightarrow \quad b^{-1}aK = K \quad \Longrightarrow \quad b^{-1}a \in K\,.$$

By definition of $b^{-1}a \in \operatorname{Ker}(\phi)$,

$$1_H = \phi(b^{-1}a) = \phi(b^{-1})\,\phi(a) = \phi(b)^{-1}\,\phi(a) \quad \Longrightarrow \quad \phi(a) = \phi(b)\,.$$

By definition of i: $\quad i(aK) = \phi(a) = \phi(b) = i(bK)$. ✓

Proof of FHT (cont.) [Recall: $i\colon G/K \to \mathrm{Im}(\phi)$, $i\colon gK \mapsto \phi(g)$]

Proof (cont.)

- *Show i is a homomorphism*: We must show that $i(aK \cdot bK) = i(aK)\, i(bK)$.

$$
\begin{aligned}
i(aK \cdot bK) &= i(abK) && (aK \cdot bK := abK) \\
&= \phi(ab) && (\text{definition of } i) \\
&= \phi(a)\,\phi(b) && (\phi \text{ is a homomorphism}) \\
&= i(aK)\, i(bK) && (\text{definition of } i)
\end{aligned}
$$

Thus, i is a homomorphism. ✓

- *Show i is surjective (onto)*:

This means showing that for any element in the codomain (here, $\mathrm{Im}(\phi)$), that some element in the domain (here, G/K) gets mapped to it by i.

Pick any $\phi(a) \in \mathrm{Im}(\phi)$. By defintion, $i(aK) = \phi(a)$, hence i is surjective. ✓

Proof of FHT (cont.) [Recall: $i\colon G/K \to \mathrm{Im}(\phi)$, $i\colon gK \mapsto \phi(g)$]

Proof (cont.)

- *Show i is injective (1–1)*: We must show that $i(aK) = i(bK)$ implies $aK = bK$.

Suppose that $i(aK) = i(bK)$. Then

$$\begin{array}{rcll} i(aK) = i(bK) & \Longrightarrow & \phi(a) = \phi(b) & \text{(by definition)} \\ & \Longrightarrow & \phi(b)^{-1}\phi(a) = 1_H & \\ & \Longrightarrow & \phi(b^{-1}a) = 1_H & (\phi \text{ is a homom.}) \\ & \Longrightarrow & b^{-1}a \in K & (\text{definition of } \mathrm{Ker}(\phi)) \\ & \Longrightarrow & b^{-1}aK = K & (aH = H \iff a \in H) \\ & \Longrightarrow & aK = bK & \end{array}$$

Thus, i is injective. ✓

In summary, since $i\colon G/K \to \mathrm{Im}(\phi)$ is a well-defined homomorphism that is injective (1–1) and surjective (onto), it is an **isomorphism**.

Therefore, $G/K \cong \mathrm{Im}(\phi)$, and the FHT is proven. □

Consequences of the FHT

Corollary

If $\phi\colon G \to H$ is a homomorphism, then $\operatorname{Im}\phi \leq H$.

A few special cases

- If $\phi\colon G \to H$ is an embedding, then $\operatorname{Ker}(\phi) = \{1_G\}$. The FHT says that
$$\operatorname{Im}(\phi) \cong G/\{1_G\} \cong G\,.$$
- If $\phi\colon G \to H$ is the map $\phi(g) = 1_H$ for all $h \in G$, then $\operatorname{Ker}(\phi) = G$, so the FHT says that
$$\{1_H\} = \operatorname{Im}(\phi) \cong G/G\,.$$

Let's use the FHT to determine all homomorphisms $\phi\colon C_4 \to C_3$:

- By the FHT, $G/\operatorname{Ker}\phi \cong \operatorname{Im}\phi < C_3$, and so $|\operatorname{Im}\phi| = 1$ or 3.
- Since $\operatorname{Ker}\phi < C_4$, Lagrange's Theorem also tells us that $|\operatorname{Ker}\phi| \in \{1, 2, 4\}$, and hence $|\operatorname{Im}\phi| = |G/\operatorname{Ker}\phi| \in \{1, 2, 4\}$.

Thus, $|\operatorname{Im}\phi| = 1$, and so the *only* homomorphism $\phi\colon C_4 \to C_3$ is the trivial one.

How to show two groups are isomorphic

The standard way to show $G \cong H$ is to construct an isomorphism $\phi\colon G \to H$.

When the domain is a quotient, there is another method, due to the FHT.

Useful technique

Suppose we want to show that $G/N \cong H$. There are two approaches:

(i) Define a map $\phi\colon G/N \to H$ and prove that it is well-defined, a homomorphism, and a bijection.

(ii) Define a map $\phi\colon G \to H$ and prove that it is a homomorphism, a surjection (onto), and that $\operatorname{Ker}\phi = N$.

Usually, Method (ii) is easier. Showing well-definedness and injectivity can be tricky.

For example, each of the following are results that we will see very soon, for which (ii) works quite well:

- $\mathbb{Z}/\langle n\rangle \cong \mathbb{Z}_n$;
- $\mathbb{Q}^*/\langle -1\rangle \cong \mathbb{Q}^+$;
- $AB/B \cong A/(A \cap B)$ (assuming $A, B \lhd G$);
- $G/(A \cap B) \cong (G/A) \times (G/B)$ (assuming $G = AB$).

Cyclic groups as quotients

Consider the following normal subgroup of $\mathbb{Z}$:

$$12\mathbb{Z} = \langle 12 \rangle = \{\dots, -24, -12, 0, 12, 24, \dots\} \lhd \mathbb{Z}\,.$$

The *elements* of the quotient group $\mathbb{Z}/\langle 12 \rangle$ are the *cosets*:

$$0 + \langle 12 \rangle\,, \quad 1 + \langle 12 \rangle\,, \quad 2 + \langle 12 \rangle \quad , \ \dots\ , \quad 10 + \langle 12 \rangle\,, \quad 11 + \langle 12 \rangle\,.$$

Number theorists call these sets congruence classes modulo 12. We say that two numbers are congruent mod 12 if they are in the same coset.

Recall how to add cosets in the quotient group:

$$(a + \langle 12 \rangle) + (b + \langle 12 \rangle) := (a + b) + \langle 12 \rangle\,.$$

"(The coset containing a) + (the coset containing b) = the coset containing $a + b$."

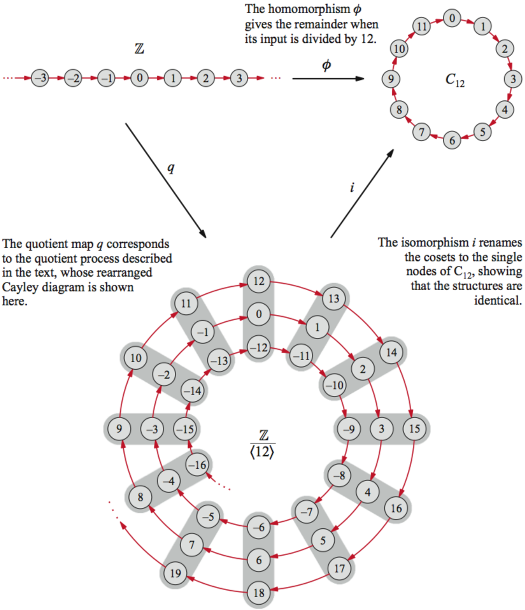
It should be clear that $\mathbb{Z}/\langle 12 \rangle$ is isomorphic to $\mathbb{Z}_{12}$. Formally, this is just the FHT applied to the following homomorphism:

$$\phi \colon \mathbb{Z} \longrightarrow \mathbb{Z}_{12}\,, \qquad \phi \colon k \longmapsto k \pmod{12}\,,$$

Clearly, $\mathrm{Ker}(\phi) = \{\dots, -24, -12, 0, 12, 24, \dots\} = \langle 12 \rangle$. By the FHT:

$$\mathbb{Z}/\,\mathrm{Ker}(\phi) = \mathbb{Z}/\langle 12 \rangle \cong \mathrm{Im}(\phi) = \mathbb{Z}_{12}\,.$$

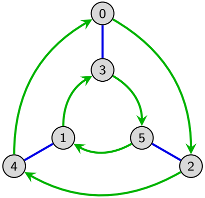
A picture of the isomorphism $i\colon \mathbb{Z}_{12} \longrightarrow \mathbb{Z}/\langle 12\rangle$ (from the VGT website)

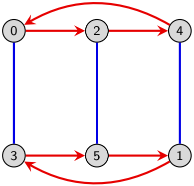


Finite abelian groups

We've seen that some cyclic groups can be expressed as a direct product, and others cannot.

Below are two ways to lay out the Cayley diagram of $\mathbb{Z}_6$ so the direct product structure is obvious: $\mathbb{Z}_6 \cong \mathbb{Z}_3 \times \mathbb{Z}_2$.





However, the group $\mathbb{Z}_8$ *cannot* be written as a direct product. No matter how we draw the Cayley graph, there *must* be an element (arrow) of order 8. (Why?)

We will answer the question of when $\mathbb{Z}_n \times \mathbb{Z}_m \cong \mathbb{Z}_{nm}$, and in doing so, completely classify all finite abelian groups.

Finite abelian groups

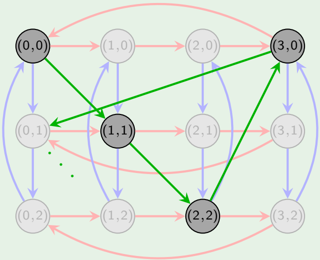
Proposition

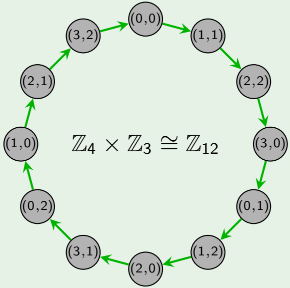
$\mathbb{Z}_{nm} \cong \mathbb{Z}_n \times \mathbb{Z}_m$ if and only if $\gcd(n, m) = 1$.

Proof (sketch)

"$\Leftarrow$": Suppose $\gcd(n, m) = 1$. We claim that $(1, 1) \in \mathbb{Z}_n \times \mathbb{Z}_m$ has order nm.

$|(1, 1)|$ is the smallest k such that "$(k, k) = (0, 0)$." This happens iff $n \mid k$ and $m \mid k$.
Thus, $k = \text{lcm}(n, m) = nm$. ✓





Finite abelian groups

Proposition

$\mathbb{Z}_{nm} \cong \mathbb{Z}_n \times \mathbb{Z}_m$ if an only if $\gcd(n, m) = 1$.

Proof (cont.)

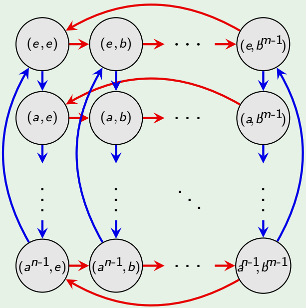
"$\Rightarrow$": Suppose $\mathbb{Z}_{nm} \cong \mathbb{Z}_n \times \mathbb{Z}_m$. Then $\mathbb{Z}_n \times \mathbb{Z}_m$ has an element (a, b) of order nm.

For convenience, we will switch to "multiplicative notation", and denote our cyclic groups by C_n.

Clearly, $\langle a \rangle = C_n$ and $\langle b \rangle = C_m$. Let's look at a Cayley diagram for $C_n \times C_m$.

The order of (a, b) must be a multiple of n (the number of rows), *and* of m (the number of columns).

By definition, this is the *least* common multiple of n and m.



But $|(a, b)| = nm$, and so $\text{lcm}(n, m) = nm$. Therefore, $\gcd(n, m) = 1$. $\square$

The Fundamental Theorem of Finite Abelian Groups

Classification theorem (by "prime powers")

Every finite abelian group A is isomorphic to a direct product of cyclic groups, i.e., for some integers $n_1, n_2, \dots, n_m$,

$$A \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_m},$$

where each n_i is a prime power, i.e., $n_i = p_i^{d_i}$, where p_i is prime and $d_i \in \mathbb{N}$.

The proof of this is more advanced, and while it is at the undergraduate level, we don't yet have the tools to do it.

However, we will be more interested in understanding and utilizing this result.

Example

Up to isomorphism, there are 6 abelian groups of order $200 = 2^3 \cdot 5^2$:

$\mathbb{Z}_8 \times \mathbb{Z}_{25}$	$\mathbb{Z}_8 \times \mathbb{Z}_5 \times \mathbb{Z}_5$
$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_{25}$	$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_5$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$

The Fundamental Theorem of Finite Abelian Groups

Finite abelian groups can be classified by their "elementary divisors." The mysterious terminology comes from the theory of modules (a graduate-level topic).

Classification theorem (by "elementary divisors")

Every finite abelian group A is isomorphic to a direct product of cyclic groups, i.e., for some integers $k_1, k_2, \ldots, k_m$,

$$A \cong \mathbb{Z}_{k_1} \times \mathbb{Z}_{k_2} \times \cdots \times \mathbb{Z}_{k_m}.$$

where each k_i is a multiple of k_{i+1}.

Example

Up to isomorphism, there are 6 abelian groups of order $200 = 2^3 \cdot 5^2$:

by "prime-powers"	by "elementary divisors"
$\mathbb{Z}_8 \times \mathbb{Z}_{25}$	$\mathbb{Z}_{200}$
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$	$\mathbb{Z}_{100} \times \mathbb{Z}_2$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$	$\mathbb{Z}_{50} \times \mathbb{Z}_2 \times \mathbb{Z}_2$
$\mathbb{Z}_8 \times \mathbb{Z}_5 \times \mathbb{Z}_5$	$\mathbb{Z}_{40} \times \mathbb{Z}_5$
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$	$\mathbb{Z}_{20} \times \mathbb{Z}_{10}$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$	$\mathbb{Z}_{10} \times \mathbb{Z}_{10} \times \mathbb{Z}_2$

The Fundamental Theorem of Finitely Generated Abelian Groups

Just for fun, here is the classification theorem for all *finitely generated* abelian groups. Note that it is not much different.

Theorem

Every finitely generated abelian group A is isomorphic to a direct product of cyclic groups, i.e., for some integers $n_1, n_2, \dots, n_m$,

$$A \cong \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{k \text{ copies}} \times \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_m} \,,$$

where each n_i is a prime power, i.e., $n_i = p_i^{d_i}$, where p_i is prime and $d_i \in \mathbb{N}$.

In other words, A is isomorphic to a (multiplicative) group with presentation:

$$A = \langle a_1, \dots, a_k, r_1, \dots, r_m \mid r_1^{n_1} = \cdots = r_m^{n_m} = 1 \rangle \,.$$

In summary, abelian groups are relatively easy to understand.

In contrast, nonabelian groups are more mysterious and complicated. Soon, we will study the *Sylow Theorems* which will help us better understand the structure of finite nonabelian groups.

The Isomorphism Theorems

The Fundamental Homomorphism Theorem (FHT) is the first of four basic theorems about homomorphism and their structure.

These are commonly called "The Isomorphism Theorems":

- First Isomorphism Theorem: "Fundamental Homomorphism Theorem"
- Second Isomorphism Theorem: "Diamond Isomorphism Theorem"
- Third Isomorphism Theorem: "Freshman Theorem"
- Fourth Isomorphism Theorem: "Correspondence Theorem"

All of these theorems have analogues in other algebraic structures: rings, vector spaces, modules, and Lie algebras, to name a few.

In this lecture, we will summarize the last three isomorphism theorems and provide visual pictures for each.

We will prove one, outline the proof of another (homework!), and encourage you to try the (very straightforward) proofs of the multiple parts of the last one.

Finally, we will introduce the concepts of a commutator and commutator subgroup, whose quotient yields the abelianization of a group.

The Second Isomorphism Theorem

Diamond isomorphism theorem

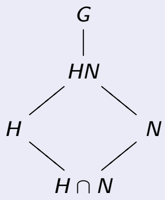
Let $H \leq G$, and $N \lhd G$. Then

(i) The product $HN = \{hn \mid h \in H,\ n \in N\}$ is a subgroup of G.

(ii) The intersection $H \cap N$ is a *normal* subgroup of G.

(iii) The following quotient groups are isomorphic:

$$HN/N \cong H/(H \cap N)$$



Proof (sketch)

Define the following map

$$\phi\colon H \longrightarrow HN/N\,, \qquad \phi\colon h \longmapsto hN\,.$$

If we can show:

1. ϕ is a homomorphism,
2. ϕ is surjective (onto),
3. $\operatorname{Ker}\phi = H \cap N$,

then the result will follow *immediately* from the FHT. The details are left as HW.

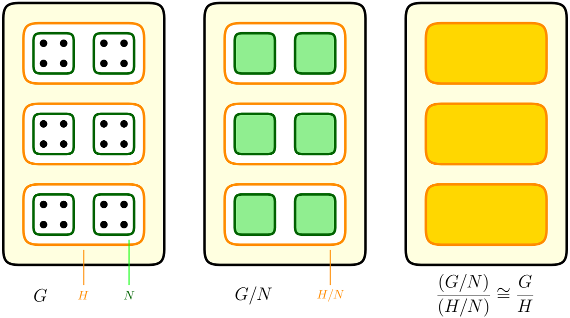
The Third Isomorphism Theorem

Freshman theorem

Consider a chain $N \leq H \leq G$ of normal subgroups of G. Then

1. The quotient H/N is a normal subgroup of G/N;
2. The following quotients are isomorphic:

$$(G/N)/(H/N) \cong G/H\,.$$



(Thanks to Zach Teitler of Boise State for the concept and graphic!)

The Third Isomorphism Theorem

Freshman theorem

Consider a chain $N \leq H \leq G$ of normal subgroups of G. Then $H/N \lhd G/N$ and $(G/N)/(H/N) \cong G/H$.

Proof

It is easy to show that $H/N \lhd G/N$ (exercise). Define the map

$$\varphi \colon G/N \longrightarrow G/H, \qquad \varphi \colon gN \longmapsto gH.$$

- *Show φ is well-defined*: Suppose $g_1 N = g_2 N$. Then $g_1 = g_2 n$ for some $n \in N$. But $n \in H$ because $N \leq H$. Thus, $g_1 H = g_2 H$, i.e., $\varphi(g_1 N) = \varphi(g_2 N)$. ✓
- *φ is clearly onto and a homomorphism*. ✓
- *Apply the FHT*:

$$\begin{aligned} \operatorname{Ker}\varphi &= \{gN \in G/N \mid \varphi(gN) = H\} \\ &= \{gN \in G/N \mid gH = H\} \\ &= \{gN \in G/N \mid g \in H\} = H/N \end{aligned}$$

By the FHT, $(G/N)/\operatorname{Ker}\varphi = (G/N)/(H/N) \cong \operatorname{Im}\varphi = G/H$. $\square$

The Fourth Isomorphism Theorem

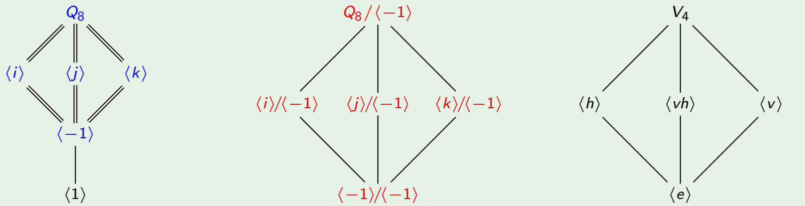
The full statement is a bit technical, so here we just state it informally.

Correspondence theorem

Let $N \lhd G$. There is a 1–1 correspondence between subgroups of G/N and subgroups of G that contain N. In particular, every subgroup of G/N has the form $\overline{A} := A/N$ for some A satisfying $N \leq A \leq G$.

This means that the corresponding subgroup lattices are identical in structure.

Example



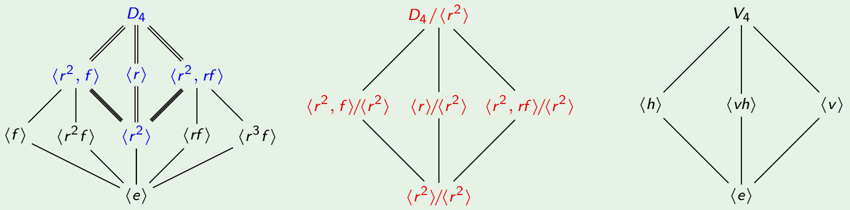
The quotient $Q_8/\langle -1 \rangle$ is isomorphic to V_4. The subgroup lattices can be visualized by "collapsing" $\langle -1 \rangle$ to the identity.

Correspondence theorem (formally)

Let $N \lhd G$. Then there is a bijection from the subgroups of G/N and subgroups of G that contain N. In particular, every subgroup of G/N has the form $\overline{A} := A/N$ for some A satisfying $N \leq A \leq G$. Moreover, if $A, B \leq G$, then

1. $A \leq B$ if and only if $\overline{A} \leq \overline{B}$,
2. If $A \leq B$, then $[B : A] = [\overline{B} : \overline{A}]$,
3. $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$,
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$,
5. $A \lhd G$ if and only if $\overline{A} \lhd \overline{G}$.

Example



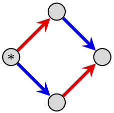
Application: commutator subgroups and abelianizations

We've seen how to divide $\mathbb{Z}$ by $\langle 12 \rangle$, thereby "forcing" all multiples of 12 to be zero. This is one way to construct the integers modulo 12: $\mathbb{Z}_{12} \cong \mathbb{Z}/\langle 12 \rangle$.

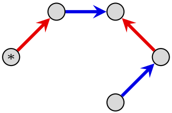
Now, suppose G is nonabelian. We would like to divide G by its "non-abelian parts," making them zero and leaving only "abelian parts" in the resulting quotient.

A commutator is an element of the form $aba^{-1}b^{-1}$. Since G is nonabelian, *there are non-identity commutators:* $aba^{-1}b^{-1} \neq e$ *in* G.

$ab = ba$



$ab \neq ba$



In this case, the set $C := \{aba^{-1}b^{-1} \mid a, b \in G\}$ contains *more* than the identity.

Define the commutator subgroup G' of G to be

$$G' := \langle aba^{-1}b^{-1} \mid a, b \in G \rangle .$$

This is a normal subgroup of G (homework exercise). If we quotient out by it, we get an abelian group! (Because we have killed every instance of the "$ab \neq ba$ pattern" shown above.)

Commutator subgroups and abelianizations

Definition

The abelianization of G is the quotient group G/G'. This is the group that one gets by "killing off" all nonabelian parts of G.

In some sense, the commutator subgroup G' is the smallest normal subgroup N of G such that G/N is abelian. [Note that G would be the "largest" such subgroup.]

Equivalently, the quotient G/G' is the largest abelian quotient of G. [Note that $G/G \cong \langle e \rangle$ would be the "smallest" such quotient.]

Universal property of commutator subgroups

Suppose $f\colon G \to A$ is a homomorphism to an abelian group A. Then there is a unique homomorphism $h\colon G/G' \to A$ such that $f = hq$:

$$\begin{tikzcd} G \arrow[rr, "f"] \arrow[dr, "q"'] & & A \\ & G/G' \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

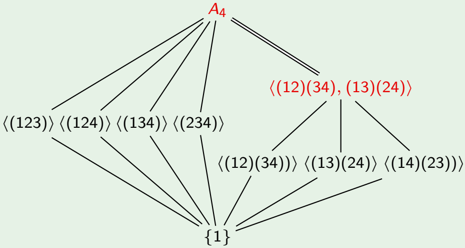
We say that f "factors through" the abelianization, G/G'.

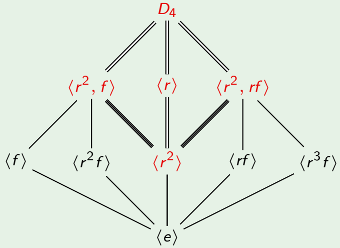
Commutator subgroups and abelianizations

Examples

Consider the groups A_4 and D_4. It is easy to check that

$$G'_{A_4} = \langle xyx^{-1}y^{-1} \mid x, y \in A_4 \rangle \cong V_4\,, \qquad G'_{D_4} = \langle xyx^{-1}y^{-1} \mid x, y \in D_4 \rangle = \langle r^2 \rangle\,.$$





By the *Correspondence Theorem*, the abelianization of A_4 is $A_4/V_4 \cong C_3$, and the abelianization of D_4 is $D_4/\langle r^2 \rangle \cong V_4$.

Notice that G/G' is abelian, and moreover, taking the quotient of G by *anything above* G' will yield an abelian group.

Automorphisms

Definition

An automorphism is an isomorphism from a group to itself.

The set of all automorphisms of G forms a group, called the automorphism group of G, and denoted $\mathrm{Aut}(G)$.

Remarks.

- An automorphism is determined by where it sends the generators.
- An automorphism ϕ must send generators to generators. In particular, if G is cyclic, then it determines a permutation of the set of (all possible) generators.

Examples

1. There are two automorphisms of $\mathbb{Z}$: the identity, and the mapping $n \mapsto -n$. Thus, $\mathrm{Aut}(\mathbb{Z}) \cong C_2$.
2. There is an automorphism $\phi\colon \mathbb{Z}_5 \to \mathbb{Z}_5$ for each choice of $\phi(1) \in \{1, 2, 3, 4\}$. Thus, $\mathrm{Aut}(\mathbb{Z}_5) \cong C_4$ or V_4. (Which one?)
3. An automorphism ϕ of $V_4 = \langle h, v\rangle$ is determined by the image of h and v. There are 3 choices for $\phi(h)$, and then 2 choices for $\phi(v)$. Thus, $|\mathrm{Aut}(V_4)| = 6$, so it is either $C_6 \cong C_2 \times C_3$, or S_3. (Which one?)

Automorphism groups of $\mathbb{Z}_n$

Definition

The multiplicative group of integers modulo n, denoted $\mathbb{Z}_n^*$ or $U(n)$, is the group

$$U(n) := \{k \in \mathbb{Z}_n \mid \gcd(n,k) = 1\}$$

where the binary operation is multiplication, modulo n.

$U(5) = \{1,2,3,4\} \cong C_4$

	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

$U(6) = \{1,5\} \cong C_2$

	1	5
1	1	5
5	5	1

$U(8) = \{1,3,5,7\} \cong C_2 \times C_2$

	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

Proposition (homework)

The automorphism group of $\mathbb{Z}_n$ is $\mathrm{Aut}(\mathbb{Z}_n) = \{\sigma_a \mid a \in U(n)\} \cong U(n)$, where

$$\sigma_a \colon \mathbb{Z}_n \longrightarrow \mathbb{Z}_n\,, \qquad \sigma_a(1) = a\,.$$

Automorphisms of D_3

Let's find all automorphisms of $D_3 = \langle r, f \rangle$. We'll see a very similar example to this when we study Galois theory.

Clearly, every automorphism ϕ is completely determined by $\phi(r)$ and $\phi(f)$.

Since automorphisms preserve order, if $\phi \in \mathrm{Aut}(D_3)$, then

$$\phi(e) = e\,, \qquad \phi(r) = \underbrace{r \text{ or } r^2}_{\text{2 choices}}, \qquad \phi(f) = \underbrace{f,\ rf, \text{ or } r^2 f}_{\text{3 choices}}\,.$$

Thus, there are *at most* $2 \cdot 3 = 6$ automorphisms of D_3.

Let's try to define two maps, (i) $\alpha\colon D_3 \to D_3$ fixing r, and (ii) $\beta\colon D_3 \to D_3$ fixing f:

$$\begin{cases} \alpha(r) = r \\ \alpha(f) = rf \end{cases} \qquad\qquad \begin{cases} \beta(r) = r^2 \\ \beta(f) = f \end{cases}$$

I claim that:

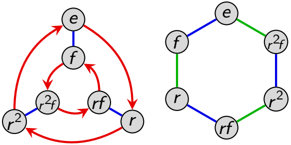
- these both define automorphisms (check this!)
- these generate six *different* automorphisms, and thus $\langle \alpha, \beta \rangle = \mathrm{Aut}(D_3)$.

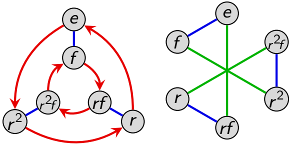
To determine what group this is isomorphic to, find these six automorphisms, and make a group presentation and/or multiplication table. Is it abelian?

Automorphisms of D_3

An automorphism can be thought of as a re-wiring of the Cayley diagram.

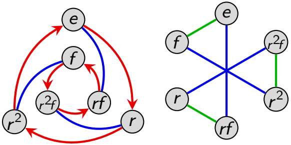
$$r \xmapsto{\ id\ } r \qquad f \longmapsto f$$

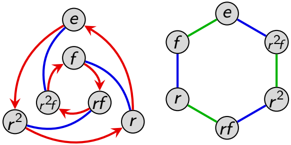




$$r \xmapsto{\ \beta\ } r^2 \qquad f \longmapsto f$$

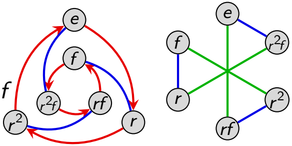
$$r \xmapsto{\ \alpha\ } r \qquad f \longmapsto rf$$

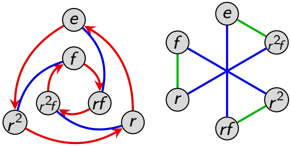




$$r \xmapsto{\ \alpha\beta\ } r^2 \qquad f \longmapsto r^2 f$$

$$r \xmapsto{\ \alpha^2\ } r \qquad f \longmapsto r^2 f$$



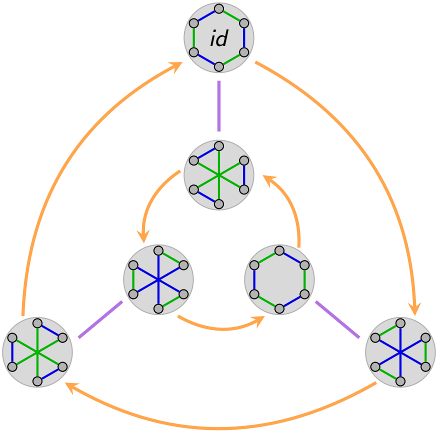


$$r \xmapsto{\ \alpha^2\beta\ } r^2 \qquad f \longmapsto rf$$

Automorphisms of D_3

Here is the multiplication table and Cayley diagram of $\mathrm{Aut}(D_3) = \langle \alpha, \beta \rangle$.

	id	α	α^2	β	$\alpha\beta$	$\alpha^2\beta$
id	id	α	α^2	β	$\alpha\beta$	$\alpha^2\beta$
α	α	α^2	id	$\alpha\beta$	$\alpha^2\beta$	β
α^2	α^2	id	α	$\alpha^2\beta$	β	$\alpha\beta$
β	β	$\alpha^2\beta$	$\alpha\beta$	id	α^2	α
$\alpha\beta$	$\alpha\beta$	β	$\alpha^2\beta$	α	id	α^2
$\alpha^2\beta$	$\alpha^2\beta$	$\alpha\beta$	β	α^2	α	id



It is purely coincidence that $\mathrm{Aut}(D_3) \cong D_3$. For example, we've already seen that

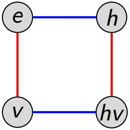
$$\mathrm{Aut}(\mathbb{Z}_5) \cong U(5) \cong C_4 \,, \qquad \mathrm{Aut}(\mathbb{Z}_6) \cong U(6) \cong C_2 \,, \qquad \mathrm{Aut}(\mathbb{Z}_8) \cong U(8) \cong C_2 \times C_2 \,.$$

Automorphisms of $V_4 = \langle h, v \rangle$

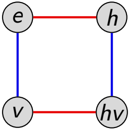
The following permutations are both automorphisms:

$\alpha : h \quad v \quad hv$ and $\beta : h \quad v \quad hv$

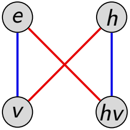
$$\begin{aligned} h &\overset{id}{\longmapsto} h \\ v &\longmapsto v \\ hv &\longmapsto hv \end{aligned}$$



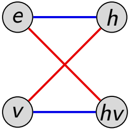
$$\begin{aligned} h &\overset{\beta}{\longmapsto} v \\ v &\longmapsto h \\ hv &\longmapsto hv \end{aligned}$$



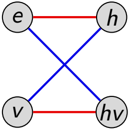
$$\begin{aligned} h &\overset{\alpha}{\longmapsto} v \\ v &\longmapsto hv \\ hv &\longmapsto h \end{aligned}$$



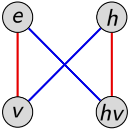
$$\begin{aligned} h &\overset{\alpha\beta}{\longmapsto} h \\ v &\longmapsto hv \\ hv &\longmapsto v \end{aligned}$$



$$\begin{aligned} h &\overset{\alpha^2}{\longmapsto} hv \\ v &\longmapsto h \\ hv &\longmapsto v \end{aligned}$$



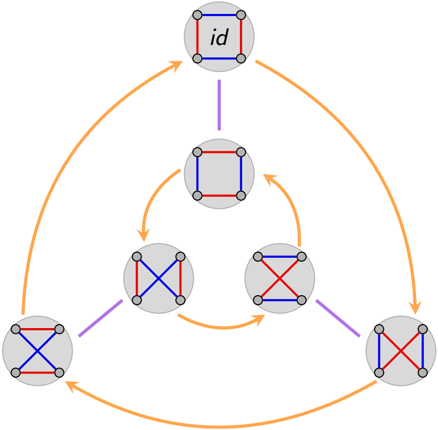
$$\begin{aligned} h &\overset{\alpha^2\beta}{\longmapsto} hv \\ v &\longmapsto v \\ hv &\longmapsto h \end{aligned}$$



Automorphisms of $V_4 = \langle h, v \rangle$

Here is the multiplication table and Cayley diagram of $\mathrm{Aut}(V_4) = \langle \alpha, \beta \rangle \cong S_3 \cong D_3$.

	id	α	α^2	β	$\alpha\beta$	$\alpha^2\beta$
id	id	α	α^2	β	$\alpha\beta$	$\alpha^2\beta$
α	α	α^2	id	$\alpha\beta$	$\alpha^2\beta$	β
α^2	α^2	id	α	$\alpha^2\beta$	β	$\alpha\beta$
β	β	$\alpha^2\beta$	$\alpha\beta$	id	α^2	α
$\alpha\beta$	$\alpha\beta$	β	$\alpha^2\beta$	α	id	α^2
$\alpha^2\beta$	$\alpha^2\beta$	$\alpha\beta$	β	α^2	α	id



Recall that α and β can be thought of as the permutations $h \quad v \quad hv$ and $h \quad v \quad hv$ and so $\mathrm{Aut}(G) \hookrightarrow \mathrm{Perm}(G) \cong S_n$ always holds.

Section 5: Groups acting on sets

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Math 4120, Modern Algebra

Overview

Intuitively, a group action occurs when a group G "naturally permutes" a set S *of states*.

For example:

- The "Rubik's cube group" consists of the 4.3×10^{19} actions that *permutated* the 4.3×10^{19} configurations of the cube.
- The group D_4 consists of the 8 symmetries of the square. These symmetries are *actions* that *permuted* the 8 configurations of the square.

Group actions help us understand the interplay between the actual group of actions and sets of objects that they "rearrange."

There are many other examples of groups that "act on" sets of objects. We will see examples when the group and the set have different sizes.

There is a rich theory of group actions, and it can be used to prove many deep results in group theory.

Actions vs. configurations

The group D_4 can be thought of as the 8 symmetries of the square:

There is a subtle but *important* distinction to make, between the actual 8 symmetries of the square, and the 8 configurations.

For example, the 8 symmetries (alternatively, "actions") can be thought of as

$$e, \qquad r, \qquad r^2, \qquad r^3, \qquad f, \qquad rf, \qquad r^2 f, \qquad r^3 f\,.$$

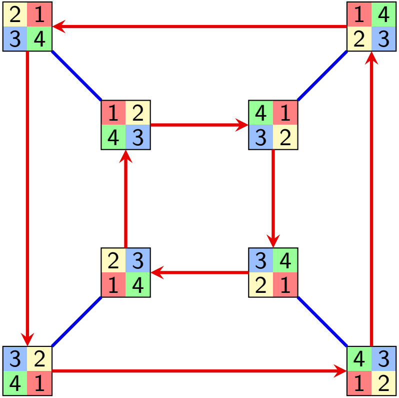
The 8 configurations (or *states*) of the square are the following:

When we were just learning about groups, we made an action diagram.

- The vertices correspond to the states.
- The edges correspond to generators.
- The paths corresponded to actions (group elements).

Action diagrams

Here is the action diagram of the group $D_4 = \langle r, f \rangle$:



In the beginning of this course, we picked a configuration to be the "solved state," and this gave us a *bijection* between configurations and actions (group elements).

The resulting diagram was a Cayley diagram. In this section, we'll skip this step.

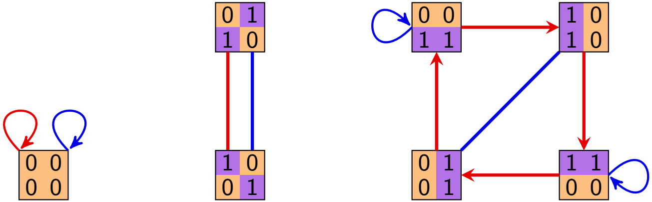
Action diagrams

In all of the examples we saw in the beginning of the course, we had a bijective correspondence between actions and states. *This need not always happen!*

Suppose we have a size-7 set consisting of the following "binary squares."

$$S = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

The group $D_4 = \langle r, f \rangle$ "acts on S" as follows:



The action diagram above has some properties of Cayley diagrams, but there are some fundamental differences as well.

A "group switchboard"

Suppose we have a "switchboard" for G, with every element $g \in G$ having a "button."

If $a \in G$, then pressing the a-button rearranges the objects in our set S. In fact, it is a permutation of S; call it $\phi(a)$.

If $b \in G$, then pressing the b-button rearranges the objects in S a different way. Call this permutation $\phi(b)$.

The element $ab \in G$ also has a button. We require that pressing the ab-button yields the same result as pressing the a-button, followed by the b-button. That is,

$$\phi(ab) = \phi(a)\phi(b)\,, \qquad \text{for all } a, b \in G\,.$$

Let $\mathrm{Perm}(S)$ be the group of permutations of S. Thus, if $|S| = n$, then $\mathrm{Perm}(S) \cong S_n$. (We typically think of S_n as the permutations of $\{1, 2, \ldots, n\}$.)

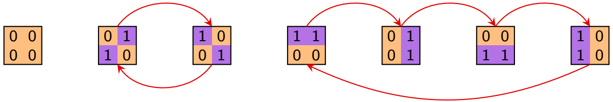
Definition

A group G acts on a set S if there is a homomorphism $\phi\colon G \to \mathrm{Perm}(S)$.

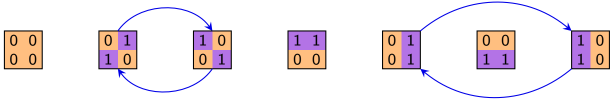
A "group switchboard"

Returning to our binary square example, pressing the r-button and f-button permutes the set S as follows:

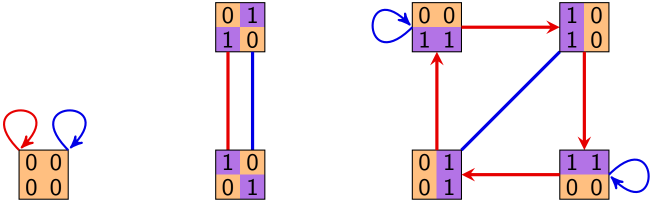
$\phi(r)$:



$\phi(f)$:



Observe how these permutations are encoded in the action diagram:



Left actions vs. right actions (an annoyance we can deal with)

As we've defined group actions, "*pressing the a-button followed by the b-button should be the same as pressing the ab-button*."

However, sometimes it has to be the same as "*pressing the ba-button*."

This is best seen by an example. Suppose our action is conjugation:

"Left group action"

$$\begin{tikzcd} H \arrow[r, "\text{conjugate by } a"] \arrow[rr, bend right, "\text{conjugate by } ba"'] & aHa^{-1} \arrow[r, "\text{conjugate by } b"] & baHa^{-1}b^{-1} \end{tikzcd}$$

$$\phi(a)\phi(b) = \phi(ba)$$

"Right group action"

$$\begin{tikzcd} H \arrow[r, "\text{conjugate by } a"] \arrow[rr, bend right, "\text{conjugate by } ab"'] & a^{-1}Ha \arrow[r, "\text{conjugate by } b"] & b^{-1}a^{-1}Hab \end{tikzcd}$$

$$\phi(a)\phi(b) = \phi(ab)$$

Some books forgo our "ϕ-notation" and use the following notation to distinguish left vs. right group actions:

$$g.(h.s) = (gh).s\,, \qquad (s.g).h = s.(gh)\,.$$

We'll usually keep the ϕ, and write $\phi(g)\phi(h)s = \phi(gh)s$ and $s.\phi(g)\phi(h) = s.\phi(gh)$. As with groups, the "dot" will be optional.

Left actions vs. right actions (an annoyance we can deal with)

Alternative definition (other textbooks)

A right group action is a mapping

$$G \times S \longrightarrow S\,, \qquad (a, s) \longmapsto s.a$$

such that

- $s.(ab) = (s.a).b$, for all $a, b \in G$ and $s \in S$
- $s.e = s$, for all $s \in S$.

A left group action can be defined similarly.

Pretty much all of the theorems for left actions hold for right actions.

Usually if there is a left action, there is a related right action. We will usually use right actions, and we will write

$$s.\phi(g)$$

for "the element of S that the permutation $\phi(g)$ sends s to," i.e., where pressing the g-button sends s.

If we have a left action, we'll write $\phi(g).s$.

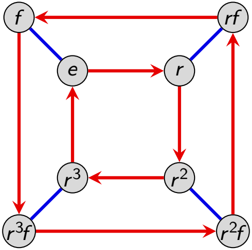
Cayley diagrams as action diagrams

Every Cayley diagram can be thought of as the action diagram of a particular (right) group action.

For example, consider the group $G = D_4 = \langle r, f \rangle$ acting on itself. That is, $S = D_4 = \{e, r, r^2, r^3, f, rf, r^2f, r^3f\}$.

Suppose that pressing the g-button on our "group switchboard" multiplies every element *on the right* by g.

Here is the action diagram:



We say that "*G acts on itself by right-multiplication*."

Orbits, stabilizers, and fixed points

Suppose G acts on a set S. Pick a configuration $s \in S$. We can ask two questions about it:

(i) What other states (in S) are reachable from s? (We call this the orbit of s.)

(ii) What group elements (in G) fix s? (We call this the stabilizer of s.)

Definition

Suppose that G acts on a set S (on the right) via $\phi\colon G \to \mathrm{Perm}(S)$.

(i) The orbit of $s \in S$ is the set

$$\mathrm{Orb}(s) = \{s.\phi(g) \mid g \in G\}\,.$$

(ii) The stabilizer of s in G is

$$\mathrm{Stab}(s) = \{g \in G \mid s.\phi(g) = s\}\,.$$

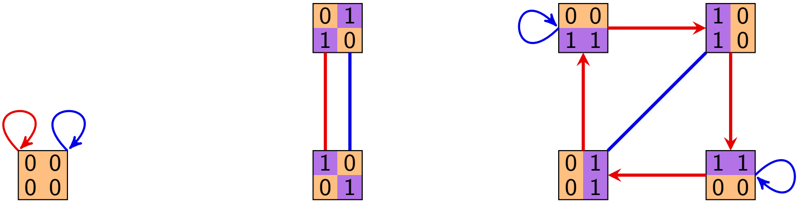
(iii) The fixed points of the action are the orbits of size 1:

$$\mathrm{Fix}(\phi) = \{s \in S \mid s.\phi(g) = s \text{ for all } g \in G\}\,.$$

Note that the orbits of ϕ are the connected components in the action diagram.

Orbits, stabilizers, and fixed points

Let's revisit our running example:



The orbits are the 3 connected components. There is only one fixed point of ϕ. The stabilizers are:

$$\mathrm{Stab}\left(\boxed{\begin{smallmatrix}0&0\\0&0\end{smallmatrix}}\right)=D_4,$$

$$\mathrm{Stab}\left(\boxed{\begin{smallmatrix}0&1\\1&0\end{smallmatrix}}\right)=\{e,r^2,rf,r^3f\},\qquad \mathrm{Stab}\left(\boxed{\begin{smallmatrix}1&0\\0&1\end{smallmatrix}}\right)=\{e,r^2,rf,r^3f\},$$

$$\mathrm{Stab}\left(\boxed{\begin{smallmatrix}0&0\\1&1\end{smallmatrix}}\right)=\{e,f\},\qquad \mathrm{Stab}\left(\boxed{\begin{smallmatrix}1&0\\1&0\end{smallmatrix}}\right)=\{e,r^2f\},$$

$$\mathrm{Stab}\left(\boxed{\begin{smallmatrix}1&1\\0&0\end{smallmatrix}}\right)=\{e,f\},\qquad \mathrm{Stab}\left(\boxed{\begin{smallmatrix}0&1\\0&1\end{smallmatrix}}\right)=\{e,r^2f\}.$$

Observations?

Orbits and stabilizers

Proposition

For any $s \in S$, the set $\mathrm{Stab}(s)$ is a subgroup of G.

Proof (outline)

To show $\mathrm{Stab}(s)$ is a group, we need to show three things:

(i) *Contains the identity*. That is, $s.\phi(e) = s$.

(ii) *Inverses exist*. That is, if $s.\phi(g) = s$, then $s.\phi(g^{-1}) = s$.

(iii) *Closure*. That is, if $s.\phi(g) = s$ and $s.\phi(h) = s$, then $s.\phi(gh) = s$.

You'll do this on the homework.

Remark

The kernel of the action ϕ is the set of all group elements that fix everything in S:

$$\mathrm{Ker}\,\phi = \{g \in G \mid \phi(g) = e\} = \{g \in G \mid s.\phi(g) = s \text{ for all } s \in S\}.$$

Notice that

$$\mathrm{Ker}\,\phi = \bigcap_{s \in S} \mathrm{Stab}(s).$$

The Orbit-Stabilizer Theorem

The following result is another one of the central results of group theory.

Orbit-Stabilizer theorem

For any group action $\phi \colon G \to \mathrm{Perm}(S)$, and any $s \in S$,

$$|\,\mathrm{Orb}(s)| \cdot |\,\mathrm{Stab}(s)| = |G|\,.$$

Proof

Since $\mathrm{Stab}(s) < G$, Lagrange's theorem tells us that

$$\underbrace{[G : \ \mathrm{Stab}(s)]}_{\text{number of cosets}} \cdot \underbrace{|\,\mathrm{Stab}(s)|}_{\text{size of subgroup}} = |G|.$$

Thus, it suffices to show that $|\,\mathrm{Orb}(s)| = [G : \ \mathrm{Stab}(s)]$.

Goal: Exhibit a bijection between elements of $\mathrm{Orb}(s)$, and right cosets of $\mathrm{Stab}(s)$.

That is, *two elements in G send s to the same place iff they're in the same coset*.

The Orbit-Stabilizer Theorem: $|\mathrm{Orb}(s)| \cdot |\mathrm{Stab}(s)| = |G|$

Proof (cont.)

Let's look at our previous example to get some intuition for why this should be true.

We are seeking a bijection between $\mathrm{Orb}(s)$, and the right cosets of $\mathrm{Stab}(s)$.

That is, two elements in G send s to the same place iff they're in the same coset.

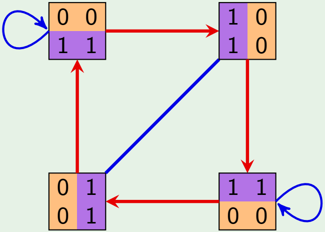
Let $s = \begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}$

Then $\mathrm{Stab}(s) = \langle f \rangle$.

Partition of D_4 by the right cosets of H :

$G = D_4$ and $H = \langle f \rangle$

e	r	r^2	r^3
f	fr	fr^2	fr^3
H	Hr	Hr^2	Hr^3



Note that $s.\phi(g) = s.\phi(k)$ iff g and k are in the same right coset of H in G.

The Orbit-Stabilizer Theorem: $|\mathrm{Orb}(s)| \cdot |\mathrm{Stab}(s)| = |G|$

Proof (cont.)

Throughout, let $H = \mathrm{Stab}(s)$.

"$\Rightarrow$" *If two elements send s to the same place, then they are in the same coset.*

Suppose $g, k \in G$ both send s to the same element of S. This means:

$$\begin{array}{rcll}
s.\phi(g) = s.\phi(k) & \Longrightarrow & s.\phi(g)\phi(k)^{-1} = s & \\
& \Longrightarrow & s.\phi(g)\phi(k^{-1}) = s & \\
& \Longrightarrow & s.\phi(gk^{-1}) = s & \text{(i.e., } gk^{-1} \text{ stabilizes } s) \\
& \Longrightarrow & gk^{-1} \in H & \text{(recall that } H = \mathrm{Stab}(s)) \\
& \Longrightarrow & Hgk^{-1} = H & \\
& \Longrightarrow & Hg = Hk &
\end{array}$$

"$\Leftarrow$" *If two elements are in the same coset, then they send s to the same place.*

Take two elements $g, k \in G$ in the same right coset of H. This means $Hg = Hk$.

This is the last line of the proof of the forward direction, above. We can change each $\Longrightarrow$ into $\Longleftrightarrow$, and thus conclude that $s.\phi(g) = s.\phi(k)$. $\square$

If we have instead, a left group action, the proof carries through but using left cosets.

Groups acting on elements, subgroups, and cosets

It is frequently of interest to analyze the action of a group G on its elements, subgroups, or cosets of some fixed $H \leq G$.

Sometimes, the orbits and stabilizers of these actions are actually familiar algebraic objects.

Also, sometimes a deep theorem has a slick proof via a clever group action.

For example, we will see how Cayley's theorem (every group G is isomorphic to a group of permutations) follows immediately once we look at the correct action.

Here are common examples of group actions:

- G acts on itself by right-multiplication (or left-multiplication).
- G acts on itself by conjugation.
- G acts on its subgroups by conjugation.
- G acts on the right-cosets of a fixed subgroup $H \leq G$ by right-multiplication.

For each of these, we'll analyze the orbits, stabilizers, and fixed points.

Groups acting on themselves by right-multiplication

We've seen how groups act on themselves by right-multiplication. While this action is boring (any Cayley diagram is an action diagram!), it leads to a slick proof of Cayley's theorem.

Cayley's theorem

If $|G| = n$, then there is an embedding $G \hookrightarrow S_n$.

Proof.

The group G acts on itself (that is, $S = G$) by **right-multiplication**:

$$\phi \colon G \longrightarrow \mathrm{Perm}(S) \cong S_n\,, \qquad \phi(g) = \text{the permutation that sends each } x \mapsto xg.$$

There is only one orbit: $G = S$. The stabilizer of any $x \in G$ is just the identity element:

$$\mathrm{Stab}(x) = \{g \in G \mid xg = x\} = \{e\}\,.$$

Therefore, the kernel of this action is $\mathrm{Ker}\,\phi = \bigcap_{x \in G} \mathrm{Stab}(x) = \{e\}$.

Since $\mathrm{Ker}\,\phi = \{e\}$, the homomorphism ϕ is an embedding. $\square$

Groups acting on themselves by conjugation

Another way a group G can act on itself (that is, $S = G$) is by **conjugation**:

$$\phi\colon G \longrightarrow \mathrm{Perm}(S)\,, \qquad \phi(g) = \text{the permutation that sends each } x \mapsto g^{-1}xg.$$

- The orbit of $x \in G$ is its conjugacy class:

$$\mathrm{Orb}(x) = \{x.\phi(g) \mid g \in G\} = \{g^{-1}xg \mid g \in G\} = \mathrm{cl}_G(x)\,.$$

- The stabilizer of x is the set of elements that commute with x; called its centralizer:

$$\mathrm{Stab}(x) = \{g \in G \mid g^{-1}xg = x\} = \{g \in G \mid xg = gx\} := C_G(x)$$

- The fixed points of ϕ are precisely those in the center of G:

$$\mathrm{Fix}(\phi) = \{x \in G \mid g^{-1}xg = x \text{ for all } g \in G\} = Z(G)\,.$$

By the Orbit-Stabilizer theorem, $|G| = |\,\mathrm{Orb}(x)| \cdot |\,\mathrm{Stab}(x)| = |\,\mathrm{cl}_G(x)| \cdot |C_G(x)|$. Thus, we immediately get the following new result about conjugacy classes:

Theorem

For any $x \in G$, the size of the conjugacy class $\mathrm{cl}_G(x)$ divides the size of G.

Groups acting on themselves by conjugation

As an example, consider the action of $G = D_6$ on itself by **conjugation**.

The orbits of the action are the conjugacy classes:

e	r	r^2	f	$r^2 f$	$r^4 f$
r^3	r^5	r^4	rf	$r^3 f$	$r^5 f$

The fixed points of ϕ are the size-1 conjugacy classes. These are the elements in the center: $Z(D_6) = \{e\} \cup \{r^3\} = \langle r^3 \rangle$.

By the Orbit-Stabilizer theorem:

$$|\operatorname{Stab}(x)| = \frac{|D_6|}{|\operatorname{Orb}(x)|} = \frac{12}{|\operatorname{cl}_G(x)|}.$$

The stabilizer subgroups are as follows:

- $\operatorname{Stab}(e) = \operatorname{Stab}(r^3) = D_6$,
- $\operatorname{Stab}(r) = \operatorname{Stab}(r^2) = \operatorname{Stab}(r^4) = \operatorname{Stab}(r^5) = \langle r \rangle = C_6$,
- $\operatorname{Stab}(f) = \{e, r^3, f, r^3 f\} = \langle r^3, f \rangle$,
- $\operatorname{Stab}(rf) = \{e, r^3, rf, r^4 f\} = \langle r^3, rf \rangle$,
- $\operatorname{Stab}(r^i f) = \{e, r^3, r^i f, r^i f\} = \langle r^3, r^i f \rangle$.

Groups acting on subgroups by conjugation

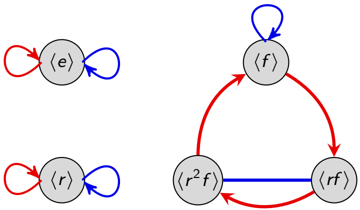
Let $G = D_3$, and let S be the set of proper subgroups of G:

$$S = \Big\{ \langle e \rangle,\ \langle r \rangle,\ \langle f \rangle,\ \langle rf \rangle,\ \langle r^2 f \rangle \Big\}.$$

There is a right group action of $D_3 = \langle r, f \rangle$ on S by conjugation:

$$\tau \colon D_3 \longrightarrow \mathrm{Perm}(S)\,, \qquad \tau(g) = \text{the permutation that sends each } H \text{ to } g^{-1}Hg.$$

$\tau(e)$	$=$	$\langle e \rangle$	$\langle r \rangle$	$\langle f \rangle$	$\langle rf \rangle$	$\langle r^2 f \rangle$
$\tau(r)$	$=$	$\langle e \rangle$	$\langle r \rangle$	$\langle f \rangle$	$\langle rf \rangle$	$\langle r^2 f \rangle$
$\tau(r^2)$	$=$	$\langle e \rangle$	$\langle r \rangle$	$\langle f \rangle$	$\langle rf \rangle$	$\langle r^2 f \rangle$
$\tau(f)$	$=$	$\langle e \rangle$	$\langle r \rangle$	$\langle f \rangle$	$\langle rf \rangle$	$\langle r^2 f \rangle$
$\tau(rf)$	$=$	$\langle e \rangle$	$\langle r \rangle$	$\langle f \rangle$	$\langle rf \rangle$	$\langle r^2 f \rangle$
$\tau(r^2 f)$	$=$	$\langle e \rangle$	$\langle r \rangle$	$\langle f \rangle$	$\langle rf \rangle$	$\langle r^2 f \rangle$



The action diagram.

$$\mathrm{Stab}(\langle e \rangle) = \mathrm{Stab}(\langle r \rangle) = D_3 = N_{D_3}(\langle r \rangle)$$
$$\mathrm{Stab}(\langle f \rangle) = \langle f \rangle = N_{D_3}(\langle f \rangle),$$
$$\mathrm{Stab}(\langle rf \rangle) = \langle rf \rangle = N_{D_3}(\langle rf \rangle),$$
$$\mathrm{Stab}(\langle r^2 f \rangle) = \langle r^2 f \rangle = N_{D_3}(\langle r^2 f \rangle).$$

Groups acting on subgroups by conjugation

More generally, any group G acts on its set S of subgroups by **conjugation**:

$$\phi\colon G \longrightarrow \mathrm{Perm}(S), \qquad \phi(g) = \text{the permutation that sends each } H \text{ to } g^{-1}Hg.$$

This is a right action, but there is an associated left action: $H \mapsto gHg^{-1}$.

Let $H \leq G$ be an element of S.

- The orbit of H consists of all conjugate subgroups:

$$\mathrm{Orb}(H) = \{g^{-1}Hg \mid g \in G\}.$$

- The stabilizer of H is the normalizer of H in G:

$$\mathrm{Stab}(H) = \{g \in G \mid g^{-1}Hg = H\} = N_G(H).$$

- The fixed points of ϕ are precisely the normal subgroups of G:

$$\mathrm{Fix}(\phi) = \{H \leq G \mid g^{-1}Hg = H \text{ for all } g \in G\}.$$

- The kernel of this action is G iff every subgroup of G is normal. In this case, ϕ is the trivial homomorphism: pressing the g-button fixes (i.e., normalizes) every subgroup.

Groups acting on cosets of H by right-multiplication

Fix a subgroup $H \leq G$. Then G acts on its **right cosets** by **right-multiplication**:

$$\phi\colon G \longrightarrow \mathrm{Perm}(S), \qquad \phi(g) = \text{the permutation that sends each } Hx \text{ to } Hxg.$$

Let Hx be an element of $S = G/H$ (the right cosets of H).

- There is only one orbit. For example, given two cosets Hx and Hy,

$$\phi(x^{-1}y) \text{ sends } Hx \longmapsto Hx(x^{-1}y) = Hy.$$

- The stabilizer of Hx is the conjugate subgroup $x^{-1}Hx$:

$$\mathrm{Stab}(Hx) = \{g \in G \mid Hxg = Hx\} = \{g \in G \mid Hxgx^{-1} = H\} = x^{-1}Hx\,.$$

- Assuming $H \neq G$, there are no fixed points of ϕ. The only orbit has size $[G : H] > 1$.
- The kernel of this action is the intersection of all conjugate subgroups of H:

$$\mathrm{Ker}\,\phi = \bigcap_{x \in G} x^{-1}Hx$$

Notice that $\langle e \rangle \leq \mathrm{Ker}\,\phi \leq H$, and $\mathrm{Ker}\,\phi = H$ iff $H \lhd G$.

Fixed points of group actions

Recall the subtle difference between fixed points and stabilizers:

- The fixed points of an action $\phi \colon G \to \mathrm{Perm}(S)$ are the elements of S fixed by every $g \in G$.
- The stabilizer of an element $s \in S$ is the set of elements of G that fix s.

Lemma

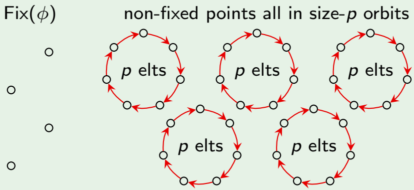
If a group G of prime order p acts on a set S via $\phi \colon G \to \mathrm{Perm}(S)$, then

$$|\operatorname{Fix}(\phi)| \equiv |S| \pmod{p}.$$

Proof (sketch)

By the Orbit-Stabilizer theorem, all orbits have size 1 or p.

I'll let you fill in the details.



Cauchy's Theorem

Cauchy's theorem

If p is a prime number dividing $|G|$, then G has an element g of order p.

Proof

Let P be the set of ordered p-tuples of elements from G whose product is e, i.e.,

$$(x_1, x_2, \ldots, x_p) \in P \quad \text{iff} \quad x_1 x_2 \cdots x_p = e\,.$$

Observe that $|P| = |G|^{p-1}$. (We can choose $x_1, \ldots, x_{p-1}$ freely; then x_p is forced.)

The group $\mathbb{Z}_p$ acts on P by cyclic shift:

$$\phi\colon \mathbb{Z}_p \longrightarrow \mathrm{Perm}(P), \qquad (x_1, x_2, \ldots, x_p) \xmapsto{\phi(1)} (x_2, x_3 \ldots, x_p, x_1)\,.$$

(This is because if $x_1 x_2 \cdots x_p = e$, then $x_2 x_3 \cdots x_p x_1 = e$ as well.)

The elements of P are partitioned into orbits. By the orbit-stabilizer theorem, $|\operatorname{Orb}(s)| = [\mathbb{Z}_p : \operatorname{Stab}(s)]$, which divides $|\mathbb{Z}_p| = p$. Thus, $|\operatorname{Orb}(s)| = 1$ or p.

Observe that the only way that an orbit of $(x_1, x_2, \ldots, x_p)$ could have size 1 is if $x_1 = x_2 = \cdots = x_p$.

Cauchy's Theorem

Proof (cont.)

Clearly, $(e, e, \ldots, e) \in P$, and the orbit containing it has size 1.

Excluding $(e, \ldots, e)$, there are $|G|^{p-1} - 1$ other elements in P, and these are partitioned into orbits of size 1 or p.

Since $p \nmid |G|^{p-1} - 1$, there must be some other orbit of size 1.

Thus, there is some $(x, x, \ldots, x) \in P$, with $x \neq e$ such that $x^p = e$. $\square$

Corollary

If p is a prime number dividing $|G|$, then G has a subgroup of order p.

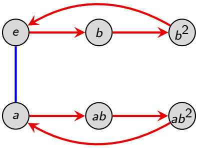
Note that just by using the theory of group actions, and the orbit-stabilzer theorem, we have already proven:

- Cayley's theorem: Every group G is isomorphic to a group of permutations.
- The size of a conjugacy class divides the size of G.
- Cauchy's theorem: If p divides $|G|$, then G has an element of order p.

Classification of groups of order 6

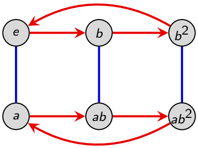
By Cauchy's theorem, every group of order 6 must have an element a of order 2, and an element b of order 3.

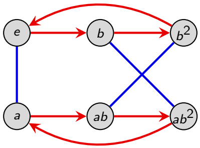
Clearly, $G = \langle a, b \rangle$ for two such elements. Thus, G must have a Cayley diagram that looks like the following:



It is now easy to see that up to isomorphism, there are only 2 groups of order 6:

$C_6 \cong C_2 \times C_3$





D_3

p-groups and the Sylow theorems

Definition

A *p-group* is a group whose order is a power of a prime p. A p-group that is a subgroup of a group G is a *p-subgroup* of G.

Notational convention

Throughout, G will be a group of order $|G| = p^n \cdot m$, with $p \nmid m$. That is, p^n is the *highest power* of p dividing $|G|$.

There are three Sylow theorems, and loosely speaking, they describe the following about a group's p-subgroups:

1. **Existence**: In every group, p-subgroups of all possible sizes exist.
2. **Relationship**: All maximal p-subgroups are conjugate.
3. **Number**: There are strong restrictions on the number of p-subgroups a group can have.

Together, these place strong restrictions on the structure of a group G with a fixed order.

p-groups

Before we introduce the Sylow theorems, we need to better understand p-groups.

Recall that a p-group is any group of order p^n. For example, C_1, C_4, V_4, D_4 and Q_8 are all 2-groups.

p-group Lemma

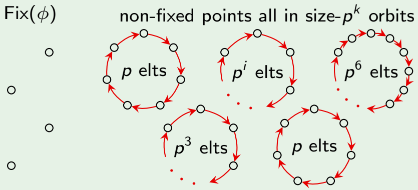
If a p-group G acts on a set S via $\phi \colon G \to \mathrm{Perm}(S)$, then

$$|\operatorname{Fix}(\phi)| \equiv_p |S|.$$

Proof (sketch)

Suppose $|G| = p^n$.

By the Orbit-Stabilizer theorem, the only possible orbit sizes are $1, p, p^2, \dots, p^n$.



Normalizer lemma, Part 1

If H is a p-subgroup of G, then

$$[N_G(H) \colon H] \equiv_p [G \colon H] \,.$$

Proof

Let $S = G/H = \{Hx \mid x \in G\}$. The group H acts on S by **right-multiplication**, via $\phi \colon H \to \mathrm{Perm}(S)$, where

$$\phi(h) = \text{the permutation sending each } Hx \text{ to } Hxh.$$

The fixed points of ϕ are the cosets Hx in the normalizer $N_G(H)$:

$$\begin{aligned} Hxh = Hx, \quad \forall h \in H \qquad &\Longleftrightarrow \qquad Hxhx^{-1} = H, \quad \forall h \in H \\ &\Longleftrightarrow \qquad xhx^{-1} \in H, \quad \forall h \in H \\ &\Longleftrightarrow \qquad x \in N_G(H) \,. \end{aligned}$$

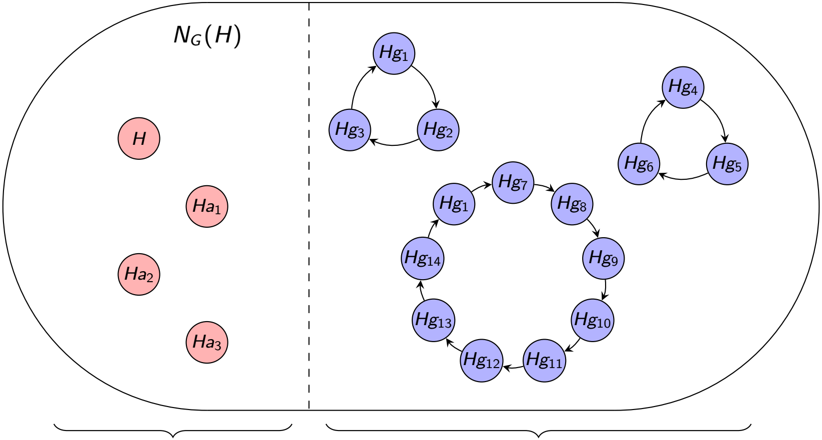
Therefore, $|\,\mathrm{Fix}(\phi)| = [N_G(H) \colon H]$, and $|S| = [G : H]$. By our p-group Lemma,

$$|\,\mathrm{Fix}(\phi)| \equiv_p |S| \quad \Longrightarrow \quad [N_G(H) \colon H] \equiv_p [G \colon H] \,. \qquad \square$$

p-groups

Here is a picture of the action of the p-subgroup H on the set $S = G/H$, from the proof of the Normalizer Lemma.

$S = G/H =$ set of right cosets of H in G



The fixed points are precisely the cosets in $N_G(H)$

Orbits of size > 1 are of various sizes dividing $|H|$, but all lie outside $N_G(H)$

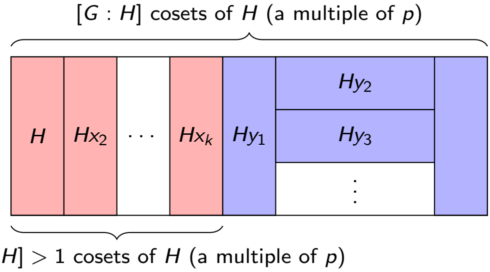
p-subgroups

The following result will be useful in proving the first Sylow theorem.

The Normalizer lemma, Part 2

Suppose $|G| = p^n m$, and $H \leq G$ with $|H| = p^i < p^n$. Then $H \lneq N_G(H)$, and the index $[N_G(H) : H]$ is a multiple of p.

$$H \lneq N_G(H) \leq G$$



Conclusions:

- $H = N_G(H)$ is impossible!
- p^{i+1} divides $|N_G(H)|$.

Proof of the normalizer lemma

The Normalizer lemma, Part 2

Suppose $|G| = p^n m$, and $H \leq G$ with $|H| = p^i < p^n$. Then $H \lneq N_G(H)$, and the index $[N_G(H) : H]$ is a multiple of p.

Proof

Since $H \lhd N_G(H)$, we can create the quotient map

$$q\colon N_G(H) \longrightarrow N_G(H)/H\,, \qquad\qquad q\colon g \longmapsto gH\,.$$

The size of the quotient group is $[N_G(H)\colon H]$, the number of cosets of H in $N_G(H)$.

By The Normalizer lemma Part 1, $[N_G(H)\colon H] \equiv_p [G\colon H]$. By Lagrange's theorem,

$$[N_G(H)\colon H] \equiv_p [G\colon H] = \frac{|G|}{|H|} = \frac{p^n m}{p^i} = p^{n-i} m \equiv_p 0\,.$$

Therefore, $[N_G(H)\colon H]$ is a multiple of p, so $N_G(H)$ must be strictly larger than H. $\square$

The Sylow theorems

The Sylow theorems are about one question:

What finite groups are there?

Early on, we saw five families of groups: cyclic, dihedral, abelian, symmetric, alternating.

Later, we classified all (finitely generated) *abelian* groups.

But what *other* groups are there, and what do they look like? For example, for a fixed order $|G|$, we may ask the following questions about G:

1. How big are its subgroups?
2. How are those subgroups related?
3. How many subgroups are there?
4. Are any of them normal?

There is no one general method to answer this for any given order.

However, the Sylow Theorems, developed by Norwegian mathematician Peter Sylow (1832–1918), are powerful tools that help us attack this question.

p-subgroups

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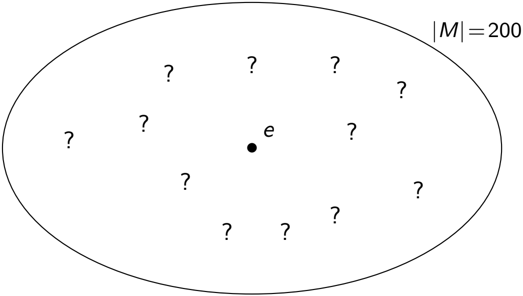
There are three Sylow theorems, and loosely speaking, they describe the following about a group's p-subgroups:

1. **Existence**: In every group, p-subgroups of all possible sizes exist.
2. **Relationship**: All maximal p-subgroups are conjugate.
3. **Number**: There are strong restrictions on the number of p-subgroups a group can have.

Together, these place strong restrictions on the structure of a group G with a fixed order.

Our unknown group of order 200

Throughout our two lectures on the Sylow theorems, we will have a running example, a "mystery group" M of order 200.



Using *only* the fact that $|M| = 200$, we will unconver as much about the structure of M as we can.

We actually already know a little bit. Recall Cauchy's theorem:

Cauchy's theorem

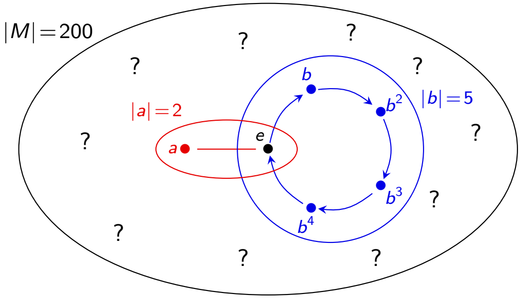
If p is a prime number dividing $|G|$, then G has an element g of order p.

Our mystery group of order 200

Since our mystery group M has order $|M| = 2^3 \cdot 5^2 = 200$, Cauchy's theorem tells us that:

- M has an element a of order 2;
- M has an element b of order 5;

Also, by Lagrange's theorem, $\langle a \rangle \cap \langle b \rangle = \{e\}$.



The 1^{st} Sylow Theorem: Existence of p-subgroups

First Sylow Theorem

G has a subgroup of order p^k, for each p^k dividing $|G|$. Also, every p-subgroup with fewer than p^n elements sits inside one of the larger p-subgroups.

The First Sylow Theorem is in a sense, a generalization of Cauchy's theorem. Here is a comparison:

Cauchy's Theorem	First Sylow Theorem
If p divides $|G|$, then ...	*If p^k divides $|G|$, then ...*
There is a subgroup of order p	There is a subgroup of order p^k
which is cyclic and has no non-trivial proper subgroups.	which has subgroups of order $1, p, p^2 \ldots p^k$.
G contains an element of order p	G might not contain an element of order p^k.

The 1^{st} Sylow Theorem: Existence of p-subgroups

Proof

The trivial subgroup $\{e\}$ has order $p^0 = 1$.

Big idea: Suppose we're given a subgroup $H < G$ of order $p^i < p^n$. We will construct a subgroup H' of order p^{i+1}.

By the normalizer lemma, $H \lneq N_G(H)$, and the order of the quotient group $N_G(H)/H$ is a multiple of p.

By Cauchy's Theorem, $N_G(H)/H$ contains an element (a coset!) of order p. Call this element aH. Note that $\langle aH \rangle$ is cyclic of order p.

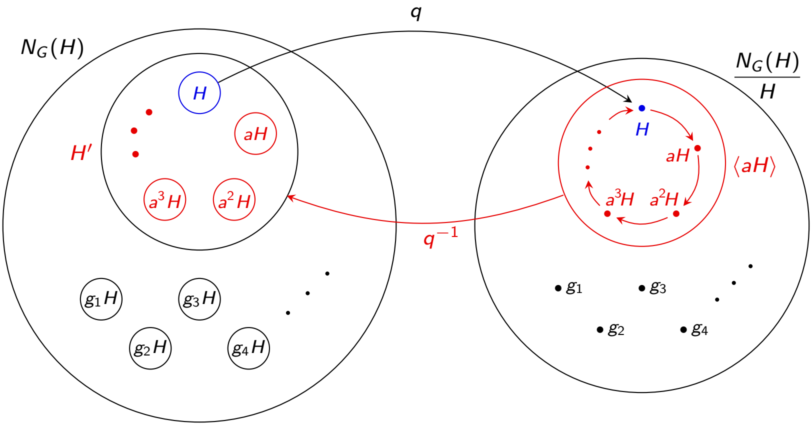
Claim: The preimage of $\langle aH \rangle$ under the quotient $q \colon N_G(H) \to N_G(H)/H$ is the subgroup H' we seek.

The preimages $q^{-1}(H)$, $q^{-1}(aH)$, $q^{-1}(a^2H)$, $\dots$, $q^{-1}(a^{p-1}H)$ are all distinct cosets of H in $N_G(H)$, each of size p^i.

Thus, the preimage $H' = q^{-1}(\langle aH \rangle)$ contains $p \cdot |H| = p^{i+1}$ elements. $\square$

The 1^{st} Sylow Theorem: Existence of p-subgroups

Here is a picture of how we found the group $H' = q^{-1}(\langle aH \rangle)$.

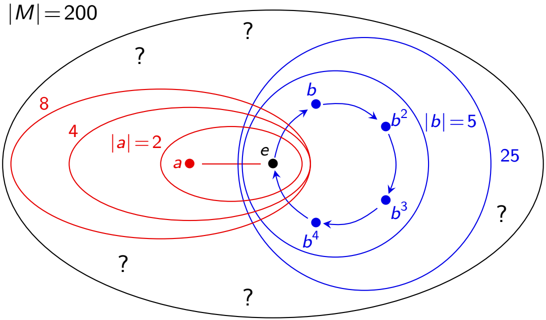


Since $|H| = p^i$, the subgroup $H' = \bigcup_{k=0}^{p-1} a^k H$ contains $p \cdot |H| = p^{i+1}$ elements.

Our unknown group of order 200

We now know a little bit more about the structure of our mystery group of order $|M| = 2^3 \cdot 5^2$:

- M has a 2-subgroup P_2 of order $2^3 = 8$;
- M has a 5-subgroup P_5 of order $25 = 5^2$;
- Each of these subgroups contains a nested chain of p-subgroups, down to the trivial group, $\{e\}$.



The 2nd Sylow Theorem: Relationship among p-subgroups

Definition

A subgroup $H < G$ of order p^n, where $|G| = p^n \cdot m$ with $p \nmid m$ is called a Sylow p-subgroup of G. Let $\mathrm{Syl}_p(G)$ denote the set of Sylow p-subgroups of G.

Second Sylow Theorem

Any two Sylow p-subgroups are conjugate (and hence isomorphic).

Proof

Let $H < G$ be any Sylow p-subgroup of G, and let $S = G/H = \{Hg \mid g \in G\}$, the set of right cosets of H.

Pick *any other* Sylow p-subgroup K of G. (If there is none, the result is trivial.)

The group K acts on S by **right-multiplication**, via $\phi\colon K \to \mathrm{Perm}(S)$, where

$$\phi(k) = \text{the permutation sending each } Hg \text{ to } Hgk.$$

The 2^{nd} Sylow Theorem: All Sylow p-subgroups are conjugate

Proof

A fixed point of ϕ is a coset $Hg \in S$ such that

$$\begin{aligned} Hgk = Hg\,, \quad \forall k \in K \qquad &\Longleftrightarrow \qquad Hgkg^{-1} = H\,, \quad \forall k \in K \\ &\Longleftrightarrow \qquad gkg^{-1} \in H\,, \quad \forall k \in K \\ &\Longleftrightarrow \qquad gKg^{-1} \subset H \\ &\Longleftrightarrow \qquad gKg^{-1} = H\,. \end{aligned}$$

Thus, if ϕ has a fixed point Hg, then H and K are conjugate by g, and we're done!

All we need to do is show that $|\operatorname{Fix}(\phi)| \not\equiv_p 0$.

By the p-group Lemma, $|\operatorname{Fix}(\phi)| \equiv_p |S|$. Recall that $|S| = [G, H]$.

Since H is a Sylow p-subgroup, $|H| = p^n$. By Lagrange's Theorem,

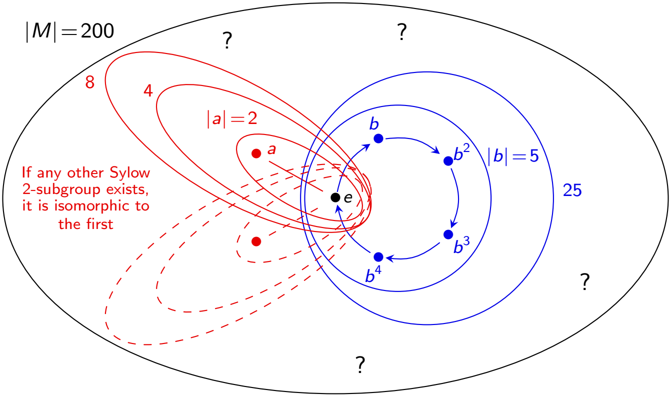
$$|S| = [G : H] = \frac{|G|}{|H|} = \frac{p^n m}{p^n} = m, \qquad p \nmid m\,.$$

Therefore, $|\operatorname{Fix}(\phi)| \equiv_p m \not\equiv_p 0$. $\square$

Our unknown group of order 200

We now know even more about the structure of our mystery group M, of order $|M| = 2^3 \cdot 5^2$:

- If M has any other Sylow 2-subgroup, it is isomorphic to P_2;
- If M has any other Sylow 5-subgroup, it is isomorphic to P_5.



The 3rd Sylow Theorem: Number of p-subgroups

Third Sylow Theorem

Let n_p be the number of Sylow p-subgroups of G. Then

$$n_p \text{ divides } |G| \qquad \text{and} \qquad n_p \equiv_p 1\,.$$

(Note that together, these imply that $n_p \mid m$, where $|G| = p^n \cdot m$.)

Proof

The group G acts on $S = \mathrm{Syl}_p(G)$ by conjugation, via $\phi\colon G \to \mathrm{Perm}(S)$, where

$$\phi(g) = \text{the permutation sending each } H \text{ to } g^{-1}Hg.$$

By the Second Sylow Theorem, all Sylow p-subgroups are conjugate! Thus there is only one orbit, $\mathrm{Orb}(H)$, of size $n_p = |S|$.

By the Orbit-Stabilizer Theorem,

$$\underbrace{|\,\mathrm{Orb}(H)|}_{=n_p} \cdot |\,\mathrm{Stab}(H)| = |G| \qquad \Longrightarrow \qquad n_p \text{ divides } |G|\,.$$

The 3rd Sylow Theorem: Number of p-subgroups

Proof (cont.)

Now, pick any $H \in \mathrm{Syl}_p(G) = S$. The group H acts on S by conjugation, via $\theta \colon H \to \mathrm{Perm}(S)$, where

$$\theta(h) = \text{the permutation sending each } K \text{ to } h^{-1}Kh.$$

Let $K \in \mathrm{Fix}(\theta)$. Then $K \leq G$ is a Sylow p-subgroup satisfying

$$h^{-1}Kh = K\,, \quad \forall h \in H \quad \Longleftrightarrow \quad H \leq N_G(K) \leq G\,.$$

We know that:

- H and K are Sylow p-subgroups of G, but also of $N_G(K)$.
- Thus, H and K are conjugate in $N_G(K)$. (2nd Sylow Thm.)
- $K \lhd N_G(K)$, thus the only conjugate of K in $N_G(K)$ is itself.

Thus, $K = H$. That is, $\mathrm{Fix}(\theta) = \{H\}$ contains only 1 element.

By the p-group Lemma, $n_p := |S| \equiv_p |\mathrm{Fix}(\theta)| = 1$. $\square$

Summary of the proofs of the Sylow Theorems

For the 1st Sylow Theorem, we started with $H = \{e\}$, and inductively created larger subgroups of size $p, p^2, \dots, p^n$.

For the 2^{nd} and 3^{rd} Sylow Theorems, we used a clever group action and then applied one or both of the following:

(i) *Orbit-Stabilizer Theorem*. If G acts on S, then $|\operatorname{Orb}(s)| \cdot |\operatorname{Stab}(s)| = |G|$.

(ii) *p-group Lemma*. If a p-group acts on S, then $|S| \equiv_p |\operatorname{Fix}(\phi)|$.

To summarize, we used:

S2 The action of $K \in \operatorname{Syl}_p(G)$ on $S = G/H$ by right multiplication for some other $H \in \operatorname{Syl}_p(G)$.

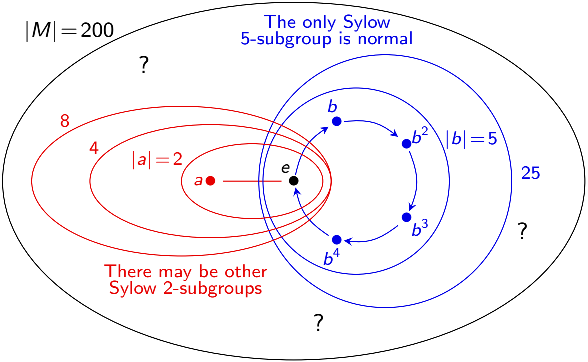
S3a The action of G on $S = \operatorname{Syl}_p(G)$, by conjugation.

S3b The action of $H \in \operatorname{Syl}_p(G)$ on $S = \operatorname{Syl}_p(G)$, by conjugation.

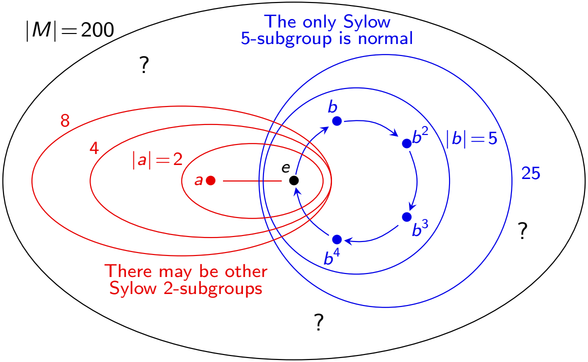
Our unknown group of order 200

We now know a little bit more about the structure of our mystery group M, of order $|M| = 2^3 \cdot 5^2 = 200$:

- $n_5 \mid 8$, thus $n_5 \in \{1, 2, 4, 8\}$. But $n_5 \equiv_5 1$, so $n_5 = 1$.
- $n_2 \mid 25$ and is odd. Thus $n_2 \in \{1, 5, 25\}$.
- We conclude that M has a unique (and hence normal) Sylow 5-subgroup P_5 (of order $5^2 = 25$), and either 1, 5, or 25 Sylow 2-subgroups (of order $2^3 = 8$).



Our unknown group of order 200



Suppose M has a subgroup isomorphic to D_4.

This would be a Sylow 2-subgroup. Since all of them are conjugate, M *cannot* contain a subgroup isomorphic to Q_8, $C_4 \times C_2$, or C_8!

In particular, M cannot even contain an element of order 8. (Why?)

Simple groups and the Sylow theorems

Definition

A group G is simple if its only normal subgroups are G and $\langle e \rangle$.

Since all Sylow p-subgroups are conjugate, the following result is straightforward:

Proposition (HW)

A Sylow p-subgroup is normal in G if and only if it is the unique Sylow p-subgroup (that is, if $n_p = 1$).

The Sylow theorems are very useful for establishing statements like:

There are no simple groups of order k (for some k).

To do this, we usually just need to show that $n_p = 1$ for some p dividing $|G|$.

Since we established $n_5 = 1$ for our running example of a group of size $|M| = 200 = 2^3 \cdot 5^2$, there are no simple groups of order 200.

An easy example

Tip

When trying to show that $n_p = 1$, it's usually more helpful to analyze the largest primes first.

Proposition

There are no simple groups of order 84.

Proof

Since $|G| = 84 = 2^2 \cdot 3 \cdot 7$, the Third Sylow Theorem tells us:

- n_7 divides $2^2 \cdot 3 = 12$ (so $n_7 \in \{1, 2, 3, 4, 6, 12\}$)
- $n_7 \equiv_7 1$.

The only possibility is that $n_7 = 1$, so the Sylow 7-subgroup must be normal. $\square$

Observe why it is beneficial to use the largest prime first:

- n_3 divides $2^2 \cdot 7 = 28$ and $n_3 \equiv_3 1$. Thus $n_3 \in \{1, \cancel{2}, 4, 7, \cancel{14}, 28\}$.
- n_2 divides $3 \cdot 7 = 21$ and $n_2 \equiv_2 1$. Thus $n_2 \in \{1, 3, 7, 21\}$.

A harder example

Proposition

There are no simple groups of order 351.

Proof

Since $|G| = 351 = 3^3 \cdot 13$, the Third Sylow Theorem tells us:

- n_{13} divides $3^3 = 27$ (so $n_{13} \in \{1, 3, 9, 27\}$)
- $n_{13} \equiv_{13} 1$.

The only possibilies are $n_{13} = 1$ or 27.

A Sylow 13-subgroup P has order 13, and a Sylow 3-subgroup Q has order $3^3 = 27$. Therefore, $P \cap Q = \{e\}$.

Suppose $n_{13} = 27$. Every Sylow 13-subgroup contains 12 non-identity elements, and so G must contain $27 \cdot 12 = 324$ elements of order 13.

This leaves $351 - 324 = 27$ elements in G not of order 13. Thus, G contains only one Sylow 3-subgroup (i.e., $n_3 = 1$) and so G cannot be simple. $\square$

The hardest example

Proposition

If $H \lneq G$ and $|G|$ does not divide $[G:H]!$, then G cannot be simple.

Proof

Let G act on the **right cosets** of H (i.e., $S = G/H$) by **right-multiplication**:

$$\phi \colon G \longrightarrow \mathrm{Perm}(S) \cong S_n\,, \qquad \phi(g) = \text{the permutation that sends each } Hx \text{ to } Hxg.$$

Recall that the kernel of ϕ is the intersection of all conjugate subgroups of H:

$$\mathrm{Ker}\,\phi = \bigcap_{x \in G} x^{-1} H x.$$

Notice that $\langle e \rangle \leq \mathrm{Ker}\,\phi \leq H \lneq G$, and $\mathrm{Ker}\,\phi \lhd G$.

If $\mathrm{Ker}\,\phi = \langle e \rangle$ then $\phi \colon G \hookrightarrow S_n$ is an embedding. But this is *impossible* because $|G|$ does not divide $|S_n| = [G:H]!$. $\square$

Corollary

There are no simple groups of order 24.

Theorem (classification of finite simple groups)

Every finite simple group is isomorphic to one of the following groups:

- A cyclic group $\mathbb{Z}_p$, with p prime;
- An alternating group A_n, with $n \geq 5$;
- A Lie-type Chevalley group: $\mathrm{PSL}(n, q)$, $\mathrm{PSU}(n, q)$, $\mathrm{PsP}(2n, p)$, and $P\Omega^{\epsilon}(n, q)$;
- A Lie-type group (twisted Chevalley group or the Tits group): $D_4(q)$, $E_6(q)$, $E_7(q)$, $E_8(q)$, $F_4(q)$, ${}^2F_4(2^n)'$, $G_2(q)$, ${}^2G_2(3^n)$, ${}^2B(2^n)$;
- One of 26 exceptional "sporadic groups."

The two largest sporadic groups are the:

- "baby monster group" B, which has order

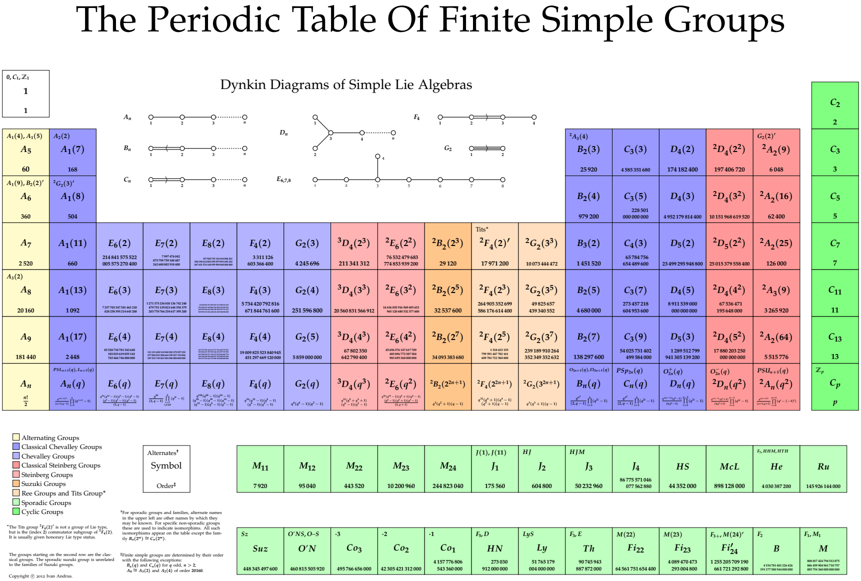
$$|B| = 2^{41} \cdot 3^{13} \cdot 5^{6} \cdot 7^{2} \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47 \approx 4.15 \times 10^{33};$$

- "monster group" M, which has order

$$|M| = 2^{46} \cdot 3^{20} \cdot 5^{9} \cdot 7^{6} \cdot 11^{2} \cdot 13^{3} \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71 \approx 8.08 \times 10^{53}.$$

The proof of this classification theorem is spread across $\approx$ 15,000 pages in $\approx$ 500 journal articles by over 100 authors, published between 1955 and 2004.

Image by *Ivan Andrus*, 2012



Finite Simple Group (of Order Two), by The Klein Four$^{\mathrm{TM}}$

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12	Stefanie (The Ballad of Galois)	Klein Four	4:51	$0.99	View In iTunes ▸
13	Musical Fruitcake (Pass it Around)	Klein Four	2:50	$0.99	View In iTunes ▸
14	Abandon Soap	Klein Four	2:17	$0.99	View In iTunes ▸

14 Songs

Section 6: Field and Galois theory

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Math 4120, Modern Algebra

Some history and the search for the quintic

The quadradic formula is well-known. It gives us the two roots of a degree-2 polynomial $ax^2 + bx + c = 0$:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

There are formulas for cubic and quartic polynomials, but they are *very* complicated. For years, people wondered if there was a quintic formula. Nobody could find one.

In the 1830s, 19-year-old political activist Évariste Galois, with no formal mathematical training proved that no such formula existed.

He invented the concept of a group to solve this problem.

After being challenged to a dual at age 20 that he knew he would lose, Galois spent the last few days of his life frantically writing down what he had discovered.

In a final letter Galois wrote, "*Later there will be, I hope, some people who will find it to their advantage to decipher all this mess.*"

Hermann Weyl (1885–1955) described Galois' final letter as: "*if judged by the novelty and profundity of ideas it contains, is perhaps the most substantial piece of writing in the whole literature of mankind.*" Thus was born the field of group theory!

Arithmetic

Most people's first exposure to mathematics comes in the form of counting.

At first, we only know about the natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$, and how to add them.

Soon after, we learn how to subtract, and we learn about negative numbers as well. At this point, we have the integers, $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.

Then, we learn how to divide numbers, and are introducted to fractions. This brings us to the rational numbers, $\mathbb{Q} = \{\frac{a}{b} \mid a, b \in \mathbb{Z},\ b \neq 0\}$.

Though there are other numbers out there (irrational, complex, etc.), we don't need these to do basic arithmetic.

Key point

To do arithmetic, we need *at least* the rational numbers.

Fields

Definition

A set F with addition and multiplication operations is a field if the following three conditions hold:

- F is an abelian group under addition.
- $F \setminus \{0\}$ is an abelian group under multiplication.
- The distributive law holds: $a(b+c) = ab + ac$.

Examples

- The following sets are fields: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ (prime p).
- The following sets are *not* fields: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Z}_n$ (composite n).

Definition

If F and E are fields with $F \subset E$, we say that E is an extension of F.

For example, $\mathbb{C}$ is an extension of $\mathbb{R}$, which is an extension of $\mathbb{Q}$.

In this chapter, we will explore some more unusual fields and study their automorphisms.

An extension field of $\mathbb{Q}$

Question

What is the smallest extension field F of $\mathbb{Q}$ that contains $\sqrt{2}$?

This field must contain all sums, differences, and quotients of numbers we can get from $\sqrt{2}$. For example, it must include:

$$-\sqrt{2}, \qquad \tfrac{1}{\sqrt{2}}, \qquad 6+\sqrt{2}, \qquad \left(\sqrt{2}+\tfrac{3}{2}\right)^3, \qquad \tfrac{\sqrt{2}}{16+\sqrt{2}}\,.$$

However, these can be simplified. For example, observe that

$$\left(\sqrt{2}+\tfrac{3}{2}\right)^3 = (\sqrt{2})^3 + \tfrac{9}{2}(\sqrt{2})^2 + \tfrac{27}{4}\sqrt{2} + \tfrac{27}{8} = \tfrac{99}{8} + \tfrac{35}{4}\sqrt{2}\,.$$

In fact, *all* of these numbers can be written as $a+b\sqrt{2}$, for some $a, b \in \mathbb{Q}$.

Key point

The smallest extension of $\mathbb{Q}$ that contains $\sqrt{2}$ is called "$\mathbb{Q}$ adjoin $\sqrt{2}$," and denoted:

$$\mathbb{Q}(\sqrt{2}) = \{a+b\sqrt{2} : a, b \in \mathbb{Q}\} = \left\{\tfrac{p}{q} + \tfrac{r}{s}\sqrt{2} : p, q, r, s \in \mathbb{Z},\ q, s \neq 0\right\}\,.$$

$\mathbb{Q}(i)$: Another extension field of $\mathbb{Q}$

Question

What is the smallest extension field F of $\mathbb{Q}$ that contains $i = \sqrt{-1}$?

This field must contain

$$-i, \qquad \tfrac{2}{i}, \qquad 6+i, \qquad \left(i+\tfrac{3}{2}\right)^3, \qquad \tfrac{i}{16+i}\,.$$

As before, we can write all of these as $a + bi$, where $a, b \in \mathbb{Q}$. Thus, the field "$\mathbb{Q}$ adjoin i" is

$$\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\} = \left\{ \frac{p}{q} + \frac{r}{s}\, i : p, q, r, s \in \mathbb{Z},\ q, s \neq 0 \right\}.$$

Remarks

- $\mathbb{Q}(i)$ is much smaller than $\mathbb{C}$. For example, it does *not* contain $\sqrt{2}$.
- $\mathbb{Q}(\sqrt{2})$ is a subfield of $\mathbb{R}$, but $\mathbb{Q}(i)$ is not.
- $\mathbb{Q}(\sqrt{2})$ contains all of the roots of $f(x) = x^2 - 2$. It is called the splitting field of $f(x)$. Similarly, $\mathbb{Q}(i)$ is the splitting field of $g(x) = x^2 + 1$.

$\mathbb{Q}(\sqrt{2}, i)$: Another extension field of $\mathbb{Q}$

Question

What is the smallest extension field F of $\mathbb{Q}$ that contains $\sqrt{2}$ and $i = \sqrt{-1}$?

We can do this in two steps:

(i) Adjoin the roots of the polynomial $x^2 - 2$ to $\mathbb{Q}$, yielding $\mathbb{Q}(\sqrt{2})$;

(ii) Adjoin the roots of the polynomial $x^2 + 1$ to $\mathbb{Q}(\sqrt{2})$, yielding $\mathbb{Q}(\sqrt{2})(i)$;

An element in $\mathbb{Q}(\sqrt{2}, i) := \mathbb{Q}(\sqrt{2})(i)$ has the form

$$\begin{array}{ll} \quad \alpha + \beta i & \qquad \alpha, \beta \in \mathbb{Q}(\sqrt{2}) \\ = (a + b\sqrt{2}) + (c + d\sqrt{2})i & \qquad a, b, c, d \in \mathbb{Q} \\ = a + b\sqrt{2} + ci + d\sqrt{2}i & \qquad a, b, c, d \in \mathbb{Q} \end{array}$$

We say that $\{1, \sqrt{2}, i, \sqrt{2}i\}$ is a basis for the extension $\mathbb{Q}(\sqrt{2}, i)$ over $\mathbb{Q}$. Thus,

$$\mathbb{Q}(\sqrt{2}, i) = \{a + b\sqrt{2} + ci + d\sqrt{2}i : a, b, c, d \in \mathbb{Q}\}$$

In summary, $\mathbb{Q}(\sqrt{2}, i)$ is constructed by starting with $\mathbb{Q}$, and adjoining all roots of $h(x) = (x^2 - 2)(x^2 + 1) = x^4 - x^2 - 2$. It is the splitting field of $h(x)$.

$\mathbb{Q}(\sqrt{2}, \sqrt{3})$: Another extension field of $\mathbb{Q}$

Question

What is the smallest extension field F of $\mathbb{Q}$ that contains $\sqrt{2}$ and $\sqrt{3}$?

This time, our field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, constructed by starting with $\mathbb{Q}$, and adjoining all roots of the polynomial $h(x) = (x^2 - 2)(x^2 - 3) = x^4 - 5x^2 + 6$.

It is not difficult to show that $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ is a basis for this field, i.e.,

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \{a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} : a, b, c, d \in \mathbb{Q}\}.$$

Like with did with a group and its subgroups, we can arrange the subfields of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ in a lattice.

I've labeled each extension with the degree of the polynomial whose roots I need to adjoin.

Just for fun: What *group* has a subgroup lattice that looks like this?

$$\begin{tikzcd}
& \mathbb{Q}(\sqrt{2}, \sqrt{3}) \arrow[dl, dash, "2"'] \arrow[d, dash, "2"] \arrow[dr, dash, "2"] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash, "2"'] & \mathbb{Q}(\sqrt{6}) \arrow[d, dash, "2"] & \mathbb{Q}(\sqrt{3}) \arrow[dl, dash, "2"] \\
& \mathbb{Q} &
\end{tikzcd}$$

Field automorphisms

Recall that an automorphism of a group G was an isomorphism $\phi\colon G \to G$.

Definition

Let F be a field. A field automorphism of F is a bijection $\phi\colon F \to F$ such that for all $a, b \in F$,

$$\phi(a+b) = \phi(a) + \phi(b) \qquad \text{and} \qquad \phi(ab) = \phi(a)\,\phi(b)\,.$$

In other words, ϕ must preserve the structure of the field.

For example, let $F = \mathbb{Q}(\sqrt{2})$. Verify (HW) that the function

$$\phi\colon \mathbb{Q}(\sqrt{2}) \longrightarrow \mathbb{Q}(\sqrt{2})\,, \qquad \phi\colon a + b\sqrt{2} \longmapsto a - b\sqrt{2}\,.$$

is an automorphism. That is, show that

- $\phi((a+b\sqrt{2}) + (c+d\sqrt{2})) = \cdots = \phi(a+b\sqrt{2}) + \phi(c+d\sqrt{2})$
- $\phi((a+b\sqrt{2})(c+d\sqrt{2})) = \cdots = \phi(a+b\sqrt{2})\,\phi(c+d\sqrt{2})$.

What other field automorphisms of $\mathbb{Q}(\sqrt{2})$ are there?

A defining property of field automorphisms

Field automorphisms are central to Galois theory! We'll see why shortly.

Proposition

If ϕ is an automorphism of an extension field F of $\mathbb{Q}$, then

$$\phi(q) = q \qquad \text{for all } q \in \mathbb{Q}.$$

Proof

Suppose that $\phi(1) = q$. Clearly, $q \neq 0$. (Why?) Observe that

$$q = \phi(1) = \phi(1 \cdot 1) = \phi(1)\,\phi(1) = q^2\,.$$

Similarly,

$$q = \phi(1) = \phi(1 \cdot 1 \cdot 1) = \phi(1)\,\phi(1)\,\phi(1) = q^3\,.$$

And so on. It follows that $q^n = q$ for every $n \geq 1$. Thus, $q = 1$. $\square$

Corollary

$\sqrt{2}$ is irrational. $\square$

The Galois group of a field extension

The set of all automorphisms of a field forms a group under composition.

Definition

Let F be an extension field of $\mathbb{Q}$. The Galois group of F is the group of automorphisms of F, denoted $\mathrm{Gal}(F)$.

Here are some examples (without proof):

- The Galois group of $\mathbb{Q}(\sqrt{2})$ is C_2:

$$\mathrm{Gal}(\mathbb{Q}(\sqrt{2})) = \langle f \rangle \cong C_2\,, \qquad \text{where } f\colon \sqrt{2} \longmapsto -\sqrt{2}$$

- An automorphism of $F = \mathbb{Q}(\sqrt{2}, i)$ is completely determined by where it sends $\sqrt{2}$ and i. There are four possibilities: the identity map e, and

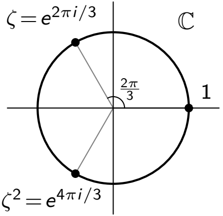
$$\begin{cases} h(\sqrt{2}) = -\sqrt{2} \\ h(i) = i \end{cases} \qquad \begin{cases} v(\sqrt{2}) = \sqrt{2} \\ v(i) = -i \end{cases} \qquad \begin{cases} r(\sqrt{2}) = -\sqrt{2} \\ r(i) = -i \end{cases}$$

Thus, the Galois group of F is $\mathrm{Gal}(\mathbb{Q}(\sqrt{2}, i)) = \langle h, v \rangle \cong V_4$.

$\mathbb{Q}(\zeta, \sqrt[3]{2})$: Another extension field of $\mathbb{Q}$

Question

What is the smallest extension field F of $\mathbb{Q}$ that contains all roots of $g(x) = x^3 - 2$?



Let $\zeta = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$. This is a 3rd root of unity; the roots of $x^3 - 1 = (x-1)(x^2 + x + 1)$ are $1, \zeta, \zeta^2$.

Note that the roots of $g(x)$ are

$$z_1 = \sqrt[3]{2}\,, \qquad z_2 = \zeta\sqrt[3]{2}\,, \qquad z_3 = \zeta^2\sqrt[3]{2}\,.$$

Thus, the field we seek is $F = \mathbb{Q}(z_1, z_2, z_3)$.

I claim that $F = \mathbb{Q}(\zeta, \sqrt[3]{2})$. Note that this field contains z_1, z_2, and z_3. Conversely, we can construct ζ and $\sqrt[3]{2}$ from z_1 and z_2, using arithmetic.

A little algebra can show that

$$\mathbb{Q}(\zeta, \sqrt[3]{2}) = \{a + b\sqrt[3]{2} + c\sqrt[3]{4} + d\zeta + e\zeta\sqrt[3]{2} + f\zeta\sqrt[3]{4} : a, b, c, d, e, f \in \mathbb{Q}\}\,.$$

Since $\zeta = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ lies in $\mathbb{Q}(\zeta, \sqrt[3]{2})$, so does $2(\zeta + \frac{1}{2}) = \sqrt{3}i = \sqrt{-3}$. Thus,

$$\mathbb{Q}(\zeta, \sqrt[3]{2}) = \mathbb{Q}(\sqrt{-3}, \sqrt[3]{2}) = \mathbb{Q}(\sqrt{3}i, \sqrt[3]{2})\,.$$

Subfields of $\mathbb{Q}(\zeta, \sqrt[3]{2})$

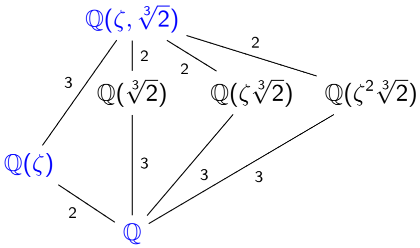
What are the subfields of

$$\mathbb{Q}(\zeta, \sqrt[3]{2}) = \{a + b\sqrt[3]{2} + c\sqrt[3]{4} + d\zeta + e\zeta\sqrt[3]{2} + f\zeta\sqrt[3]{4} : a, b, c, d, e, f \in \mathbb{Q}\}\,?$$

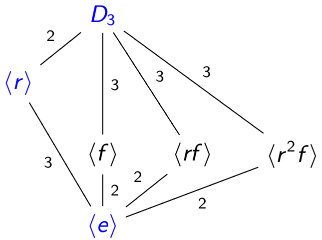
Note that $(\zeta^2)^2 = \zeta^4 = \zeta$, and so $\mathbb{Q}(\zeta^2) = \mathbb{Q}(\zeta) = \{a + b\zeta : a, b \in \mathbb{Q}\}$.

Similarly, $(\sqrt[3]{4})^2 = 2\sqrt[3]{2}$, and so $\mathbb{Q}(\sqrt[3]{4}) = \mathbb{Q}(\sqrt[3]{2}) = \{a + b\sqrt[3]{2} + c\sqrt[3]{4} : a, b, c \in \mathbb{Q}\}$.

There are two more subfields. As we did before, we can arrange them in a lattice:



Look familiar?



Compare this to the subgroup lattice of D_3.

Summary so far

Roughly speaking, a field is a group under both addition and multiplication (if we exclude 0), with the distributive law connecting these two operations.

We are mostly interested in the field $\mathbb{Q}$, and certain extension fields: $F \supseteq \mathbb{Q}$. Some of the extension fields we've encountered:

$$\mathbb{Q}(\sqrt{2}), \qquad \mathbb{Q}(i), \qquad \mathbb{Q}(\sqrt{2}, i), \qquad \mathbb{Q}(\sqrt{2}, \sqrt{3}), \qquad \mathbb{Q}(\zeta, \sqrt[3]{2}).$$

An automorphism of a field $F \supset \mathbb{Q}$ is a structure-preserving map that fixes $\mathbb{Q}$.

The set of all automorphisms of $F \supseteq \mathbb{Q}$ forms a group, called the Galois group of F, denoted $\mathrm{Gal}(F)$.

There is an intriguing but mysterious connection between subfields of F and subgroups of $\mathrm{Gal}(F)$. This is at the heart of Galois theory!

Something to ponder

How does this all relate to solving polynomials with radicals?

Polynomials

Definition

Let x be an unknown variable. A polynomial is a function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0 \,.$$

The highest non-zero power of n is called the degree of f.

We can assume that all of our coefficients a_i lie in a field F.

For example, if each $a_i \in \mathbb{Z}$ (not a field), we could alternatively say that $a_i \in \mathbb{Q}$.

Let $F[x]$ denote the set of polynomials with coefficients in F. We call this the set of polynomials over F.

Remark

Even though $\mathbb{Z}$ is not a field, we can still write $\mathbb{Z}[x]$ to be the set of polynomials with integer coefficients. Most polynomials we encounter have integer coeffients anyways.

Radicals

The roots of low-degree polynomials can be expressed using arithmetic and radicals.

For example, the roots of the polynomial $f(x) = 5x^4 - 18x^2 - 27$ are

$$x_{1,2} = \pm\sqrt{\frac{6\sqrt{6}+9}{5}}, \qquad x_{3,4} = \pm\sqrt{\frac{9-6\sqrt{6}}{5}}.$$

Remark

The operations of arithmetic, and radicals, are really the "only way" we have to write down generic complex numbers.

Thus, if there is some number that cannot be expressed using radicals, we have no way to express it, unless we invent a special symbol for it (e.g., π or e).

Even weirder, since a computer program is just a string of 0s and 1s, there are only countably infinite many possible programs.

Since $\mathbb{R}$ is an uncountable set, there are numbers (in fact, "almost all" numbers) that can *never* be expressed algorithmically by a computer program! Such numbers are called "uncomputable."

Algebraic numbers

Definition

A complex number is algebraic (over $\mathbb{Q}$) if it is the root of some polynomial in $\mathbb{Z}[x]$. The set $\mathbb{A}$ of all algebraic numbers forms a field (this is not immediately obvious).

A number that is not algebraic over $\mathbb{Q}$ (e.g., π, e) is called transcendental.

Every number that can be expressed from the natural numbers using arithmetic and radicals is algebraic. For example, consider

$$x = \sqrt[5]{1+\sqrt{-3}} \qquad \begin{aligned} &\iff \quad x^5 = 1+\sqrt{-3} \\ &\iff \quad x^5 - 1 = \sqrt{-3} \\ &\iff \quad (x^5-1)^2 = -3 \\ &\iff \quad x^{10} - 2x^5 + 4 = 0\,. \end{aligned}$$

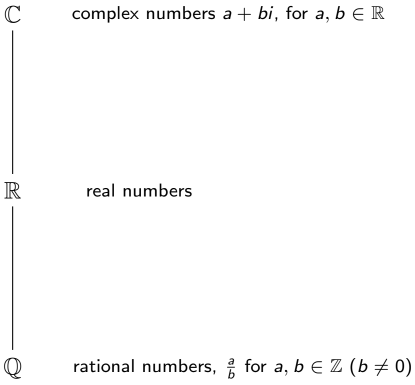
Question

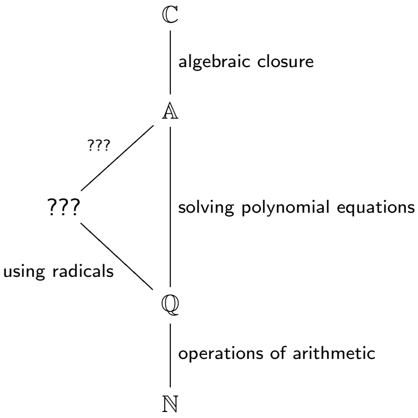
Can *all* algebraic numbers be expressed using radicals?

This question was unsolved until the early 1800s.

Hasse diagrams

The relationship between the natural numbers $\mathbb{N}$, and the fields $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{A}$, and $\mathbb{C}$, is shown in the following Hasse diagrams.





Some basic facts about the complex numbers

Definition

A field F is algebraically closed if for any polynomial $f(x) \in F[x]$, all of the roots of $f(x)$ lie in F.

Non-examples

- $\mathbb{Q}$ is not algebraically closed because $f(x) = x^2 - 2 \in \mathbb{Q}[x]$ has a root $\sqrt{2} \notin \mathbb{Q}$.
- $\mathbb{R}$ is not algebraically closed because $f(x) = x^2 + 1 \in \mathbb{R}[x]$ has a root $\sqrt{-1} \notin \mathbb{R}$.

Fundamental theorem of algebra

The field $\mathbb{C}$ is algebraically closed.

Thus, every polynomial $f(x) \in \mathbb{Z}[x]$ completely factors, or splits over $\mathbb{C}$:

$$f(x) = (x - r_1)(x - r_2) \cdots (x - r_n)\,, \qquad r_i \in \mathbb{C}\,.$$

Conversely, if F is *not* algebraically closed, then there are polynomials $f(x) \in F[x]$ that do *not* split into linear factors over F.

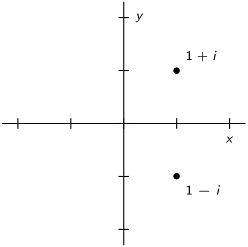
Complex conjugates

Recall that complex roots of $f(x) \in \mathbb{Q}[x]$ come in conjugate pairs: If $r = a + bi$ is a root, then so is $\overline{r} := a - bi$.

For example, here are the roots of some polynomials (degrees 2 through 5) plotted in the complex plane. All of them exhibit symmetry across the x-axis.

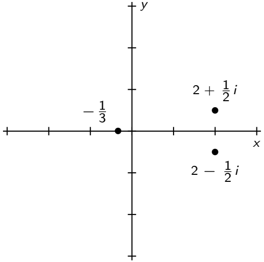
$f(x) = x^2 - 2x + 2$

Roots: $1 \pm i$



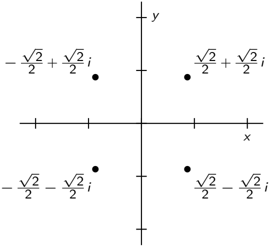
$f(x) = 12x^3 - 44x^2 + 35x + 17$

Roots: $-\frac{1}{3}$, $2 \pm \frac{1}{2}i$



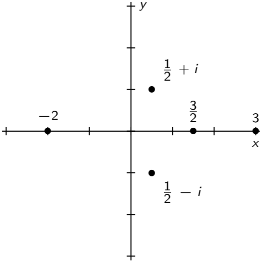
$f(x) = x^4 + 1$

Roots: $\pm \frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}$



$f(x) = 8x^5 - 28x^4 - 6x^3 + 83x^2 - 117x + 90$

Roots: -2, $\frac{3}{2}$, 3, $\frac{1}{2}i \pm i$



Irreducibility

Definition

A polynomial $f(x) \in F[x]$ is reducible over F if we can factor it as $f(x) = g(x)h(x)$ for some $g(x), h(x) \in F[x]$ of strictly lower degree. If $f(x)$ is not reducible, we say it is irreducible over F.

Examples

- $x^2 - x - 6 = (x+2)(x-3)$ is reducible over $\mathbb{Q}$.
- $x^4 + 5x^2 + 4 = (x^2+1)(x^2+4)$ is reducible over $\mathbb{Q}$, but it has no roots in $\mathbb{Q}$.
- $x^3 - 2$ is irreducible over $\mathbb{Q}$. If we could factor it, then one of the factors would have degree 1. But $x^3 - 2$ has no roots in $\mathbb{Q}$.

Facts

- If $\deg(f) > 1$ and has a root in F, then it is reducible over F.
- Every polynomial in $\mathbb{Z}[x]$ is reducible over $\mathbb{C}$.
- If $f(x) \in F[x]$ is a degree-2 or 3 polynomial, then $f(x)$ is reducible over F if and only if $f(x)$ has a root in F.

Eisenstein's criterion for irreducibility

Lemma

Let $f \in \mathbb{Z}[x]$ be irreducible. Then f is also irreducible over $\mathbb{Q}$.

Equivalently, if $f \in \mathbb{Z}[x]$ factors over $\mathbb{Q}$, then it factors over $\mathbb{Z}$.

Theorem (Eisenstein's criterion)

A polynomial $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0 \in \mathbb{Z}[x]$ is irreducible if for some prime p, the following all hold:

1. $p \nmid a_n$;
2. $p \mid a_k$ for $k = 0, \ldots, n-1$;
3. $p^2 \nmid a_0$.

For example, Eisenstein's criterion tells us that $x^{10} + 4x^7 + 18x + 14$ is irreducible.

Remark

If Eisenstein's criterion fails for all primes p, that does *not* necessarily imply that f is reducible. For example, $f(x) = x^2 + x + 1$ is irreducible over $\mathbb{Q}$, but Eisenstein cannot detect this.

Extension fields as vector spaces

Recall that a vector space over $\mathbb{Q}$ is a set of vectors V such that

- If $u, v \in V$, then $u + v \in V$ (closed under addition)
- If $v \in V$, then $cv \in V$ for all $c \in \mathbb{Q}$ (closed under scalar multiplication).

The field $\mathbb{Q}(\sqrt{2})$ is a 2-dimensional vector space over $\mathbb{Q}$:

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}.$$

This is why we say that $\{1, \sqrt{2}\}$ is a basis for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$.

Notice that the other field extensions we've seen are also vector spaces over $\mathbb{Q}$:

$$\begin{aligned} \mathbb{Q}(\sqrt{2}, i) &= \{a + b\sqrt{2} + ci + d\sqrt{2}i : a, b, c, d \in \mathbb{Q}\}, \\ \mathbb{Q}(\zeta, \sqrt[3]{2}) &= \{a + b\sqrt[3]{2} + c\sqrt[3]{4} + d\zeta + e\zeta\sqrt[3]{2} + f\zeta\sqrt[3]{4} : a, b, c, d, e, f \in \mathbb{Q}\}. \end{aligned}$$

As $\mathbb{Q}$-vector spaces, $\mathbb{Q}(\sqrt{2}, i)$ has dimension 4, and $\mathbb{Q}(\zeta, \sqrt[3]{2})$ has dimension 6.

Definition

If $F \subseteq E$ are fields, then the degree of the extension, denoted $[E : F]$, is the dimension of E as a vector space over F.

Equivalently, this is the number of terms in the expression for a general element for E using coefficients from F.

Minimial polynomials

Definition

Let $r \notin F$ be algebraic. The minimal polynomial of r over F is the irreducible polynomial in $F[x]$ of which r is a root. It is unique up to scalar multiplication.

Examples

- $\sqrt{2}$ has minimal polynomial $x^2 - 2$ over $\mathbb{Q}$, and $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.
- $i = \sqrt{-1}$ has minimal polynomial $x^2 + 1$ over $\mathbb{Q}$, and $[\mathbb{Q}(i) : \mathbb{Q}] = 2$.
- $\zeta = e^{2\pi i/3}$ has minimal polynomial $x^2 + x + 1$ over $\mathbb{Q}$, and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 2$.
- $\sqrt[3]{2}$ has minimal polynomial $x^3 - 2$ over $\mathbb{Q}$, and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.

What are the minimal polynomials of the following numbers over $\mathbb{Q}$?

$$-\sqrt{2}\,, \qquad -i\,, \qquad \zeta^2\,, \qquad \zeta\sqrt[3]{2}\,, \qquad \zeta^2\sqrt[3]{2}\,.$$

Degree theorem

The degree of the extension $\mathbb{Q}(r)$ is the degree of the minimal polynomial of r.

The Galois group of a polynomial

Definition

Let $f \in \mathbb{Z}[x]$ be a polynomial, with roots $r_1, \ldots, r_n$. The splitting field of f is the field

$$\mathbb{Q}(r_1, \ldots, r_n)\,.$$

The splitting field F of $f(x)$ has several equivalent characterizations:

- the smallest field that contains all of the roots of $f(x)$;
- the smallest field in which $f(x)$ splits into linear factors:

$$f(x) = (x - r_1)(x - r_2) \cdots (x - r_n) \in F[x]\,.$$

Recall that the Galois group of an extension $F \supseteq \mathbb{Q}$ is the group of automorphisms of F, denoted $\mathrm{Gal}(F)$.

Definition

The Galois group of a polynomial $f(x)$ is the Galois group of its splitting field, denoted $\mathrm{Gal}(f(x))$.

A few examples of Galois groups

- The polynomial $x^2 - 2$ splits in $\mathbb{Q}(\sqrt{2})$, so
$$\mathrm{Gal}(x^2 - 2) = \mathrm{Gal}(\mathbb{Q}(\sqrt{2})) \cong C_2 \, .$$
- The polynomial $x^2 + 1$ splits in $\mathbb{Q}(i)$, so
$$\mathrm{Gal}(x^2 + 1) = \mathrm{Gal}(\mathbb{Q}(i)) \cong C_2 \, .$$
- The polynomial $x^2 + x + 1$ splits in $\mathbb{Q}(\zeta)$, where $\zeta = e^{2\pi i/3}$, so
$$\mathrm{Gal}(x^2 + x + 1) = \mathrm{Gal}(\mathbb{Q}(\zeta)) \cong C_2 \, .$$
- The polynomial $x^3 - 1 = (x-1)(x^2 + x + 1)$ also splits in $\mathbb{Q}(\zeta)$, so
$$\mathrm{Gal}(x^3 - 1) = \mathrm{Gal}(\mathbb{Q}(\zeta)) \cong C_2 \, .$$
- The polynomial $x^4 - x^2 - 2 = (x^2 - 2)(x^2 + 1)$ splits in $\mathbb{Q}(\sqrt{2}, i)$, so
$$\mathrm{Gal}(x^4 - x^2 - 2) = \mathrm{Gal}(\mathbb{Q}(\sqrt{2}, i)) \cong V_4 \, .$$
- The polynomial $x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$ splits in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, so
$$\mathrm{Gal}(x^4 - 5x^2 + 6) = \mathrm{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})) \cong V_4 \, .$$
- The polynomial $x^3 - 2$ splits in $\mathbb{Q}(\zeta, \sqrt[3]{2})$, so
$$\mathrm{Gal}(x^3 - 2) = \mathrm{Gal}(\mathbb{Q}(\zeta, \sqrt[3]{2})) \cong D_3 \text{ ???}$$

The tower law of field extensions

Recall that if we had a chain of subgroups $K \leq H \leq G$, then the index satisfies a tower law: $[G : K] = [G : H][H : K]$.

Not surprisingly, the degree of field extensions obeys a similar tower law:

Theorem (Tower law)

For any chain of field extensions, $F \subset E \subset K$,

$$[K : F] = [K : E][E : F]\,.$$

We have already observed this in our subfield lattices:

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\underbrace{\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})}_{\text{min. poly: } x^2-3}][\underbrace{\mathbb{Q}(\sqrt{2}) : \mathbb{Q}}_{\text{min. poly: } x^2-2}] = 2 \cdot 2 = 4\,.$$

Here is another example:

$$[\mathbb{Q}(\zeta, \sqrt[3]{2}) : \mathbb{Q}] = [\underbrace{\mathbb{Q}(\zeta, \sqrt[3]{2}) : \mathbb{Q}(\sqrt[3]{2})}_{\text{min. poly: } x^2+x+1}][\underbrace{\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}}_{\text{min. poly: } x^3-2}] = 2 \cdot 3 = 6\,.$$

Primitive elements

Primitive element theorem

If F is an extension of $\mathbb{Q}$ with $[F:\mathbb{Q}]<\infty$, then F has a primitive element: some $\alpha \notin \mathbb{Q}$ for which $F=\mathbb{Q}(\alpha)$.

How do we find a primitive element α of $F=\mathbb{Q}(\zeta,\sqrt[3]{2})=\mathbb{Q}(i\sqrt{3},\sqrt[3]{2})$?

Let's try $\alpha = i\sqrt{3}\sqrt[3]{2} \in F$. Clearly, $[\mathbb{Q}(\alpha):\mathbb{Q}]\leq 6$. Observe that

$$\alpha^2=-3\sqrt[3]{4},\quad \alpha^3=-6i\sqrt{3},\quad \alpha^4=-18\sqrt[3]{2},\quad \alpha^5=18i\sqrt[3]{4}\sqrt{3},\quad \alpha^6=-108.$$

Thus, α is a root of x^6+108. The following are equivalent (why?):

(i) α is a primitive element of F;
(ii) $[\mathbb{Q}(\alpha):\mathbb{Q}]=6$;
(iii) the minimal polynomial $m(x)$ of α has degree 6;
(iv) x^6+108 is irreducible (and hence must be $m(x)$).

In fact, $[\mathbb{Q}(\alpha):\mathbb{Q}]=6$ holds because both 2 and 3 divide $[\mathbb{Q}(\alpha):\mathbb{Q}]$:

$$[\mathbb{Q}(\alpha):\mathbb{Q}]=[\mathbb{Q}(\alpha):\mathbb{Q}(i\sqrt{3})]\underbrace{[\mathbb{Q}(i\sqrt{3}):\mathbb{Q}]}_{=2},\qquad [\mathbb{Q}(\alpha):\mathbb{Q}]=[\mathbb{Q}(\alpha):\mathbb{Q}(\sqrt[3]{2})]\underbrace{[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}]}_{=3}.$$

An example: The Galois group of $x^4 - 5x^2 + 6$

The polynomial $f(x) = (x^2 - 2)(x^2 - 3) = x^4 - 5x^2 + 6$ has splitting field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

We already know that its Galois group should be V_4. Let's compute it explicitly; this will help us understand it better.

We need to determine all automorphisms ϕ of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. We know:

- ϕ is determined by where it sends the basis elements $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.
- ϕ must fix 1.
- If we know where ϕ sends two of $\{\sqrt{2}, \sqrt{3}, \sqrt{6}\}$, then we know where it sends the third, because
$$\phi(\sqrt{6}) = \phi(\sqrt{2}\sqrt{3}) = \phi(\sqrt{2})\,\phi(\sqrt{3})\,.$$

In addition to the identity automorphism e, we have

$$\begin{cases} \phi_2(\sqrt{2}) = -\sqrt{2} \\ \phi_2(\sqrt{3}) = \sqrt{3} \end{cases} \qquad \begin{cases} \phi_3(\sqrt{2}) = \sqrt{2} \\ \phi_3(\sqrt{3}) = -\sqrt{3} \end{cases} \qquad \begin{cases} \phi_4(\sqrt{2}) = -\sqrt{2} \\ \phi_4(\sqrt{3}) = -\sqrt{3} \end{cases}$$

Question

What goes wrong if we try to make $\phi(\sqrt{2}) = \sqrt{3}$?

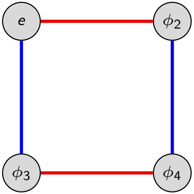
An example: The Galois group of $x^4 - 5x^2 + 6$

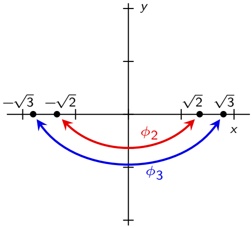
There are 4 automorphisms of $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, the splitting field of $x^4 - 5x^2 + 6$:

$$\begin{aligned}
e\colon a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \quad &\longmapsto \quad a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \\
\phi_2\colon a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \quad &\longmapsto \quad a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6} \\
\phi_3\colon a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \quad &\longmapsto \quad a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6} \\
\phi_4\colon a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \quad &\longmapsto \quad a - b\sqrt{2} - c\sqrt{3} + d\sqrt{6}
\end{aligned}$$

They form the Galois group of $x^4 - 5x^2 + 6$. The multiplication table and Cayley diagram are shown below.

	e	ϕ_2	ϕ_3	ϕ_4
e	e	ϕ_2	ϕ_3	ϕ_4
ϕ_2	ϕ_2	e	ϕ_4	ϕ_3
ϕ_3	ϕ_3	ϕ_4	e	ϕ_2
ϕ_4	ϕ_4	ϕ_3	ϕ_2	e





Remarks

- $\alpha = \sqrt{2} + \sqrt{3}$ is a primitive element of F, i.e., $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.
- There is a group action of $\mathrm{Gal}(f(x))$ on the set of roots $S = \{\pm\sqrt{2}, \pm\sqrt{3}\}$ of $f(x)$.

The Galois group acts on the roots

Theorem

If $f \in \mathbb{Z}[x]$ is a polynomial with a root in a field extension F of $\mathbb{Q}$, then any automorphism of F permutes the roots of f.

Said differently, we have a group action of $\mathrm{Gal}(f(x))$ on the set $S = \{r_1, \ldots, r_n\}$ of roots of $f(x)$.

That is, we have a homomorphism

$$\psi\colon\ \mathrm{Gal}(f(x)) \longrightarrow \mathrm{Perm}(\{r_1, \ldots, r_n\})\,.$$

If $\phi \in \mathrm{Gal}(f(x))$, then $\psi(\phi)$ is a permutation of the roots of $f(x)$.

This permutation is what results by "pressing the ϕ-button" – it permutes the roots of $f(x)$ via the automorphism ϕ of the splitting field of $f(x)$.

Corollary

If the degree of $f \in \mathbb{Z}[x]$ is n, then the Galois group of f is a subgroup of S_n.

The Galois group acts on the roots

The next results says that "$\mathbb{Q}$ can't tell apart the roots of an irreducible polynomial."

The "One orbit theorem"

Let r_1 and r_2 be roots of an irreducible polynomial over $\mathbb{Q}$. Then

(a) There is an isomorphism $\phi\colon \mathbb{Q}(r_1) \longrightarrow \mathbb{Q}(r_2)$ that fixes $\mathbb{Q}$ and with $\phi(r_1) = r_2$.

(b) This remains true when $\mathbb{Q}$ is replaced with any extension field F, where $\mathbb{Q} \subset F \subset \mathbb{C}$.

Corollary

If $f(x)$ is irreducible over $\mathbb{Q}$, then for any two roots r_1 and r_2 of $f(x)$, the Galois group $\mathrm{Gal}(f(x))$ contains an automorphism $\phi\colon r_1 \longmapsto r_2$.

In other words, if $f(x)$ is irreducible, then the action of $\mathrm{Gal}(f(x))$ on the set $S = \{r_1, \ldots, r_n\}$ of roots has only one orbit.

Normal field extensions

Definition

An extension field E of F is normal if it is the splitting field of some polynomial $f(x)$.

If E is a normal extension over F, then every irreducible polynomial in $F[x]$ that has a root in E splits over F.

Thus, if you can find an irreducible polynomial that has *one, but not all of its roots* in E, then E is *not* a normal extension.

Normal extension theorem

The degree of a normal extension is the order of its Galois group.

Corollary

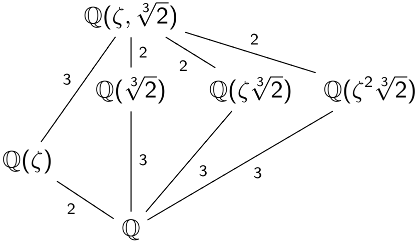
The order of the Galois group of a polynomial $f(x)$ is the degree of the extension of its splitting field over $\mathbb{Q}$.

Normal field extensions: Examples

Consider $\mathbb{Q}(\zeta, \sqrt[3]{2}) = \mathbb{Q}(\alpha)$, the splitting field of $f(x) = x^3 - 2$.

It is also the splitting field of $m(x) = x^6 + 108$, the minimal polynomial of $\alpha = \sqrt[3]{2}\sqrt{-3}$.

Let's see which of its intermediate subfields are normal extensions of $\mathbb{Q}$.



- $\mathbb{Q}$: Trivially normal.
- $\mathbb{Q}(\zeta)$: Splitting field of $x^2 + x + 1$; roots are $\zeta, \zeta^2 \in \mathbb{Q}(\zeta)$. Normal.
- $\mathbb{Q}(\sqrt[3]{2})$: Contains only one root of $x^3 - 2$, not the other two. Not normal.
- $\mathbb{Q}(\zeta\sqrt[3]{2})$: Contains only one root of $x^3 - 2$, not the other two. Not normal.
- $\mathbb{Q}(\zeta^2\sqrt[3]{2})$: Contains only one root of $x^3 - 2$, not the other two. Not normal.
- $\mathbb{Q}(\zeta, \sqrt[3]{2})$: Splitting field of $x^3 - 2$. Normal.

By the normal extension theorem,

$$|\operatorname{Gal}(\mathbb{Q}(\zeta))| = [\mathbb{Q}(\zeta) : \mathbb{Q}] = 2\,, \qquad |\operatorname{Gal}(\mathbb{Q}(\zeta, \sqrt[3]{2}))| = [\mathbb{Q}(\zeta, \sqrt[3]{2}) : \mathbb{Q}] = 6\,.$$

Moreover, you can check that $|\operatorname{Gal}(\mathbb{Q}(\sqrt[3]{2}))| = 1 < [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.

The Galois group of $x^3 - 2$

We can now conclusively determine the Galois group of $x^3 - 2$.

By definition, the Galois group of a polynomial is the Galois group of its splitting field, so $\mathrm{Gal}(x^3 - 2) = \mathrm{Gal}(\mathbb{Q}(\zeta, \sqrt[3]{2}))$.

By the normal extension theorem, the order of the Galois group of $f(x)$ is the degree of the extension of its splitting field:

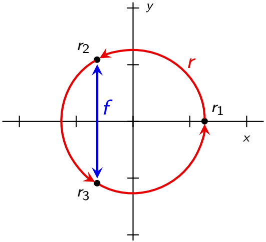
$$|\,\mathrm{Gal}(\mathbb{Q}(\zeta, \sqrt[3]{2}))| = [\mathbb{Q}(\zeta, \sqrt[3]{2}) : \mathbb{Q}] = 6\,.$$

Since the Galois group acts on the roots of $x^3 - 2$, it must be a subgroup of $S_3 \cong D_3$.

There is only one subgroup of S_3 of order 6, so $\mathrm{Gal}(x^3 - 2) \cong S_3$. Here is the action diagram of $\mathrm{Gal}(x^3 - 2)$ acting on the set $S = \{r_1, r_2, r_3\}$ of roots of $x^3 - 2$:

$$\begin{cases} r\colon \sqrt[3]{2} \longmapsto \zeta\sqrt[3]{2} \\ r\colon \zeta \longmapsto \zeta \end{cases}$$

$$\begin{cases} f\colon \sqrt[3]{2} \longmapsto \sqrt[3]{2} \\ f\colon \zeta \longmapsto \zeta^2 \end{cases}$$



Paris, May 31, 1832

The night before a duel that Évariste Galois knew he would lose, the 20-year-old stayed up late preparing his mathematical findings in a letter to Auguste Chevalier.

Hermann Weyl (1885–1955) said "*This letter, if judged by the novelty and profundity of ideas it contains, is perhaps the most substantial piece of writing in the whole literature of mankind.*"

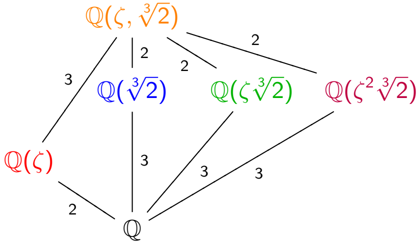
Fundamental theorem of Galois theory

Given $f \in \mathbb{Z}[x]$, let F be the splitting field of f, and G the Galois group. Then the following hold:

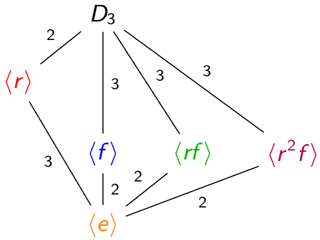
(a) The subgroup lattice of G is identical to the subfield lattice of F, but upside-down. Moreover, $H \lhd G$ if and only if the corresponding subfield is a normal extension of $\mathbb{Q}$.

(b) Given an intermediate field $\mathbb{Q} \subset K \subset F$, the corresponding subgroup $H < G$ contains precisely those automorphisms that fix K.

An example: the Galois correspondence for $f(x) = x^3 - 2$



Subfield lattice of $\mathbb{Q}(\zeta, \sqrt[3]{2})$



Subgroup lattice of $\mathrm{Gal}(\mathbb{Q}(\zeta, \sqrt[3]{2})) \cong D_3$.

- The automorphisms that fix $\mathbb{Q}$ are precisely those in D_3.
- The automorphisms that fix $\mathbb{Q}(\zeta)$ are precisely those in $\langle r \rangle$.
- The automorphisms that fix $\mathbb{Q}(\sqrt[3]{2})$ are precisely those in $\langle f \rangle$.
- The automorphisms that fix $\mathbb{Q}(\zeta\sqrt[3]{2})$ are precisely those in $\langle rf \rangle$.
- The automorphisms that fix $\mathbb{Q}(\zeta^2\sqrt[3]{2})$ are precisely those in $\langle r^2 f \rangle$.
- The automorphisms that fix $\mathbb{Q}(\zeta, \sqrt[3]{2})$ are precisely those in $\langle e \rangle$.

The normal field extensions of $\mathbb{Q}$ are: $\mathbb{Q}$, $\mathbb{Q}(\zeta)$, and $\mathbb{Q}(\zeta, \sqrt[3]{2})$.

The normal subgroups of D_3 are: D_3, $\langle r \rangle$ and $\langle e \rangle$.

An example: the Galois correspondence for $f(x) = x^8 - 2$

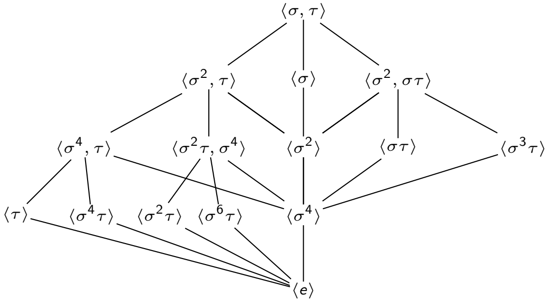
The splitting field of $x^8 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[8]{2}, i)$, a degree-16 extension over $\mathbb{Q}$. Its Galois group is the quasidihedral group $G = QD_8$:

$$QD_8 = \langle \sigma, \tau \mid \sigma^8 = 1,\ \tau^2 = 1,\ \sigma\tau = \tau\sigma^3 \rangle .$$

Let $\zeta = e^{2\pi i/8}$

$$\sqrt[8]{2} \overset{\sigma}{\longmapsto} \zeta\sqrt[8]{2}$$
$$i \longmapsto i$$

$$\sqrt[8]{2} \overset{\tau}{\longmapsto} \sqrt[8]{2}$$
$$i \longmapsto -i$$



Exercise

The subfields of $\mathbb{Q}(\sqrt[8]{2}, i)$ are: $\mathbb{Q}$, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt[4]{2})$, $\mathbb{Q}(\sqrt[8]{2})$, $\mathbb{Q}(\sqrt{2}i)$, $\mathbb{Q}(\sqrt[4]{2}i)$, $\mathbb{Q}(\sqrt[8]{2}i)$, $\mathbb{Q}(\sqrt{2}, i)$, $\mathbb{Q}(\sqrt[4]{2}, i)$, $\mathbb{Q}((1+i)\sqrt[4]{2})$, $\mathbb{Q}((1-i)\sqrt[4]{2})$, $\mathbb{Q}(\zeta\sqrt[8]{2})$, $\mathbb{Q}(\zeta^3\sqrt[8]{2})$. Construct the subfield lattice.

Solvability

Definition

A group G is solvable if it has a chain of subgroups:

$$\{e\} = N_0 \lhd N_1 \lhd N_2 \lhd \cdots \lhd N_{k-1} \lhd N_k = G\,.$$

such that each quotient N_i/N_{i-1} is abelian.

Note: Each subgroup N_i need not be normal in G, just in N_{i+1}.

Examples

- $D_4 = \langle r, f \rangle$ is solvable. There are many possible chains:

$$\langle e \rangle \lhd \langle f \rangle \lhd \langle r^2, f \rangle \lhd D_4\,, \qquad \langle e \rangle \lhd \langle r \rangle \lhd D_4\,, \qquad \langle e \rangle \lhd \langle r^2 \rangle \lhd D_4.$$

- Any abelian group A is solvable: take $N_0 = \{e\}$ and $N_1 = A$.

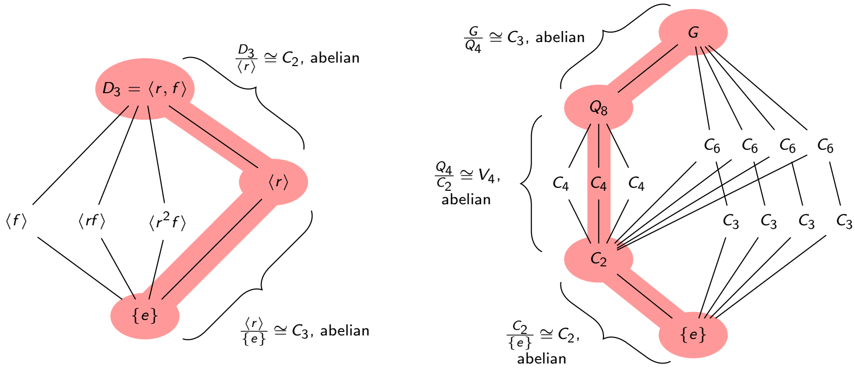
- For $n \geq 5$, the group A_n is simple and non-abelian. Thus, the only chain of normal subgroups is

$$N_0 = \{e\} \lhd A_n = N_1\,.$$

  Since $N_1/N_0 \cong A_n$ is non-abelian, A_n is not solvable for $n \geq 5$.

Some more solvable groups

$D_3 \cong S_3$ is solvable: $\{e\} \lhd \langle r \rangle \lhd D_3$.



The group above at right is denoted $G = \mathrm{SL}(2,3)$. It consists of all 2×2 matrices with determinant 1 over the field $\mathbb{Z}_3 = \{0, 1, -1\}$.

$\mathrm{SL}(2,3)$ has order 24, and is the smallest solvable group that requires a three-step chain of normal subgroups.

The hunt for an unsolvable polynomial

The following lemma follows from the Correspondence Theorem. (Why?)

Lemma

If $N \lhd G$, then G is solvable if and only if both N and G/N are solvable.

Corollary

S_n is not solvable for all $n \geq 5$. (Since $A_n \lhd S_n$ is not solvable).

Galois' theorem

A field extension $E \supset \mathbb{Q}$ contains only elements expressible by radicals if and only if its Galois group is solvable.

Corollary

$f(x)$ is solvable by radicals if and only if it has a solvable Galois group.

Thus, any polynomial with Galois group S_5 is not solvable by radicals!

An unsolvable quintic!

To find a polynomial not solvable by radicals, we'll look for a polynomial $f(x)$ with $\text{Gal}(f(x)) \cong S_5$.

We'll restrict our search to degree-5 polynomials, because $\text{Gal}(f(x)) \leq S_5$ for any degree-5 polynomial $f(x)$.

Key observation

Recall that for any 5-cycle σ and 2-cycle (=transposition) τ,

$$S_5 = \langle \sigma, \tau \rangle .$$

Moreover, the *only* elements in S_5 of order 5 are 5-cycles, e.g., $\sigma = (a\ b\ c\ d\ e)$.

Let $f(x) = x^5 + 10x^4 - 2$. It is irreducible by Eisenstein's criterion (use $p = 2$). Let $F = \mathbb{Q}(r_1, \ldots, r_5)$ be its splitting field.

Basic calculus tells us that f exactly has 3 real roots. Let $r_1, r_2 = a \pm bi$ be the complex roots, and r_3, r_4, and r_5 be the real roots.

Since f has distinct complex conjugate roots, complex conjugation is an automorphism $\tau \colon F \longrightarrow F$ that transposes r_1 with r_2, and fixes the three real roots.

An unsolvable quintic!

We just found our transposition $\tau = (r_1\ r_2)$. All that's left is to find an element (i.e., an automorphism) σ of order 5.

Take any root r_i of $f(x)$. Since $f(x)$ is irreducible, it is the minimal polynomial of r_i. By the Degree Theorem,

$$[\mathbb{Q}(r_i) : \mathbb{Q}] = \deg(\text{minimum polynomial of } r_i) = \deg f(x) = 5\,.$$

The splitting field of $f(x)$ is $F = \mathbb{Q}(r_1, \dots, r_5)$, and by the normal extension theorem, the degree of this extension over $\mathbb{Q}$ is the order of the Galois group $\mathrm{Gal}(f(x))$.

Applying the tower law to this yields

$$|\,\mathrm{Gal}(f(x))| = [\mathbb{Q}(r_1, r_2, r_3, r_4, r_5) : \mathbb{Q}] = [\mathbb{Q}(r_1, r_2, r_3, r_4, r_5) : \mathbb{Q}(r_1)]\,\underbrace{[\mathbb{Q}(r_1) : \mathbb{Q}]}_{=5}.$$

Thus, $|\,\mathrm{Gal}(f(x))|$ is a multiple of 5, so Cauchy's theorem guarantees that G has an element σ of order 5.

Since $\mathrm{Gal}(f(x))$ has a 2-cycle τ and a 5-cycle σ, it must be all of S_5.

$\mathrm{Gal}(f(x))$ is an unsolvable group, so $f(x) = x^5 + 10x^4 - 2$ is unsolvable by radicals!

Summary of Galois' work

Let $f(x)$ be a degree-n polynomial in $\mathbb{Z}[x]$ (or $\mathbb{Q}[x]$). The roots of $f(x)$ lie in some splitting field $F \supseteq \mathbb{Q}$.

The Galois group of $f(x)$ is the automorphism group of F. Every such automorphism fixes $\mathbb{Q}$ and permutes the roots of $f(x)$.

This is a group action of $\mathrm{Gal}(f(x))$ on the set of n roots! Thus, $\mathrm{Gal}(f(x)) \leq S_n$.

There is a 1–1 correspondence between subfields of F and subgroups of $\mathrm{Gal}(f(x))$.

A polynomial is solvable by radicals iff its Galois group is a solvable group.

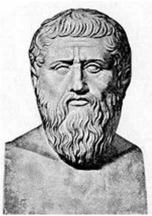
The symmetric group S_5 is not a solvable group.

Since $S_5 = \langle \tau, \sigma \rangle$ for a 2-cycle τ and 5-cycle σ, all we need to do is find a degree-5 polynomial whose Galois group contains a 2-cycle and an element of order 5.

If $f(x)$ is an irreducible degree-5 polynomial with 3 real roots, then complex conjugation is an automorphism that transposes the 2 complex roots. Moreover, Cauchy's theorem tells us that $\mathrm{Gal}(f(x))$ must have an element of order 5.

Thus, $f(x) = x^5 + 10x^4 - 2$ is not solvable by radicals!

Geometry and the Ancient Greeks



Plato (5th century B.C.) believed that the only "perfect" geometric figures were the straight line and the circle.

In Ancient Greek geometry, this philosophy meant that there were only two instruments available to perform geometric constructions:

1. the ruler: a single unmarked straight edge.
2. the compass: collapses when lifted from the page

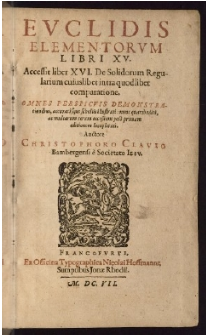
Formally, this means that the only permissible constructions are those granted by Euclid's first three postulates.

Geometry and the Ancient Greeks

Around 300 BC, ancient Greek mathematician Euclid wrote a series of thirteen books that he called The Elements.

It is a collection of definitions, postulates (axioms), and theorems & proofs, covering geometry, elementary number theory, and the Greek's "geometric algebra."

Book 1 contained Euclid's famous *10 postulates*, and other basic propositions of geometry.



Euclid's first three postulates

1. A straight line segment can be drawn joining any two points.
2. Any straight line segment can be extended indefinitely in a straight line.
3. Given any straight line segment, a circle can be drawn having the segment as radius and one endpoint as center.

Using only these tools, lines can be divided into equal segments, angles can be bisected, parallel lines can be drawn, n-gons can be "squared," and so on.

Geometry and the Ancient Greeks

One of the chief purposes of Greek mathematics was to find exact constructions for various lengths, using only the basic tools of a ruler and compass.

The ancient Greeks were unable to find constructions for the following problems:

Problem 1: Squaring the circle

Draw a square with the same area as a given circle.

Problem 2: Doubling the cube

Draw a cube with twice the volume of a given cube.

Problem 3: Trisecting an angle

Divide an angle into three smaller angles all of the same size.

For over 2000 years, these problems remained unsolved.

Alas, in 1837, Pierre Wantzel used field theory to prove that these constructions were impossible.

What does it mean to be "constructible"?

Assume P_0 is a set of points in $\mathbb{R}^2$ (or equivalently, in the complex plane $\mathbb{C}$).

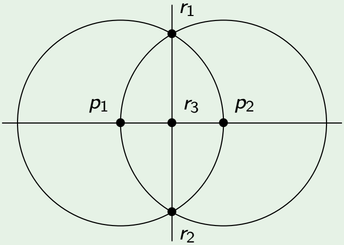
Definition

The points of intersection of any two distinct lines or circles are constructible in one step.

A point $r \in \mathbb{R}^2$ is constructible from P_0 if there is a finite sequence $r_1, \ldots, r_n = r$ of points in $\mathbb{R}^2$ such that for each $i = 1, \ldots, n$, the point r_i is constructible in one step from $P_0 \cup \{r_1, \ldots, r_{i-1}\}$.

Example: bisecting a line

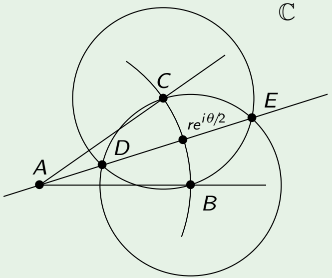
1. Start with a line p_1p_2;
2. Draw the circle of center p_1 of radius p_1p_2;
3. Draw the circle of center p_2 of radius p_1p_2;
4. Let r_1 and r_2 be the points of intersection;
5. Draw the line r_1r_2;
6. Let r_3 be the intersection of p_1p_2 and r_1r_2.



Bisecting an angle

Example: bisecting an angle

1. Start with an angle at A;
2. Draw a circle centered at A;
3. Let B and C be the points of intersection;
4. Draw a circle of radius BC centered at B;
5. Draw a circle of radius BC centered at C;
6. Let D and E be the intersections of these 2 circles;
7. Draw a line through DE.



Suppose A is at the origin in the complex plane. Then $B = r$ and $C = re^{i\theta}$.

Bisecting an angle means that we can construct $re^{i\theta/2}$ from $re^{i\theta}$.

Constructible numbers: Real vs. complex

Henceforth, we will say that a point is constructible if it is constructible from the set

$$P_0 = \{(0,0),\ (1,0)\} \subset \mathbb{R}^2 .$$

Say that $z = x + yi \in \mathbb{C}$ is constructible if $(x, y) \in \mathbb{R}^2$ is constructible. Let $K \subseteq \mathbb{C}$ denote the constructible numbers.

Lemma

A complex number $z = x + yi$ is constructible if x and y are constructible.

By the following lemma, we can restrict our focus on real constructible numbers.

Lemma

1. $K \cap \mathbb{R}$ is a subfield of $\mathbb{R}$ if and only if K is a subfield of $\mathbb{C}$.
2. Moreover, $K \cap \mathbb{R}$ is closed under (nonnegative) square roots if and only if K is closed under (all) square roots.

$K \cap \mathbb{R}$ closed under square roots means that $a \in K \cap \mathbb{R}^+$ implies $\sqrt{a} \in K \cap \mathbb{R}$.

K closed under square roots means that $z = re^{i\theta} \in K$ implies $\sqrt{z} = \sqrt{r}e^{i\theta/2} \in K$.

The field of constructible numbers

Theorem

The set of constructible numbers K is a subfield of $\mathbb{C}$ that is closed under taking square roots and complex conjugation.

Proof (sketch)

Let a and b be constructible real numbers, with $a > 0$. It is elementary to check that each of the following hold:

1. $-a$ is constructible;
2. $a + b$ is constructible;
3. ab is constructible;
4. a^{-1} is constructible;
5. $\sqrt{a}$ is constructible;
6. $a - bi$ is constructible provided that $a + bi$ is.

Corollary

If $a, b, c \in \mathbb{C}$ are constructible, then so are the roots of $ax^2 + bx + c$.

Constructions as field extensions

Let $F \subset K$ be a field generated by ruler and compass constructions.

Suppose α is constructible from F in one step. We wish to determine $[F(\alpha) : F]$.

The three ways to construct new points from F

1. Intersect two lines. The solution to $ax + by = c$ and $dx + ey = f$ lies in F.
2. Intersect a circle and a line. The solution to
$$\begin{cases} ax + by = c \\ (x-d)^2 + (y-e)^2 = r^2 \end{cases}$$
lies in (at most) a quadratic extension of F.
3. Intersect two circles. We need to solve the system
$$\begin{cases} (x-a)^2 + (y-b)^2 = s^2 \\ (x-d)^2 + (y-e)^2 = r^2 \end{cases}$$
Multiply this out and subtract. The x^2 and y^2 terms cancel, leaving the equation of a line. Intersecting this line with one of the circles puts us back in Case 2.

In all of these cases, $[F(\alpha) : F] \leq 2$.

Constructions as field extensions

In others words, constructing a number $\alpha \notin F$ in one step amounts to taking a degree-2 extension of F.

Theorem

A complex number α is constructible if and only if there is a tower of field extensions

$$\mathbb{Q} = K_0 \subset K_1 \subset \cdots \subset K_n \subset \mathbb{C}$$

where $\alpha \in K_n$ and $[K_{i+1} : K_i] \leq 2$ for each i.

Corollary

The set $K \subset \mathbb{C}$ of constructible numbers is a field. Moreover, if $\alpha \in K$, then $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^n$ for some integer n.

Next, we will show that the ancient Greeks' classical construction problems are impossible by demonstrating that each would yield a number $\alpha \in \mathbb{R}$ such that $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is not a power of two.

Classical constructibility problems, rephrased

Problem 1: Squaring the circle

Given a circle of radius r (and hence of area πr^2), construct a square of area πr^2 (and hence of side-length $\sqrt{\pi} r$).

If one could square the circle, then $\sqrt{\pi} \in K \subset \mathbb{C}$, the field of constructible numbers.

However,

$$\mathbb{Q} \subset \mathbb{Q}(\pi) \subset \mathbb{Q}(\sqrt{\pi})$$

and so $[\mathbb{Q}(\sqrt{\pi}) : \mathbb{Q}] \geq [\mathbb{Q}(\pi) : \mathbb{Q}] = \infty$. Hence $\sqrt{\pi}$ is not constructible.

Problem 2: Doubling the cube

Given a cube of length ℓ (and hence of volume ℓ^3), construct a cube of volume $2\ell^3$ (and hence of side-length $\sqrt[3]{2}\ell$).

If one could double the cube, then $\sqrt[3]{2} \in K$.

However, $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ is not a power of two. Hence $\sqrt[3]{2}$ is not constructible.

Classical constructibility problems, rephrased

Problem 3: Trisecting an angle

Given $e^{i\theta}$, construct $e^{i\theta/3}$. Or equivalently, construct $\cos(\theta/3)$ from $\cos(\theta)$.

We will show that $\theta = 60^\circ$ cannot be trisected. In other words, that $\alpha = \cos(20^\circ)$ cannot be constructed from $\cos(60^\circ)$.

The triple angle formula yields

$$\cos(\theta) = 4\cos^3(\theta/3) - 3\cos(\theta/3)\,.$$

Set $\theta = 60^\circ$. Plugging in $\cos(\theta) = 1/2$ and $\alpha = \cos(20^\circ)$ gives

$$4\alpha^3 - 3\alpha - \frac{1}{2} = 0\,.$$

Changing variables by $u = 2\alpha$, and then multiplying through by 2:

$$u^3 - 3u - 1 = 0\,.$$

Thus, u is the root of the (irreducible!) polynomial $x^3 - 3x - 1$. Therefore, $[\mathbb{Q}(u) : \mathbb{Q}] = 3$, which is not a power of 2.

Hence, $u = 2\cos(20^\circ)$ is not constructible, so neither is $\alpha = \cos(20^\circ)$.

Classical constructibility problems, resolved

The three classical ruler-and-compass constructions that stumped the ancient Greeks, when translated in the language of field theory, are as follows:

Problem 1: Squaring the circle

Construct $\sqrt{\pi}$ from 1.

Problem 2: Doubling the cube

Construct $\sqrt[3]{2}$ from 1.

Problem 3: Trisecting an angle

Construct $\cos(\theta/3)$ from $\cos(\theta)$. [Or $\cos(20^\circ)$ from 1.]

Since none of these numbers these lie in an extension of $\mathbb{Q}$ of degree 2^n, they are not constructible.

If one is allowed a "marked ruler," then these constructions become possible, which the ancient Greeks were aware of.

Construction of regular polygons

The ancient Greeks were also interested in constructing regular polygons. They knew constructions for 3-, 5-, and 15-gons.

In 1796, nineteen-year-old Carl Friedrich Gauß, who was undecided about whether to study mathematics or languages, discovered how to construct a regular 17-gon.

Gauß was so pleased with his discovery that he dedicated his life to mathematics.

He also proved the following theorem about which n-gons are constructible.

Theorem (Gauß, Wantzel)

Let p be an odd prime. A regular p-gon is constructible if and only if $p = 2^{2^n} + 1$ for some $n \geq 0$.

The next question to ask is for which n is $2^{2^n} + 1$ prime?

Construction of regular polygons and Fermat primes

Definition

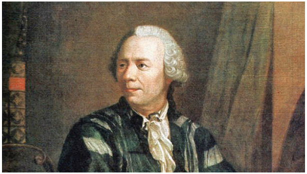
The n^{th} Fermat number is $F_n := 2^{2^n} + 1$. If F_n is prime, then it is a Fermat prime.

The first few Fermat primes are $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$, and $F_4 = 65537$.

They are named after Pierre Fermat (1601–1665), who conjectured in the 1600s that all Fermat numbers $F_n = 2^{2^n} + 1$ are prime.

Construction of regular polygons and Fermat primes

In 1732, Leonhard Euler disproved Fermat's conjecture by demonstrating



$$F_5 = 2^{2^5}+1 = 2^{32}+1 = 4294967297 = 641 \cdot 6700417\,.$$

It is not known if any other Fermat primes exist!

So far, every F_n is known to be composite for $5 \leq n \leq 32$. In 2014, a computer showed that $193 \times 2^{3329782} + 1$ is a prime factor of

$$F_{3329780} = 2^{2^{3329780}} + 1 > 10^{10^{10^6}}\,.$$

Theorem (Gauß, Wantzel)

A regular n-gon is constructible if and only if $n = 2^k p_1 \cdots p_m$, where $p_1, \ldots, p_m$ are distinct Fermat primes.

If these type of problems interest you, take Math 4100! (Number theory)

Section 7: Ring theory

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Math 4120, Modern Algebra

What is a ring?

Definition

A ring is an additive (abelian) group R with an additional binary operation (multiplication), satisfying the distributive law:

$$x(y+z) = xy + xz \qquad \text{and} \qquad (y+z)x = yx + zx \quad \forall x, y, z \in R\,.$$

Remarks

- There need not be multiplicative inverses.
- Multiplication need not be commutative (it may happen that $xy \neq yx$).

A few more terms

If $xy = yx$ for all $x, y \in R$, then R is commutative.

If R has a multiplicative identity $1 = 1_R \neq 0$, we say that "R has identity" or "unity", or "R is a ring with 1."

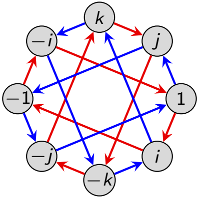
A subring of R is a subset $S \subseteq R$ that is also a ring.

What is a ring?

Examples

1. $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ are all commutative rings with 1.
2. $\mathbb{Z}_n$ is a commutative ring with 1.
3. For any ring R with 1, the set $M_n(R)$ of $n \times n$ matrices over R is a ring. It has identity $1_{M_n(R)} = I_n$ iff R has 1.
4. For any ring R, the set of functions $F = \{f : R \to R\}$ is a ring by defining
$$(f+g)(r) = f(r) + g(r), \qquad (fg)(r) = f(r)g(r)\,.$$
5. The set $S = 2\mathbb{Z}$ is a subring of $\mathbb{Z}$ but it does *not* have 1.
6. $S = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} : a \in \mathbb{R} \right\}$ is a subring of $R = M_2(\mathbb{R})$. However, note that
$$1_R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \text{but} \qquad 1_S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$
7. If R is a ring and x a variable, then the set
$$R[x] = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in R\}$$
is called the polynomial ring over R.

Another example: the quaternions



Recall the (unit) quaternion group:

$$Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = -1,\ ij = k \rangle.$$

Allowing addition makes them into a ring $\mathbb{H}$, called the quaternions, or Hamiltonians:

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}.$$

The set $\mathbb{H}$ is isomorphic to a subring of $M_4(\mathbb{R})$, the real-valued 4×4 matrices:

$$\mathbb{H} = \left\{ \begin{bmatrix} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & -c & b & a \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\} \subseteq M_4(\mathbb{R}).$$

Formally, we have an embedding $\phi \colon \mathbb{H} \hookrightarrow M_4(\mathbb{R})$ where

$$\phi(i) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \phi(j) = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \quad \phi(k) = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

We say that $\mathbb{H}$ is represented by a set of matrices.

Units and zero divisors

Definition

Let R be a ring with 1. A unit is any $x \in R$ that has a multiplicative inverse. Let $U(R)$ be the set (a multiplicative group) of units of R.

An element $x \in R$ is a left zero divisor if $xy = 0$ for some $y \neq 0$. (Right zero divisors are defined analogously.)

Examples

1. Let $R = \mathbb{Z}$. The units are $U(R) = \{-1, 1\}$. There are no (nonzero) zero divisors.
2. Let $R = \mathbb{Z}_{10}$. Then 7 is a unit (and $7^{-1} = 3$) because $7 \cdot 3 = 1$. However, 2 is not a unit.
3. Let $R = \mathbb{Z}_n$. A nonzero $k \in \mathbb{Z}_n$ is a unit if $\gcd(n, k) = 1$, and a zero divisor if $\gcd(n, k) \geq 2$.
4. The ring $R = M_2(\mathbb{R})$ has zero divisors, such as:

$$\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The groups of units of $M_2(\mathbb{R})$ are the invertible matrices.

Group rings

Let R be a commutative ring (usually, $\mathbb{Z}$, $\mathbb{R}$, or $\mathbb{C}$) and G a finite (multiplicative) group. We can define the group ring RG as

$$RG := \{a_1 g_1 + \cdots + a_n g_n \mid a_i \in R,\ g_i \in G\},$$

where multiplication is defined in the "obvious" way.

For example, let $R = \mathbb{Z}$ and $G = D_4 = \langle r, f \mid r^4 = f^2 = rfrf = 1\rangle$, and consider the elements $x = r + r^2 - 3f$ and $y = -5r^2 + rf$ in $\mathbb{Z}D_4$. Their sum is

$$x + y = r - 4r^2 - 3f + rf,$$

and their product is

$$\begin{aligned} xy &= (r + r^2 - 3f)(-5r^2 + rf) = r(-5r^2 + rf) + r^2(-5r^2 + rf) - 3f(-5r^2 + rf) \\ &= -5r^3 + r^2 f - 5r^4 + r^3 f + 15fr^2 - 3frf = -5 - 8r^3 + 16r^2 f + r^3 f. \end{aligned}$$

Remarks

- The (real) Hamiltonians $\mathbb{H}$ is *not* the same ring as $\mathbb{R}Q_8$.
- If $g \in G$ has finite order $|g| = k > 1$, then RG always has zero divisors:

$$(1 - g)(1 + g + \cdots + g^{k-1}) = 1 - g^k = 1 - 1 = 0.$$

- RG contains a subring isomorphic to R, and the group of units $U(RG)$ contains a subgroup isomorphic to G.

Types of rings

Definition

If all nonzero elements of R have a multiplicative inverse, then R is a division ring. (Think: "field without commutativity".)

An integral domain is a commutative ring with 1 and with no (nonzero) zero divisors. (Think: "field without inverses".)

A field is just a commutative division ring. Moreover:

$$\text{fields} \subsetneq \text{division rings}$$

$$\text{fields} \subsetneq \text{integral domains} \subsetneq \text{all rings}$$

Examples

- Rings that are not integral domains: $\mathbb{Z}_n$ (composite n), $2\mathbb{Z}$, $M_n(\mathbb{R})$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{H}$.
- Integral domains that are not fields (or even division rings): $\mathbb{Z}$, $\mathbb{Z}[x]$, $\mathbb{R}[x]$, $\mathbb{R}[[x]]$ (formal power series).
- Division ring but not a field: $\mathbb{H}$.

Cancellation

When doing basic algebra, we often take for granted basic properties such as cancellation: $ax = ay \implies x = y$. However, *this need not hold in all rings!*

Examples where cancellation fails

- In $\mathbb{Z}_6$, note that $2 = 2 \cdot 1 = 2 \cdot 4$, but $1 \neq 4$.
- In $M_2(\mathbb{R})$, note that $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$.

However, everything works fine as long as there aren't any (nonzero) zero divisors.

Proposition

Let R be an integral domain and $a \neq 0$. If $ax = ay$ for some $x, y \in R$, then $x = y$.

Proof

If $ax = ay$, then $ax - ay = a(x - y) = 0$.

Since $a \neq 0$ and R has no (nonzero) zero divisors, then $x - y = 0$. $\square$

Finite integral domains

Lemma (HW)

If R is an integral domain and $0 \neq a \in R$ and $k \in \mathbb{N}$, then $a^k \neq 0$. $\square$

Theorem

Every finite integral domain is a field.

Proof

Suppose R is a finite integral domain and $0 \neq a \in R$. It suffices to show that a has a multiplicative inverse.

Consider the infinite sequence $a, a^2, a^3, a^4, \dots,$ which must repeat.

Find $i > j$ with $a^i = a^j$, which means that

$$0 = a^i - a^j = a^j(a^{i-j} - 1).$$

Since R is an integral domain and $a^j \neq 0$, then $a^{i-j} = 1$.

Thus, $a \cdot a^{i-j-1} = 1$. $\square$

Ideals

In the theory of groups, we can quotient out by a subgroup if and only if it is a normal subgroup. The analogue of this for rings are (two-sided) ideals.

Definition

A subring $I \subseteq R$ is a left ideal if

$$rx \in I \qquad \text{for all } r \in R \text{ and } x \in I.$$

Right ideals, and two-sided ideals are defined similarly.

If R is commutative, then all left (or right) ideals are two-sided.

We use the term ideal and two-sided ideal synonymously, and write $I \trianglelefteq R$.

Examples

- $n\mathbb{Z} \trianglelefteq \mathbb{Z}$.
- If $R = M_2(\mathbb{R})$, then $I = \left\{ \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} : a, c \in \mathbb{R} \right\}$ is a left, but *not* a right ideal of R.
- The set $\mathrm{Sym}_n(\mathbb{R})$ of symmetric $n \times n$ matrices is a subring of $M_n(\mathbb{R})$, but *not* an ideal.

Ideals

Remark

If an ideal I of R contains 1, then $I = R$.

Proof

Suppose $1 \in I$, and take an arbitrary $r \in R$.

Then $r1 \in I$, and so $r1 = r \in I$. Therefore, $I = R$. $\square$

It is not hard to modify the above result to show that if I contains *any* unit, then $I = R$. (HW)

Let's compare the concept of a normal subgroup to that of an ideal:

- normal subgroups are characterized by being invariant under conjugation:

$$H \leq G \text{ is normal iff } ghg^{-1} \in H \text{ for all } g \in G,\ h \in H.$$

- (left) ideals of rings are characterized by being invariant under (left) multiplication:

$$I \subseteq R \text{ is a (left) ideal iff } ri \in I \text{ for all } r \in R,\ i \in I.$$

Ideals generated by sets

Definition

The left ideal generated by a set $X \subset R$ is defined as:

$$(X) := \bigcap \{I : \; I \text{ is a left ideal s.t. } X \subseteq I \subseteq R\}.$$

This is the smallest left ideal containing X.

There are analogous definitions by replacing "left" with "right" or "two-sided".

Recall the two ways to define the subgroup $\langle X \rangle$ generated by a subset $X \subseteq G$:

- "*Bottom up*": As the set of all finite products of elements in X;
- "*Top down*": As the intersection of all subgroups containing X.

Proposition (HW)

Let R be a ring *with unity*. The (left, right, two-sided) ideal generated by $X \subseteq R$ is:

- Left: $\{r_1x_1 + \cdots + r_nx_n : n \in \mathbb{N}, \; r_i \in R, \; x_i \in X\}$,
- Right: $\{x_1r_1 + \cdots + x_nr_n : n \in \mathbb{N}, \; r_i \in R, \; x_i \in X\}$,
- Two-sided: $\{r_1x_1s_1 + \cdots + r_nx_ns_n : n \in \mathbb{N}, \; r_i, s_i \in R, \; x_i \in X\}$.

Ideals and quotients

Since an ideal I of R is an additive subgroup (and hence normal), then:

- $R/I = \{x + I \mid x \in R\}$ is the set of cosets of I in R;
- R/I is a quotient group; with the binary operation (addition) defined as

$$(x + I) + (y + I) := x + y + I.$$

It turns out that if I is also a two-sided ideal, then we can make R/I into a ring.

Proposition

If $I \subseteq R$ is a (two-sided) ideal, then R/I is a ring (called a quotient ring), where multiplication is defined by

$$(x + I)(y + I) := xy + I\,.$$

Proof

We need to show this is well-defined. Suppose $x + I = r + I$ and $y + I = s + I$. This means that $x - r \in I$ and $y - s \in I$.

It suffices to show that $xy + I = rs + I$, or equivalently, $xy - rs \in I$:

$$xy - rs = xy - ry + ry - rs = (x - r)y + r(y - s) \in I\,.$$

Finite fields

We've already seen that $\mathbb{Z}_p$ is a field if p is prime, and that finite integral domains are fields. But *what do these "other" finite fields look like?*

Let $R = \mathbb{Z}_2[x]$ be the polynomial ring over the field $\mathbb{Z}_2$. (Note: we can ignore all negative signs.)

The polynomial $f(x) = x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$ because it does not have a root. (Note that $f(0) = f(1) = 1 \neq 0$.)

Consider the ideal $I = (x^2 + x + 1)$, the set of multiples of $x^2 + x + 1$.

In the quotient ring R/I, we have the relation $x^2 + x + 1 = 0$, or equivalently, $x^2 = -x - 1 = x + 1$.

The quotient has only 4 elements:

$$0 + I\,, \qquad 1 + I\,, \qquad x + I\,, \qquad (x + 1) + I\,.$$

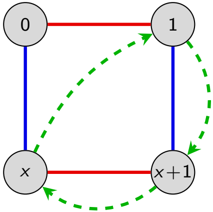
As with the quotient group (or ring) $\mathbb{Z}/n\mathbb{Z}$, we usually drop the "I", and just write

$$R/I = \mathbb{Z}_2[x]/(x^2 + x + 1) \cong \{0,\ 1,\ x,\ x + 1\}\,.$$

It is easy to check that this is a field!

Finite fields

Here is a Cayley diagram, and the operation tables for $R/I = \mathbb{Z}_2[x]/(x^2+x+1)$:



$+$	0	1	x	$x+1$
0	0	1	x	$x+1$
1	1	0	$x+1$	x
x	x	$x+1$	0	1
$x+1$	$x+1$	x	1	0

$\times$	1	x	$x+1$
1	1	x	$x+1$
x	x	$x+1$	1
$x+1$	$x+1$	1	x

Theorem

There exists a finite field $\mathbb{F}_q$ of order q, which is unique up to isomorphism, iff $q = p^n$ for some prime p. If $n > 1$, then this field is isomorphic to the quotient ring

$$\mathbb{Z}_p[x]/(f)\,,$$

where f is *any* irreducible polynomial of degree n.

Much of the error correcting techniques in coding theory are built using mathematics over $\mathbb{F}_{2^8} = \mathbb{F}_{256}$. This is what allows your CD to play despite scratches.

Motivation (spoilers!)

Many of the big ideas from group homomorphisms carry over to ring homomorphisms.

Group theory

- The quotient group G/N exists iff N is a normal subgroup.
- A homomorphism is a structure-preserving map: $f(x * y) = f(x) * f(y)$.
- The kernel of a homomorphism is a normal subgroup: $\operatorname{Ker}\phi \trianglelefteq G$.
- For every normal subgroup $N \trianglelefteq G$, there is a natural quotient homomorphism $\phi\colon G \to G/N,\ \ \phi(g) = gN$.
- There are four standard isomorphism theorems for groups.

Ring theory

- The quotient ring R/I exists iff I is a two-sided ideal.
- A homomorphism is a structure-preserving map: $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$.
- The kernel of a homomorphism is a two-sided ideal: $\operatorname{Ker}\phi \trianglelefteq R$.
- For every two-sided ideal $I \trianglelefteq R$, there is a natural quotient homomorphism $\phi\colon R \to R/I,\ \ \phi(r) = r + I$.
- There are four standard isomorphism theorems for rings.

Ring homomorphisms

Definition

A ring homomorphism is a function $f\colon R \to S$ satisfying

$$f(x+y) = f(x) + f(y) \qquad \text{and} \qquad f(xy) = f(x)f(y) \quad \text{for all } x, y \in R.$$

A ring isomorphism is a homomorphism that is bijective.

The kernel $f\colon R \to S$ is the set $\operatorname{Ker} f := \{x \in R : f(x) = 0\}$.

Examples

1. The function $\phi\colon \mathbb{Z} \to \mathbb{Z}_n$ that sends $k \mapsto k \pmod{n}$ is a ring homomorphism with $\operatorname{Ker}(\phi) = n\mathbb{Z}$.
2. For a fixed real number $\alpha \in \mathbb{R}$, the "evaluation function"

$$\phi\colon \mathbb{R}[x] \longrightarrow \mathbb{R}\,, \qquad\qquad \phi\colon p(x) \longmapsto p(\alpha)$$

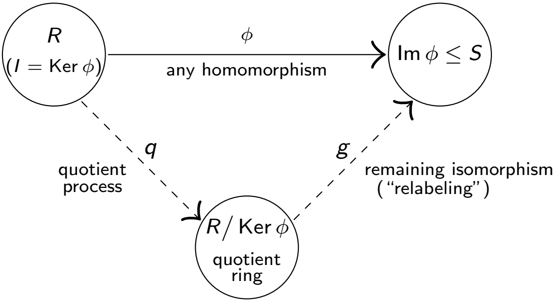
   is a homomorphism. The kernel consists of all polynomials that have α as a root.
3. The following is a homomorphism, for the ideal $I = (x^2 + x + 1)$ in $\mathbb{Z}_2[x]$:

$$\phi\colon \mathbb{Z}_2[x] \longrightarrow \mathbb{Z}_2[x]/I, \qquad\qquad f(x) \longmapsto f(x) + I\,.$$

The isomorphism theorems for rings

Fundamental homomorphism theorem

If $\phi\colon R \to S$ is a ring homomorphism, then $\operatorname{Ker}\phi$ is an ideal and $\operatorname{Im}(\phi) \cong R/\operatorname{Ker}(\phi)$.



Proof (HW)

The statement holds for the underlying additive group R. Thus, it remains to show that $\operatorname{Ker}\phi$ is a (two-sided) ideal, and the following map is a ring homomorphism:

$$g\colon R/I \longrightarrow \operatorname{Im}\phi\,, \qquad\qquad g(x+I) = \phi(x)\,.$$

The second isomorphism theorem for rings

Suppose S is a subring and I an ideal of R. Then

(i) The sum $S + I = \{s + i \mid s \in S,\ i \in I\}$ is a subring of R and the intersection $S \cap I$ is an ideal of S.

(ii) The following quotient rings are isomorphic:

$$(S + I)/I \cong S/(S \cap I)\,.$$

$$\begin{tikzcd}[arrows=-] & R \arrow[d] & \\ & S+I \arrow[dl] \arrow[dr] & \\ S \arrow[dr] & & I \arrow[dl] \\ & S \cap I & \end{tikzcd}$$

Proof (sketch)

$S + I$ is an additive subgroup, and it's closed under multiplication because

$$s_1, s_2 \in S,\ i_1, i_2 \in I \quad \Longrightarrow \quad (s_1 + i_1)(s_2 + i_2) = \underbrace{s_1 s_2}_{\in S} + \underbrace{s_1 i_2 + i_1 s_2 + i_1 i_2}_{\in I} \in S + I.$$

Showing $S \cap I$ is an ideal of S is straightforward (homework exercise).

We already know that $(S + I)/I \cong S/(S \cap I)$ as additive groups.

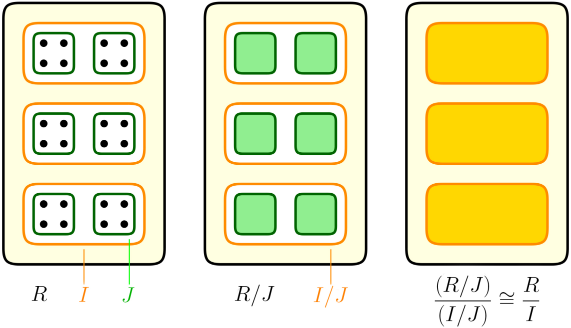
One explicit isomorphism is $\phi\colon s + (S \cap I) \mapsto s + I$. It is easy to check that $\phi\colon 1 \mapsto 1$ and ϕ preserves products. $\square$

The third isomorphism theorem for rings

Freshman theorem

Suppose R is a ring with ideals $J \subseteq I$. Then I/J is an ideal of R/J and

$$(R/J)/(I/J) \cong R/I\,.$$

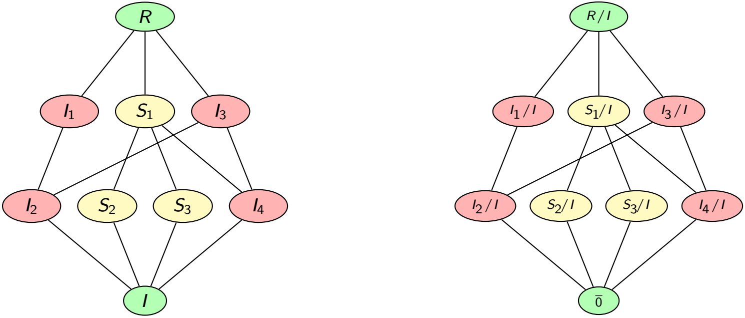


(Thanks to Zach Teitler of Boise State for the concept and graphic!)

The fourth isomorphism theorem for rings

Correspondence theorem

Let I be an ideal of R. There is a bijective correspondence between subrings (& ideals) of R/I and subrings (& ideals) of R that contain I. In particular, every ideal of R/I has the form J/I, for some ideal J satisfying $I \subseteq J \subseteq R$.



subrings & ideals that contain I

subrings & ideals of R/I

Maximal ideals

Definition

An ideal I of R is maximal if $I \neq R$ and if $I \subseteq J \subseteq R$ holds for some ideal J, then $J = I$ or $J = R$.

A ring R is simple if its only (two-sided) ideals are 0 and R.

Examples

1. If $n \neq 0$, then the ideal $M = (n)$ of $R = \mathbb{Z}$ is maximal if and only if n is prime.
2. Let $R = \mathbb{Q}[x]$ be the set of all polynomials over $\mathbb{Q}$. The ideal $M = (x)$ consisting of all polynomials with constant term zero is a maximal ideal.

   Elements in the quotient ring $\mathbb{Q}[x]/(x)$ have the form $f(x) + M = a_0 + M$.
3. Let $R = \mathbb{Z}_2[x]$, the polynomials over $\mathbb{Z}_2$. The ideal $M = (x^2 + x + 1)$ is maximal, and $R/M \cong \mathbb{F}_4$, the (unique) finite field of order 4.

In all three examples above, the quotient R/M is a field.

Maximal ideals

Theorem

Let R be a commutative ring with 1. The following are equivalent for an ideal $I \subseteq R$.

(i) I is a maximal ideal;

(ii) R/I is simple;

(iii) R/I is a field.

Proof

The equivalence (i)$\Leftrightarrow$(ii) is immediate from the Correspondence Theorem.

For (ii)$\Leftrightarrow$(iii), we'll show that an *arbitrary* ring R is simple iff R is a field.

"$\Rightarrow$": Assume R is simple. Then $(a) = R$ for any nonzero $a \in R$.

Thus, $1 \in (a)$, so $1 = ba$ for some $b \in R$, so $a \in U(R)$ and R is a field. ✓

"$\Leftarrow$": Let $I \subseteq R$ be a nonzero ideal of a field R. Take any nonzero $a \in I$.

Then $a^{-1}a \in I$, and so $1 \in I$, which means $I = R$. ✓ $\square$

Prime ideals

Definition

Let R be a commutative ring. An ideal $P \subset R$ is prime if $ab \in P$ implies either $a \in P$ or $b \in P$.

Note that $p \in \mathbb{N}$ is a prime number iff $p = ab$ implies either $a = p$ or $b = p$.

Examples

1. The ideal (n) of $\mathbb{Z}$ is a prime ideal iff n is a prime number (possibly $n = 0$).
2. In the polynomial ring $\mathbb{Z}[x]$, the ideal $I = (2, x)$ is a prime ideal. It consists of all polynomials whose constant coefficient is even.

Theorem

An ideal $P \subseteq R$ is prime iff R/P is an integral domain.

The proof is straightforward (HW). Since fields are integral domains, the following is immediate:

Corollary

In a commutative ring, every maximal ideal is prime.

Divisibility and factorization

A ring is in some sense, a generalization of the familiar number systems like $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$, where we are allowed to add, subtract, and multiply.

Two key properties about these structures are:

- multiplication is commutative,
- there are no (nonzero) zero divisors.

Blanket assumption

Throughout this lecture, unless explicitly mentioned otherwise, R is assumed to be an integral domain, and we will define $R^* := R \setminus \{0\}$.

The integers have several basic properties that we usually take for granted:

- every nonzero number can be factored uniquely into primes;
- any two numbers have a unique greatest common divisor and least common multiple;
- there is a Euclidean algorithm, which can find the gcd of two numbers.

Surprisingly, these need not always hold in integrals domains! We would like to understand this better.

Divisibility

Definition

If $a, b \in R$, say that a divides b, or b is a multiple of a if $b = ac$ for some $c \in R$. We write $a \mid b$.

If $a \mid b$ and $b \mid a$, then a and b are associates, written $a \sim b$.

Examples

- In $\mathbb{Z}$: n and $-n$ are associates.
- In $\mathbb{R}[x]$: $f(x)$ and $c \cdot f(x)$ are associates for any $c \neq 0$.
- The only associate of 0 is itself.
- The associates of 1 are the units of R.

Proposition (HW)

Two elements $a, b \in R$ are associates if and only if $a = bu$ for some unit $u \in U(R)$.

This defines an equivalence relation on R, and partitions R into equivalence classes.

Irreducibles and primes

Note that units divide everything: if $b \in R$ and $u \in U(R)$, then $u \mid b$.

Definition

If $b \in R$ is not a unit, and the only divisors of b are units and associates of b, then b is irreducible.

An element $p \in R$ is prime if p is not a unit, and $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Proposition

If $0 \neq p \in R$ is prime, then p is irreducible.

Proof

Suppose p is prime but not irreducible. Then $p = ab$ with $a, b \notin U(R)$.

Then (wlog) $p \mid a$, so $a = pc$ for some $c \in R$. Now,

$$p = ab = (pc)b = p(cb)\,.$$

This means that $cb = 1$, and thus $b \in U(R)$, a contradiction. $\square$

Irreducibles and primes

Caveat: Irreducible $\not\Rightarrow$ prime

Consider the ring $R_{-5} := \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$.

$$3 \mid (2+\sqrt{-5})(2-\sqrt{-5}) = 9 = 3 \cdot 3\,,$$

but $3 \nmid 2+\sqrt{-5}$ and $3 \nmid 2-\sqrt{-5}$.

Thus, 3 is irreducible in R_{-5} but *not* prime.

When irreducibles fail to be prime, we can lose nice properties like unique factorization.

Things can get really bad: not even the *lengths* of factorizations into irreducibles need be the same!

For example, consider the ring $R = \mathbb{Z}[x^2, x^3]$. Then

$$x^6 = x^2 \cdot x^2 \cdot x^2 = x^3 \cdot x^3.$$

The element $x^2 \in R$ is not prime because $x^2 \mid x^3 \cdot x^3$ yet $x^2 \nmid x^3$ in R (note: $x \notin R$).

Principal ideal domains

Fortunately, there is a type of ring where such "bad things" don't happen.

Definition

An ideal I generated by a single element $a \in R$ is called a principal ideal. We denote this by $I = (a)$.

If every ideal of R is principal, then R is a principal ideal domain (PID).

Examples

The following are all PIDs (stated without proof):

- The ring of integers, $\mathbb{Z}$.
- Any field F.
- The polynomial ring $F[x]$ over a field.

As we will see shortly, PIDs are "nice" rings. Here are some properties they enjoy:

- pairs of elements have a "greatest common divisor" & "least common multiple";
- irreducible $\Rightarrow$ prime;
- Every element factors uniquely into primes.

Greatest common divisors & least common multiples

Proposition

If $I \subseteq \mathbb{Z}$ is an ideal, and $a \in I$ is its smallest positive element, then $I = (a)$.

Proof

Pick any positive $b \in I$. Write $b = aq + r$, for $q, r \in \mathbb{Z}$ and $0 \leq r < a$.

Then $r = b - aq \in I$, so $r = 0$. Therefore, $b = qa \in (a)$. $\square$

Definition

A common divisor of $a, b \in R$ is an element $d \in R$ such that $d \mid a$ and $d \mid b$.

Moreover, d is a greatest common divisor (GCD) if $c \mid d$ for all other common divisors c of a and b.

A common multiple of $a, b \in R$ is an element $m \in R$ such that $a \mid m$ and $b \mid m$.

Moreover, m is a least common multiple (LCM) if $m \mid n$ for all other common multiples n of a and b.

Nice properties of PIDs

Proposition

If R is a PID, then any $a, b \in R^*$ have a GCD, $d = \gcd(a, b)$.

It is *unique up to associates*, and can be written as $d = xa + yb$ for some $x, y \in R$.

Proof

Existence. The ideal generated by a and b is

$$I = (a, b) = \{ua + vb : u, v \in R\}.$$

Since R is a PID, we can write $I = (d)$ for some $d \in I$, and so $d = xa + yb$.

Since $a, b \in (d)$, both $d \mid a$ and $d \mid b$ hold.

If c is a divisor of a & b, then $c \mid xa + yb = d$, so d is a GCD for a and b. ✓

Uniqueness. If d' is another GCD, then $d \mid d'$ and $d' \mid d$, so $d \sim d'$. ✓ □

Nice properties of PIDs

Corollary

If R is a PID, then every irreducible element is prime.

Proof

Let $p \in R$ be irreducible and suppose $p \mid ab$ for some $a, b \in R$.

If $p \nmid a$, then $\gcd(p, a) = 1$, so we may write $1 = xa + yp$ for some $x, y \in R$. Thus

$$b = (xa + yp)b = x(ab) + (yb)p\,.$$

Since $p \mid x(ab)$ and $p \mid (yb)p$, then $p \mid x(ab) + (yb)p = b$. $\square$

Not surprisingly, least common multiples also have a nice characterization in PIDs.

Proposition (HW)

If R is a PID, then any $a, b \in R^*$ have an LCM, $m = \operatorname{lcm}(a, b)$.

It is *unique up to associates*, and can be characterized as a generator of the ideal $I := (a) \cap (b)$.

Unique factorization domains

Definition

An integral domain is a unique factorization domain (UFD) if:

(i) Every nonzero element is a product of irreducible elements;

(ii) Every irreducible element is prime.

Examples

1. $\mathbb{Z}$ is a UFD: Every integer $n \in \mathbb{Z}$ can be uniquely factored as a product of irreducibles (primes):
$$n = p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k}.$$
This is the *fundamental theorem of arithmetic*.

2. The ring $\mathbb{Z}[x]$ is a UFD, because every polynomial can be factored into irreducibles. But it is not a PID because the following ideal is not principal:
$$(2, x) = \{f(x) : \text{ the constant term is even}\}.$$

3. The ring R_{-5} is not a UFD because $9 = 3 \cdot 3 = (2 + \sqrt{-5})(2 - \sqrt{-5})$.

4. We've shown that (ii) holds for PIDs. Next, we will see that (i) holds as well.

Unique factorization domains

Theorem

If R is a PID, then R is a UFD.

Proof

We need to show Condition (i) holds: every element is a product of irreducibles. A ring is Noetherian if every ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

stabilizes, meaning that $I_k = I_{k+1} = I_{k+2} = \cdots$ holds for some k.

Suppose R is a PID. It is not hard to show that R is Noetherian (HW). Define

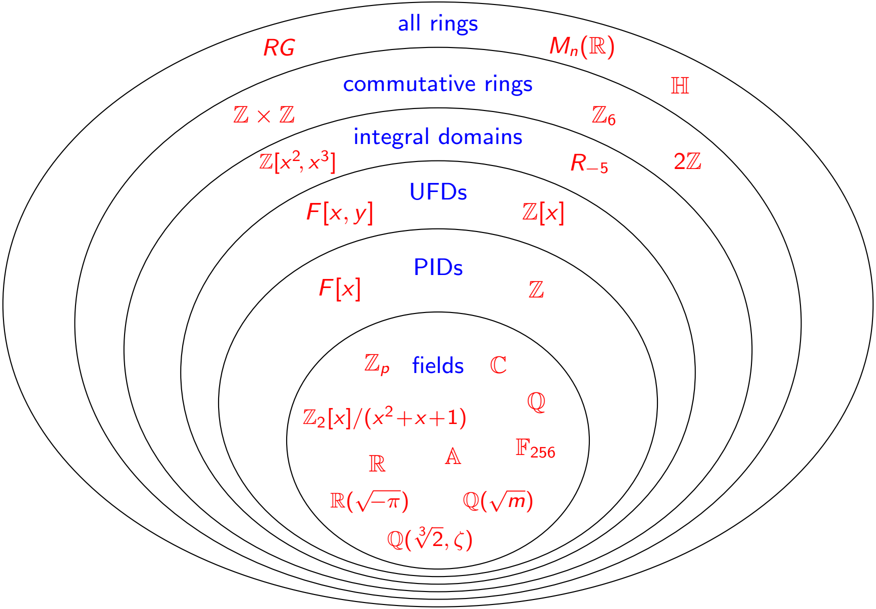
$$X = \{a \in R^* \setminus U(R) : a \text{ can't be written as a product of irreducibles}\}.$$

If $X \neq \emptyset$, then pick $a_1 \in X$. Factor this as $a_1 = a_2 b$, where $a_2 \in X$ and $b \notin U(R)$. Then $(a_1) \subsetneq (a_2) \subsetneq R$, and repeat this process. We get an ascending chain

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots$$

that does not stabilize. This is impossible in a PID, so $X = \emptyset$. $\square$

Summary of ring types



The Euclidean algorithm

Around 300 B.C., Euclid wrote his famous book, the *Elements*, in which he described what is now known as the Euclidean algorithm:

Proposition VII.2 (Euclid's *Elements*)

Given two numbers not prime to one another, to find their greatest common measure.

The algorithm works due to two key observations:

- If $a \mid b$, then $\gcd(a, b) = a$;
- If $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

This is best seen by an example: Let $a = 654$ and $b = 360$.

$$\begin{aligned}
654 &= 360 \cdot 1 + 294 & \qquad \gcd(654, 360) &= \gcd(360, 294)\\
360 &= 294 \cdot 1 + 66 & \gcd(360, 294) &= \gcd(294, 66)\\
294 &= 66 \cdot 4 + 30 & \gcd(294, 66) &= \gcd(66, 30)\\
66 &= 30 \cdot 2 + 6 & \gcd(66, 30) &= \gcd(30, 6)\\
30 &= 6 \cdot 5 & \gcd(30, 6) &= 6.
\end{aligned}$$

We conclude that $\gcd(654, 360) = 6$.

Euclidean domains

Loosely speaking, a Euclidean domain is any ring for which the Euclidean algorithm still works.

Definition

An integral domain R is Euclidean if it has a degree function $d \colon R^* \to \mathbb{Z}$ satisfying:

(i) non-negativity: $d(r) \geq 0 \quad \forall r \in R^*$.

(ii) monotonicity: $d(a) \leq d(ab)$ for all $a, b \in R^*$.

(iii) division-with-remainder property: For all $a, b \in R$, $b \neq 0$, there are $q, r \in R$ such that

$$a = bq + r \qquad \text{with} \qquad r = 0 \quad \text{or} \quad d(r) < d(b) \,.$$

Note that Property (ii) could be restated to say: *If $a \mid b$, then $d(a) \leq d(b)$*;

Examples

- $R = \mathbb{Z}$ is Euclidean. Define $d(r) = |r|$.
- $R = F[x]$ is Euclidean if F is a field. Define $d(f(x)) = \deg f(x)$.
- The Gaussian integers $R_{-1} = \mathbb{Z}[\sqrt{-1}] = \{a + bi : a, b \in \mathbb{Z}\}$ is Euclidean with degree function $d(a + bi) = a^2 + b^2$.

Euclidean domains

Proposition

If R is Euclidean, then $U(R) = \{x \in R^* : d(x) = d(1)\}$.

Proof

$\subseteq$": First, we'll show that associates have the same degree. Take $a \sim b$ in R^*:

$$\left.\begin{array}{lcl} a \mid b & \Longrightarrow & d(a) \leq d(b) \\ b \mid a & \Longrightarrow & d(b) \leq d(a) \end{array}\right. \qquad \Longrightarrow \qquad d(a) = d(b).$$

If $u \in U(R)$, then $u \sim 1$, and so $d(u) = d(1)$. ✓

"$\supseteq$": Suppose $x \in R^*$ and $d(x) = d(1)$.

Then $1 = qx + r$ for some $q \in R$ with either $r = 0$ or $d(r) < d(x) = d(1)$.

If $r \neq 0$, then $d(1) \leq d(r)$ since $1 \mid r$.

Thus, $r = 0$, and so $qx = 1$, hence $x \in U(R)$. ✓ $\square$

Euclidean domains

Proposition

If R is Euclidean, then R is a PID.

Proof

Let $I \neq 0$ be an ideal and pick some $b \in I$ with $d(b)$ minimal.

Pick $a \in I$, and write $a = bq + r$ with either $r = 0$, or $d(r) < d(b)$.

This latter case is impossible: $r = a - bq \in I$, and by minimality, $d(b) \leq d(r)$.

Therefore, $r = 0$, which means $a = bq \in (b)$. Since a was arbitrary, $I = (b)$. $\square$

Exercises.

(i) The ideal $I = (3, 2 + \sqrt{-5})$ is not principal in R_{-5}.

(ii) If R is an integral domain, then $I = (x, y)$ is not principal in $R[x, y]$.

Corollary

The rings R_{-5} (not a PID or UFD) and $R[x, y]$ (not a PID) are not Euclidean.

Algebraic integers

The algebraic integers are the roots of *monic* polynomials in $\mathbb{Z}[x]$. This is a subring of the algebraic numbers (roots of all polynomials in $\mathbb{Z}[x]$).

Assume $m \in \mathbb{Z}$ is square-free with $m \neq 0, 1$. Recall the quadratic field

$$\mathbb{Q}(\sqrt{m}) = \left\{ p + q\sqrt{m} \mid p, q \in \mathbb{Q} \right\}.$$

Definition

The ring R_m is the set of algebraic integers in $\mathbb{Q}(\sqrt{m})$, i.e., the subring consisting of those numbers that are roots of monic quadratic polynomials $x^2 + cx + d \in \mathbb{Z}[x]$.

Facts

- R_m is an integral domain with 1.
- Since m is square-free, $m \not\equiv 0 \pmod 4$. For the other three cases:

$$R_m = \begin{cases} \mathbb{Z}[\sqrt{m}] = \left\{ a + b\sqrt{m} : a, b \in \mathbb{Z} \right\} & m \equiv 2 \text{ or } 3 \pmod 4 \\ \mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = \left\{ a + b\left(\frac{1+\sqrt{m}}{2}\right) : a, b \in \mathbb{Z} \right\} & m \equiv 1 \pmod 4 \end{cases}$$

- R_{-1} is the Gaussian integers, which is a PID. (easy)
- R_{-19} is a PID. (hard)

Algebraic integers

Definition

For $x = r + s\sqrt{m} \in \mathbb{Q}(\sqrt{m})$, define the norm of x to be

$$N(x) = (r + s\sqrt{m})(r - s\sqrt{m}) = r^2 - ms^2 .$$

R_m is norm-Euclidean if it is a Euclidean domain with $d(x) = |N(x)|$.

Note that the norm is multiplicative: $N(xy) = N(x)N(y)$.

Exercises

Assume $m \in \mathbb{Z}$ is square-free, with $m \neq 0, 1$.

- $u \in U(R_m)$ iff $|N(u)| = 1$.
- If $m \geq 2$, then $U(R_m)$ is infinite.
- $U(R_{-1}) = \{\pm 1, \pm i\}$ and $U(R_{-3}) = \left\{ \pm 1, \pm \frac{1 \pm \sqrt{-3}}{2} \right\}$.
- If $m = -2$ or $m < -3$, then $U(R_m) = \{\pm 1\}$.

Euclidean domains and algebraic integers

Theorem

R_m is norm-Euclidean iff

$$m \in \{-11, -7, -3, -2, -1, 2, 3, 5, 6, 7, 11, 13, 17, 19, 21, 29, 33, 37, 41, 57, 73\}.$$

Theorem (D.A. Clark, 1994)

The ring R_{69} is a Euclidean domain that is *not* norm-Euclidean.

Let $\alpha = (1 + \sqrt{69})/2$ and $c > 25$ be an integer. Then the following degree function works for R_{69}, defined on the prime elements:

$$d(p) = \begin{cases} |N(p)| & \text{if } p \neq 10 + 3\alpha \\ c & \text{if } p = 10 + 3\alpha \end{cases}$$

Theorem

If $m < 0$ and $m \notin \{-11, -7, -3, -2, -1\}$, then R_m is not Euclidean.

Open problem

Classify which R_m's are PIDs, and which are Euclidean.

PIDs that are not Euclidean

Theorem

If $m < 0$, then R_m is a PID iff

$$m \in \{\underbrace{-1, -2, -3, -7, -11}_{\text{Euclidean}}, -19, -43, -67, -163\}.$$

Recall that R_m is norm-Euclidean iff

$$m \in \{-11, -7, -3, -2, -1, 2, 3, 5, 6, 7, 11, 13, 17, 19, 21, 29, 33, 37, 41, 57, 73\}.$$

Corollary

If $m < 0$, then R_m is a PID that is not Euclidean iff $m \in \{-19, -43, -67, -163\}$.

Algebraic integers

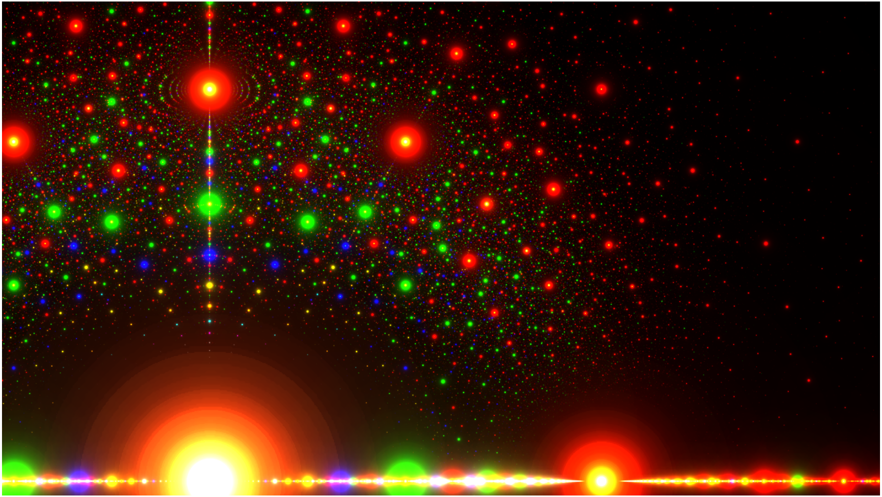


Figure: Algebraic numbers in the complex plane. Colors indicate the coefficient of the leading term: red $=1$ (algebraic integer), green $=2$, blue $=3$, yellow $=4$. Large dots mean fewer terms and smaller coefficients. Image from Wikipedia (made by Stephen J. Brooks).

Algebraic integers

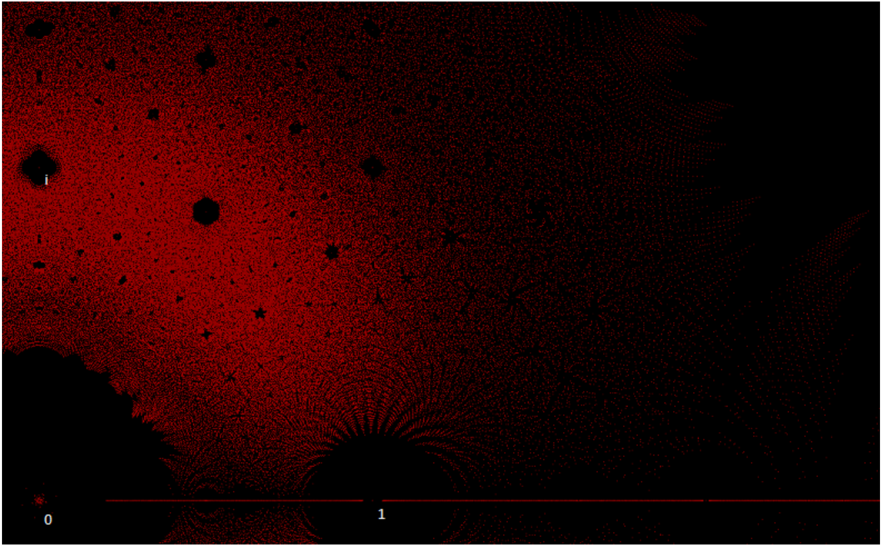


Figure: Algebraic integers in the complex plane. Each red dot is the root of a monic polynomial of degree ≤ 7 with coefficients from $\{0, \pm1, \pm2, \pm3, \pm4, \pm5\}$. From Wikipedia.

Summary of ring types